Exchange Rate Disconnect with Disagreeing Arbitrageurs

Rikuto Onishi*

Preliminary and Incomplete

Click here for the latest version

Abstract

I develop a two-country general equilibrium model in which international arbitrageurs disagree about future exchange rates. Trading gains and losses redistribute wealth across arbitrageurs, changing whose beliefs determine equilibrium prices. This mechanism amplifies exchange rate responses to standard macroeconomic shocks and generates an endogenous wedge in uncovered interest parity, without requiring a separate financial shock. Calibrated to macroeconomic, financial, and survey data, the model accounts for exchange rates' high volatility and weak relationship with observable fundamentals. The calibrated wealth-share multiplier implies that amplification accounts for approximately 70 percent of exchange rate variance.

1 Introduction

Exchange rates remain difficult to reconcile with standard international macroeconomic models. They are highly volatile and close to random walks, observable macroeconomic fundamentals explain little of their movements, interest differentials often predict currency movements in the opposite direction from uncovered interest parity (UIP), and exchange rates display much weaker comovement with relative consumption than standard risk-sharing models

*The University of Texas at Austin. Email: rikuto.onishi@utexas.edu. I am grateful for the continuous guidance and support of Olivier Coibion (co-chair), Stefano Eusepi (co-chair), and Christoph Boehm. I am especially grateful to Saroj Bhattarai, Masao Fukui (external committee), and Oliver Pfäuti for their generous time, close attention to this project, and many thoughtful discussions throughout its development. I also thank Michael Bauer, Ryan Chahrour, Jaedo Choi, Andrés Drenik, Xavier Gabaix, Oleg Itskhoki, Rohan Kekre, Husang Kim, Zheng Liu, Thomas Mertens, Nitya Pandalai-Nayar, Thuy Lan Nguyen, Damjan Pfajfar, Sanjay Singh, Mark Spiegel, and Yeji Sung for valuable comments and conversations, as well as seminar and conference participants at numerous institutions and meetings. All remaining errors are my own.

imply. Recent quantitative models have made substantial progress by combining segmented international financial markets with exogenous shocks to currency demand or intermediaries' risk-bearing capacity, which generate movements in the UIP wedge. This paper provides an alternative approach that instead endogenizes the UIP wedge through disagreement among international arbitrageurs and the resulting changes in their wealth shares.

Consider two groups of arbitrageurs who disagree about the future value of a currency. Investors who are relatively optimistic about the currency take larger long positions, while pessimistic investors take smaller or short positions. When a macroeconomic shock causes the currency to appreciate, optimistic investors earn higher returns and gain wealth relative to pessimistic investors. Since arbitrageurs' capacity to bear risk increases with their wealth, the beliefs of the optimists receive greater weight in equilibrium asset demand. Their increased influence raises demand for the currency and amplifies its initial appreciation. The additional currency movement further redistributes wealth toward the investors whose positions performed well. Through this wealth-revaluation channel, disagreement changes the transmission of macroeconomic shocks even when average beliefs are rational.

I formalize this mechanism in a two-country general equilibrium model with segmented asset markets and risk-averse international arbitrageurs. The framework builds on Gabaix and Maggiori (2015) and Itskhoki and Mukhin (2021). Home households trade bonds denominated in their own consumption bundle, Foreign households trade bonds denominated in theirs, and international arbitrageurs can hold both assets. Arbitrageurs take currency positions according to their subjective expected excess returns, and their risk tolerance is proportional to their wealth. Heterogeneous beliefs generate differences in currency positions, while realized returns change the distribution of wealth across arbitrageurs.

This environment generates an endogenous UIP wedge. Market clearing depends on the wealth-weighted average of arbitrageurs' expectations. When investors who are relatively optimistic about a currency hold a larger share of intermediary wealth, their expectations receive greater weight in equilibrium pricing. The resulting difference between wealth-weighted and average beliefs creates a wealth-disagreement wedge in expected currency returns. Since realized exchange-rate movements redistribute wealth across arbitrageurs, this wedge evolves endogenously in response to macroeconomic shocks.

A simplified version of the model provides an analytical characterization of this mechanism. The equilibrium can be reduced to three equations governing the interest-rate differential, the exchange rate, and the wealth share of relatively optimistic arbitrageurs. These equations deliver a closed-form wealth-share multiplier. Fundamental news moves the exchange rate, the resulting return redistributes arbitrageur wealth, and the change in wealth shares feeds back into equilibrium pricing. Solving this feedback yields an exchange-rate

response that exceeds the response in an economy without the wealth-revaluation channel. The strength of the multiplier increases with disagreement and decreases with arbitrageurs' effective risk aversion and the rate at which wealth differences dissipate. Goods-market adjustment places an upper bound on the multiplier.

The same mechanism generates several dimensions of exchange-rate disconnect. When macroeconomic fundamentals and arbitrageur wealth shares are sufficiently persistent, exchange rates approach random walks even though they continue to respond to macroeconomic shocks. The endogenous UIP wedge can also produce a negative Fama coefficient. At the same time, amplification of exchange-rate movements relative to consumption can generate a negative Backus-Smith correlation. These results provide conditions under which a common wealth-revaluation mechanism produces exchange-rate volatility, UIP deviations, and weak exchange-rate comovement with macroeconomic quantities.

I next examine whether the mechanism is quantitatively important when its central parameters are disciplined by independent data. I measure disagreement using individual forecasts from the Blue Chip Financial Forecasts, which provide cross-sectional information on expectations about the U.S. dollar. I discipline intermediary wealth using the equity capital of U.S. broker-dealers relative to GDP. Macroeconomic quantities and yields discipline the fundamental shock processes and the remaining parameters. In this way, survey data determine the magnitude and persistence of disagreement, while intermediary data discipline the persistence of the wealth distribution.

The central quantitative benchmark imposes full-information rational expectations on average across arbitrageurs and does not require an exogenous UIP shock. The calibrated model reproduces the main features of exchange-rate disconnect: exchange rates are highly volatile relative to consumption and interest differentials, are close to random walks, and generate negative Fama and Backus-Smith coefficients. The role of disagreement is especially clear in the counterfactuals. Holding other parameters fixed, eliminating disagreement reduces quarterly exchange-rate volatility by 62 percent and reverses the signs of both the Fama and Backus-Smith coefficients. Re-estimating the conventional macroeconomic shock processes in the economy without disagreement does not recover the same joint behavior of exchange rates.

The quantitative results also clarify the role played by the underlying shocks. In the richer specification, conventional supply and demand shocks account for approximately 80 percent of short-horizon exchange-rate variation. Disagreement affects how strongly these shocks are transmitted to currency prices. A macroeconomic shock generates a currency return, the return redistributes wealth across arbitrageurs with different beliefs, and the new wealth distribution changes equilibrium asset demand. Direct belief innovations contribute

little to unconditional exchange-rate variance. The model therefore preserves an important role for conventional macroeconomic shocks while allowing their exchange-rate effects to be substantially amplified through financial markets.

I provide additional empirical evidence for two elements of the mechanism using data separate from those used to discipline the quantitative model. First, I examine whether survey expectations are reflected in actual currency positions. Using Bloomberg forecasts and CFTC dealer positions for six major currencies, I find that revisions in consensus exchange-rate forecasts predict changes in dealer positions after controlling for contemporaneous and lagged spot-rate movements. The relationship is positive for all six currencies. Second, the model predicts that greater disagreement increases conditional exchange-rate volatility because larger differences in beliefs generate larger differences in positions and stronger wealth redistribution following currency movements. Consistent with this prediction, unusually high survey disagreement predicts greater subsequent realized currency volatility. A one-residual-standard-deviation increase in disagreement predicts approximately 2 percent higher realized volatility over the following year. These exercises provide supporting evidence for the links between expectations, currency positions, and subsequent currency risk that underlie the model.

The paper contributes first to the literature on exchange-rate determination in segmented financial markets, including Gabaix and Maggiori (2015), Itskhoki and Mukhin (2021, 2025), and Kekre and Lenel (2024). These papers show how limited international risk-bearing allows shocks to currency demand or financial intermediation to generate large exchange-rate movements. In the present framework, disagreement and endogenous wealth redistribution generate fluctuations in the UIP wedge following standard macroeconomic shocks. This provides a channel through which supply and demand shocks can produce large currency movements even when the corresponding movements in aggregate macroeconomic quantities and interest rates remain modest[1].

The paper also contributes to a long literature studying subjective and heterogeneous beliefs in exchange-rate determination, including Evans and Lyons (2002), Gourinchas and Tornell (2004), Bacchetta and Van Wincoop (2006), Du, Eusepi and Preston (2021), Candian and De Leo (2025), and Molavi, Tahbaz-Salehi and Vedolin (2024). The distinctive element here is the interaction between disagreement and endogenous investor wealth. Investors' past positions determine how shocks redistribute wealth, and these wealth changes subsequently alter the influence of different beliefs on equilibrium prices. The distribution of wealth across

[1]Other explanations for UIP deviations emphasize convenience yields (Engel, 2016; Valchev, 2020; Jiang et al., 2024) or portfolio adjustment costs (Chang et al., 2015; Bacchetta and Van Wincoop, 2021). These approaches give economic content to the wedge between expected currency returns. My contribution is to identify a distinct source of that wedge, and evaluate its quantitative implications in general equilibrium.

investors with different beliefs therefore becomes a state variable linking past currency returns to current exchange-rate determination.

This paper also relates to recently accumulating evidence that suggests that exchange rates are connected to macroeconomic fundamentals (Hassan, Mertens and Zhang, 2016; Lilley, Maggiori, Neiman and Schreger, 2022; Engel and Wu, 2024; Kekre and Lenel, 2024; Chahrour, Cormun, De Leo, Guerrón-Quintana and Valchev, 2024; Stavrakeva and Tang, 2024). For the current mechanism, Stavrakeva and Tang (2024)'s finding that lagged macroeconomic news explain majority of exchange rate variations is particularly important. Their finding suggests that UIP wedge is driven by a history-dependent component, which in my case is the wealth-disagreement wedge. Therefore, my mechanism implies explainability of exchange rates by lagged macroeconomic news almost by construction, and the relevance of average belief bias introduced in the quantitative section further confirms their point.

2 Model

This section outlines a two-country general equilibrium model with segmented asset markets and risk averse international arbitrageurs who try to make profits from carry trades. This framework follows Itskhoki and Mukhin (2021); Kekre and Lenel (2024), but deviates from their setup by (i) assuming risk tolerance proportional to arbitrageur wealth and (ii) allowing arbitrageurs to hold heterogeneous, potentially subjective beliefs.

2.1 Environment

Time is discrete, $t = 0, 1, 2, \ldots$. There are two countries, Home and Foreign, comprised of a measure 1 households. The economy features three sets of agents: Home households, Foreign households, and arbitrageurs, whose roles are detailed below. Home and Foreign households share common beliefs, while arbitrageurs may hold heterogeneous and potentially subjective beliefs.

Home households The representative Home household chooses consumption and Home bond holdings to maximize expected lifetime utility,

$$\max_{\{c_{t+s},\, b_{t+s}\}_{s\geq 0}} \mathbb{E}_t \left[\sum_{s=0}^{\infty} \left(\prod_{k=0}^{s-1} \beta_{t+k} \right) \frac{c_{t+s}^{1-\sigma}}{1-\sigma} \right],$$

subject to its budget constraint below at every date and a no-Ponzi condition, given initial bond holdings. Here β_t is the time-varying subjective discount factor from date t to $t+1$,

and $\sigma > 0$ is relative risk aversion. Expectations reflect the households' common beliefs, and the empty product for $s = 0$ equals one. The consumption basket is a CES aggregator over domestic Home goods and imported Foreign goods,

$$c_t = \left[(1-\gamma)^{\frac{1}{\theta}} (c_{Ht})^{\frac{\theta-1}{\theta}} + \gamma^{\frac{1}{\theta}} (c_{Ft})^{\frac{\theta-1}{\theta}} \right]^{\frac{\theta}{\theta-1}}.$$

Here, θ denotes the trade elasticity, which governs the elasticity of substitution across domestic goods and imports, and $\gamma \in (0, 1/2)$ is the import expenditure share at equal prices; a lower value means greater consumption home bias.

At the beginning of periods t, the representative Home household is endowed with predetermined position in Home bond holdings, b_{t-1}. This bond is denominated in the Home consumption bundle, paying a risk-free interest rate r_t at $t+1$. Intermediate production is exogenous in this economy. The Home household receives an endowment of the Home intermediate good, y_t, which it sells at producer price p_t measured in units of its consumption bundle. It also receives profit transfers π_t from the domestic final good firms it owns. It also receives net profits from intermediaries, π_t^a, which are described further below. It can either consume their wealth or save it in the form of Home bonds. Putting all together, the budget constraint of Home household is

$$c_t + \frac{b_t}{1+r_t} = p_t y_t + b_{t-1} + \pi_t + \pi_t^a. \tag{1}$$

The households minimize expenditure, resulting in the conventional constant elasticity demand schedules:

$$c_{Ht} = (1-\gamma) (p_{Ht})^{-\theta} c_t, \qquad c_{Ft} = \gamma (p_{Ft})^{-\theta} c_t.$$

Foreign households The representative Foreign household solves the symmetric intertemporal problem,

$$\max_{\{c^*_{t+s}, b^*_{t+s}\}_{s\geq 0}} \mathbb{E}_t \left[\sum_{s=0}^{\infty} \left(\prod_{k=0}^{s-1} \beta^*_{t+k} \right) \frac{(c^*_{t+s})^{1-\sigma}}{1-\sigma} \right],$$

subject to its budget constraint below at every date and a no-Ponzi condition, given initial Foreign bond holdings. The one-period subjective discount factor is β^*_t, and Foreign counterparts are denoted by a superscript $*$. The consumption basket is a CES aggregator over domestic Foreign goods and imported Home goods,

$$c^*_t = \left[(1-\gamma)^{\frac{1}{\theta}} (c^*_{Ft})^{\frac{\theta-1}{\theta}} + \gamma^{\frac{1}{\theta}} (c^*_{Ht})^{\frac{\theta-1}{\theta}} \right]^{\frac{\theta}{\theta-1}}.$$

Denoting b_t^* the household's saving in its Foreign domestic bond denominated by its overall consumption bundle which pays r_t^* at $t+1$, y_t^* its endowment of the Foreign intermediate good, and π_t^* its transfers from domestic firms, its budget constraint is

$$c_t^* + \frac{b_t^*}{1+r_t^*} = p_t^* y_t^* + \pi_t^* + b_{t-1}^*,$$

where p_{Ft}^* (p_{Ht}^*) is the price of the domestic (imported) bundle, and p_t^* the producer price of its endowment, relative to its overall consumption bundle. The CES demands are given by

$$c_{Ht}^* = \gamma \left(p_{Ht}^*\right)^{-\theta} c_t^*,\ c_{Ft}^* = (1-\gamma)\left(p_{Ft}^*\right)^{-\theta} c_t^*.$$

We have two Euler equations from households:

$$\beta_t \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \right] (1+r_t) = 1$$
$$\beta_t^* \mathbb{E}_t \left[\left(\frac{c_{t+1}^*}{c_t^*} \right)^{-\sigma} \right] (1+r_t^*) = 1$$

Arbitrageurs A third set of agents, arbitrageurs, trade in both Home and Foreign bonds. There is a continuum of arbitrageurs in the unit interval $\iota \in [0,1]$. Each individual arbitrageur is infinitely small, taking prices as exogenously given. For concreteness, they are comprised of two groups, denoted by $i \in \{A, B\}$, whose only ex-ante heterogeneity comes from their expectations. We let $\eta_i \geq 0$ denote the mass of arbitrageurs of group $i \in \{A, B\}$, where $\eta_A + \eta_B = 1$.[2]

They invest their current wealth in the risk-free Home bond and take a self-financing carry position by going long the Foreign bond and short an equal value of the Home bond, or vice versa. We denote arbitrageur ι belonging to group i's carry positions in the corresponding bonds as $b_t^{*i}(\iota)$ and $b_t^i(\iota)$. The carry positions satisfy $\frac{b_t^i(\iota)}{1+r_t} = -\frac{e_t b_t^{*i}(\iota)}{1+r_t^*}$, where e_t is the price of the Foreign consumption bundle in units of the Home bundle (the real exchange rate). Thus, the increase of e_t represents the appreciation of Foreign currency. I denote the size of this position in Home consumption units as $n_t^i(\iota) = \frac{e_t b_t^{*i}(\iota)}{1+r_t^*}$. Arbitrageurs have a discrete time version of the perpetual youth structure, with a constant exit probability ξ. When they exit, their accumulated wealth is rebated back to Home households. While they survive, they retain their wealth and rebalance their carry positions each period.

[2]Two groups suffice to study the wealth-disagreement mechanism. The portfolio aggregation argument extends to any finite number of groups; the analytical results below use two groups and the symmetry restrictions stated in Section 2.2.

Following Kekre and Lenel (2026a), arbitrageurs have a myopic mean-variance objective over next-period wealth. At each date t, arbitrageur ι in group i chooses its current position according to

$$\max_{n_t^i(\iota)} \left\{ E_t^i \left[w_{t+1}^i(\iota) \right] - \frac{\nu}{2(1+r_t) w_t^i(\iota)} \mathrm{Var}_t^i \left[w_{t+1}^i(\iota) \right] \right\},$$

where $\nu > 0$ governs risk aversion, gross current wealth $(1+r_t)w_t^i(\iota) > 0$ is taken as given, and conditional moments reflect group i's beliefs. Risk tolerance is proportional to current wealth. This objective can be motivated by a second-order certainty-equivalent approximation to CRRA utility over next-period wealth, with risk tolerance evaluated at risk-free next-period wealth. Myopic portfolio choice rules out intertemporal hedging demand by assumption. The choice is subject to the law of motion for individual wealth before exit,

$$w_{t+1}^i(\iota) = w_t^i(\iota)(1+r_t) + n_t^i(\iota) \left(\frac{e_{t+1}}{e_t}(1+r_t^*) - (1+r_t) \right).$$

Let W_t^i and N_t^i denote wealth and the Home-valued carry position per member of group i, measured after exit and entry at date t. Aggregate arbitrageur wealth and the aggregate carry position are $W_t = \sum_{i \in \{A,B\}} \eta_i W_t^i$ and $N_t = \sum_{i \in \{A,B\}} \eta_i N_t^i$. Averaging individual wealth within each group and accounting for exit and entry gives

$$W_{t+1}^i = (1-\xi) \left(W_t^i (1+r_t) + N_t^i \left(\frac{e_{t+1}}{e_t}(1+r_t^*) - (1+r_t) \right) \right) + \xi W_0^i.$$

Here, a fraction ξ of arbitrageurs exit the market at the beginning of each period, and they are replaced by the same mass of newborn arbitrageurs, each endowed with actual wealth W_0^i, where the constant endowment is $W_0^i = W_0$ common to both groups. The entrants' contribution to average group wealth, ξW_0^i, enters the law of motion for wealth, and is financed by Home households. Once they exit the market, the accumulated wealth is rebated back to Home household. On aggregate, in Home consumption units, exiting arbitrageurs rebate $\frac{\xi}{1-\xi}(W_t - \xi W_0)$ and new entrants receive ξW_0. Taken together, writing the entrant endowment averaged across groups as $W_0 = \sum_{i \in \{A,B\}} \eta_i W_0^i$, the net profits Home household receives each period is

$$\pi_t^a = \frac{\xi}{1-\xi}(W_t - W_0).$$

In the limiting case in which the exit probability approaches unity, $\xi \to 1$, the group wealth law resets wealth to the fixed entrant endowment each period, so the wealth distribution ceases to be a state variable.

Final good firms In each country, a continuum of monopolistically competitive final good firms, owned by domestic households, transforms the domestic intermediate good one-for-one into varieties sold globally at destination-specific flexible prices. As in Kekre and Lenel (2024), these firms exist in the model *only* to accommodate incomplete pass-through of the exchange rate into the prices consumers face, as in the data. This model block replaces the law of one price (LOP) assumption, motivated by empirically observed large deviations from LOP.

The bundles $c_{Ht}, c_{Ft}, c^*_{Ht}, c^*_{Ft}$ of are themselves CES aggregators over the varieties, with elasticity θ_v across *domestic* varieties. Following Alessandria (2009) and Kekre and Lenel (2024), the elasticity across *imported* varieties moves with the real exchange rate, that is, the Home households' aggregators over domestic and imported varieties are

$$c_{Ht} = \left(\int_0^1 c_{Ht}(i)^{\frac{\theta_v - 1}{\theta_v}} \, di \right)^{\frac{\theta_v}{\theta_v - 1}}, \quad c_{Ft} = \left(\int_0^1 c_{Ft}(i^*)^{\frac{\theta_{v,t} - 1}{\theta_{v,t}}} \, di^* \right)^{\frac{\theta_{v,t}}{\theta_{v,t} - 1}},$$

respectively, where the elasticity of substitution across imported varieties is $\theta_{v,t} = \theta_v \left(e_t/e\right)^{\xi_v}$, and throughout I denote steady-state values without time subscripts. Similarly, Foreign households' aggregators over imported and domestic varieties are

$$c^*_{Ht} = \left(\int_0^1 c^*_{Ht}(i)^{\frac{\theta^*_{v,t} - 1}{\theta^*_{v,t}}} \, di \right)^{\frac{\theta^*_{v,t}}{\theta^*_{v,t} - 1}}, \quad c^*_{Ft} = \left(\int_0^1 c^*_{Ft}(i^*)^{\frac{\theta_v - 1}{\theta_v}} \, di^* \right)^{\frac{\theta_v}{\theta_v - 1}},$$

where the elasticity of substitution across imported varieties is $\theta^*_{v,t} \equiv \theta_v \left(e_t/e\right)^{-\xi_v}$. This captures the pricing-to-market in a reduced form. Here, pricing complementarities make markups fall in markets where a firm's relative cost has risen. $\xi_v = 0$ restores constant elasticities and the LOP.

A Home firm i buys the intermediate at p_t (from Home households) and chooses prices $\{p_{Ht}(i),\, p^*_{Ht}(i)\}$, facing the demand schedules in the two markets, to maximize profits

$$\pi_t(i) = \big(p_{Ht}(i) - p_t\big)\Big(\frac{p_{Ht}(i)}{p_{Ht}}\Big)^{-\theta_v} c_{Ht} + \big(e_t\, p^*_{Ht}(i) - p_t\big)\Big(\frac{p^*_{Ht}(i)}{p^*_{Ht}}\Big)^{-\theta^*_{v,t}} c^*_{Ht}, \tag{2}$$

where $e_t p^*_{Ht}(i)$ converts export revenue into Home real units and c^*_{Ht} is aggregate Foreign demand. A Foreign firm i^* solves the mirror problem,

$$\pi^*_t(i^*) = \Big(\frac{p_{Ft}(i^*)}{e_t} - p^*_t\Big)\Big(\frac{p_{Ft}(i^*)}{p_{Ft}}\Big)^{-\theta_{v,t}} c_{Ft} + \big(p^*_{Ft}(i^*) - p^*_t\big)\Big(\frac{p^*_{Ft}(i^*)}{p^*_{Ft}}\Big)^{-\theta_v} c^*_{Ft}. \tag{3}$$

Per-capita profit rebates are $\pi_t = \int_0^1 \pi_t(i)\,di$ to Home households and $\pi_t^* = \int_0^1 \pi_t^*(i^*)\,di^*$ to Foreign households.

Market clearing and driving forces There are four markets: Home goods, Foreign goods, Home bonds, and Foreign bonds. Both bonds are in zero net supply, so their market-clearing conditions are

$$\frac{b_t}{1+r_t} + W_t - N_t = 0, \qquad \frac{e_t b_t^*}{1+r_t^*} + N_t = 0.$$

Here b_t and b_t^* are the face values of household bond holdings; the two equations express all positions in Home consumption units.

The economy is perturbed by demand and supply forces, each of which are modeled as discount factor and endowment shocks. For each driving variable $x_t \in \{\beta_t, \beta_t^*, y_t, y_t^*\}$ with its steady-state value x, I assume $\hat{x}_t = \log x_t - \log x$ has a persistent AR(1) component, τ_t^x, with mean reversion ρ_x, and mean zero short-run i.i.d. component, ε_t^x, with standard deviation σ_{ε^x}:

$$\hat{x}_t = \tau_t^x + \varepsilon_t^x,$$

where $\tau_t^x = \rho_x \tau_{t-1}^x + v_t^x$, with standard deviation of shocks being σ_{v^x}. Persistent components of shocks are correlated across countries to capture clear international comovement patterns observed in the data. The correlation of shocks with another driving variable x' is denoted by $\rho_{xx'}$. Agents observe realizations of $\hat{x}_t$, but not τ_t^x and ε_t^x separately. Therefore, agents need to filter out the persistent components to form expectations about future x. This is discussed in detail in a separate section below.

2.2 Information structure

The two groups of arbitrageurs interpret the same observed fundamentals differently. Following Caballero and Simsek (2022), they are opinionated: each group regards the other's interpretation as uninformative conditional on its own. Current interpretations and beliefs are mutually observed, but neither the other group's beliefs nor equilibrium prices provide additional information about the latent persistent component. Disagreement therefore persists despite common observations.

Group $i \in \{A, B\}$ treats $\hat{x}_t + \mu_t^{xi}$ as a noisy observation of τ_t^x, where $\mu_t^{xi} = \mu^{xi} + \hat{\mu}_t^{xi}$ consists of a constant interpretation and a transitory interpretation drawn from a joint normal distribution that is i.i.d. across periods. The groups have equal population shares, $\eta_A = \eta_B = 1/2$, so average interpretation bias is $\bar{\mu}_t^x = (\mu_t^{xA} + \mu_t^{xB})/2$.

Let $\tau^{xi}_{t|t} \equiv E^i_t[\tau^x_t]$ denote group i's filtered persistent state. Define consensus beliefs as $\bar{\tau}^x_{t|t} = (\tau^{xA}_{t|t} + \tau^{xB}_{t|t})/2$ and disagreement as $d^x_t = \tau^{xA}_{t|t} - \tau^{xB}_{t|t}$. With common steady-state Kalman gain G_x and $\varphi_x = \rho_x(1 - G_x)$, their laws of motion are

$$\begin{aligned}\bar{\tau}^x_{t|t} &= \varphi_x \bar{\tau}^x_{t-1|t-1} + G_x(\hat{x}_t + \bar{\mu}^x_t),\\ d^x_t &= (1-\varphi_x)d^x_{ss} + \varphi_x d^x_{t-1} + G_x \varepsilon^x_{\mu,t},\end{aligned}$$

where d^x_{ss} is the permanent belief gap and $\varepsilon^x_{\mu,t} = \hat{\mu}^{xA}_t - \hat{\mu}^{xB}_t \sim N(0, \sigma^2_{\mu,x})$ is the transitory disagreement innovation.

In the zero-average-bias benchmark, $\bar{\mu}^x_t = 0$, consensus beliefs follow the objective filter even though the groups disagree. Average bias and disagreement can therefore be varied separately. Both groups understand the same equilibrium mapping from fundamentals and the distribution of beliefs to prices and wealth, and they agree on the expected path of disagreement.

Remark 1. (Analytical benchmark.) The information structure nests the benchmark used in Section 3. With unbiased consensus beliefs, $\bar{\mu}^x_t = 0$, let transitory fundamental noise and disagreement innovations vanish, $\sigma_{\varepsilon^x} \to 0$ and $\sigma_{\mu,x} \to 0$, for each fundamental x. Then $G_x \to 1$ and $\varphi_x \to 0$. In the limiting benchmark, observed fundamentals follow $\hat{x}_t = \rho_x \hat{x}_{t-1} + v^x_t$, and the fundamental belief gap is constant, $d^x_t = d^x_{ss} = \mu^{xA} - \mu^{xB}$. Wealth redistribution nevertheless changes the influence of these beliefs on equilibrium prices. In the affine equilibrium used below, the total one-period exchange-rate forecast gap is also constant, $D_t = d_{ss}$, and hence $d_t = 0$. Appendix B.5 derives the consistency condition linking d_{ss} to the fundamental gaps d^x_{ss}.

Appendix B.1 gives the formal information structure, Proposition 4, and its proof. The next section shows how wealth weighting makes disagreement affect equilibrium pricing even when consensus beliefs are unbiased.

Remark 2. (Private information and price learning.) With heterogeneous subjective long-run models and common constant filtering gains, private signals and shared price signals can preserve the AR(1) form of centered fundamental belief gaps. Here a nonzero permanent gap comes from fixed subjective model differences; this construction does not establish a common-prior rational-expectations foundation for the current model. Appendix B.1.1 states the conditions and proves the result.

2.3 Equilibrium and solution

Appendix A collects the full equilibrium system and defines the units and approximation conventions used below.

First-order approximation We use the small noise approximation developed by Borovička, Hansen and Sargent (2023), as is common in recent segmented market models such as Itskhoki and Mukhin (2021) and Kekre and Lenel (2024). Let all stochastic deviations be scaled by δ and let risk aversion satisfy

$$\nu(\delta)\delta^2 \to \Gamma \in (0, \infty)$$

as $\delta \to 0$. In words, as variances become infinitely small, risk aversion becomes infinitely larger at the same rate as variance. In effect, the risk terms remain at the first order and portfolio positions remain bounded even as primitive risk vanishes. The portfolio rule below assumes a common conditional variance across groups and holds it fixed at its steady-state value in the first-order approximation. We normalize this steady-state conditional variance of $\Delta \log e_{t+1}$ to δ^2, absorbing its constant scale into Γ. All equilibrium conditions below are first order log-linear conditions around the possibly asymmetric deterministic steady state. The log deviation from the steady state is denoted as $\hat{z}_t = \log z_t - \log z$, where I denote steady state objects by dropping time subscript throughout the paper. For interest rates, hats denote log deviations of gross rates: $\hat{r}_t = \log[(1+r_t)/(1+r)]$ and $\hat{r}^*_t = \log[(1+r^*_t)/(1+r^*)]$. In some cases, I also use normalized level deviation from the steady state, in which case a variable is denoted as $\tilde{z}_t$, with a slight abuse of notations.

Optimal portfolio choice Writing the carry return as $R^x_{t+1} \equiv \frac{e_{t+1}}{e_t}(1+r^*_t) - (1+r_t)$, the first-order condition for the myopic mean-variance problem is

$$E^i_t\left[R^x_{t+1}\right] - \nu \frac{n^i_t(\iota)}{(1+r_t)w^i_t(\iota)} \mathrm{Var}^i_t\left[R^x_{t+1}\right] = 0.$$

Following an approximation approach similar to Campbell and Viceira (1999) and aggregating decisions within each type, we obtain the log-return approximation

$$\frac{N^i_t}{W^i_t} = \frac{E^i_t\left[\Delta \log e_{t+1}\right] + \log \frac{1+r^*_t}{1+r_t}}{\nu \mathrm{Var}^i_t\left[\Delta \log e_{t+1}\right]},$$

which is the maintained myopic portfolio rule. Appendix B.2 derives the exact optimum, the log-return approximation, and the first-order position rule. Let $L_i \equiv E^i\left[\log e\right] + \log \frac{1+r^*}{1+r}$

denote group i's steady-state expected carry return and define $UIP_t^i \equiv E_t^i[q_{t+1}^p] - L_i = E_t^i[\hat{e}_{t+1}] - \hat{e}_t + \hat{r}_t^* - \hat{r}_t$. For exchange-rate forecasts in first-order equations, $E_t^i[\hat{e}_{t+1}]$ denotes the change from group i's own steady-state forecast; the realized $\hat{e}_t$ is the common log deviation. Here $q_{t+1}^p = \Delta \log e_{t+1} + \log[(1+r_t^*)/(1+r_t)]$ is the log carry return. At first order,

$$\widetilde{N}_t^i = \frac{L_i}{\Gamma}\hat{W}_t^i + \frac{1}{\Gamma}UIP_t^i,$$

where $\tilde{N}_t^i = \frac{N_t^i - N^i}{W^i}$. Heterogeneous expectations affect positions through UIP_t^i, while permanent disagreement affects both steady-state positions and their wealth elasticities through L_i.

Modified UIP Substituting the position rule into bond market clearing, we obtain a version of UIP in this economy.

Lemma 1. *The Home-bond version of the modified UIP condition in levels is*

$$\sum_{i\in\{A,B\}} \omega_t^i \frac{E_t^i\left[\Delta \log e_{t+1}\right] + \log\frac{1+r_t^*}{1+r_t}}{\nu var_t^i\left[\Delta \log e_{t+1}\right]} = 1 + \frac{b_t}{(1+r_t)W_t}, \tag{4}$$

where $\omega_t^i = \frac{W_t^i}{2W_t}$ denotes group i's share of aggregate arbitrageur wealth. For wealth shares, $\hat{\omega}_t^i \equiv \omega_t^i - \omega^i$ denotes a level deviation from steady state. In the benchmark with unbiased consensus beliefs, the first-order condition is

$$\begin{aligned}\mathbb{E}_t\left[\hat{e}_{t+1}\right] - \hat{e}_t + \hat{r}_t^* - \hat{r}_t = &-d_{ss}\hat{\omega}_t^A - \delta_w d_t \\ &+ \frac{\beta\Gamma}{W}\left[y\widetilde{b}_t - b(\hat{W}_t + \hat{r}_t)\right],\end{aligned} \tag{5}$$

where $\mathbb{E}\left[\cdot\right]$ denotes the rational expectations operator given the information structure (2.2), $d_{ss} = L_A - L_B$ is the steady-state one-period exchange-rate forecast gap. Write the total gap as $D_t \equiv E_t^A[\log e_{t+1}] - E_t^B[\log e_{t+1}] = d_{ss} + d_t$, so $d_t = UIP_t^A - UIP_t^B$ is its centered deviation. The remaining coefficients are $\delta_w = (\omega^A - \omega^B)/2$, the steady-state wealth-share imbalance, and $\widetilde{b}_t = (b_t - b)/y$, the Home bond deviation normalized by steady-state Home endowment; b and W are the steady-state Home household bond position and aggregate arbitrageur wealth.

The modified UIP condition (4) shows that exchange rates are priced by the wealth-weighted average of the two groups' expectations. To first order, this wealth-weighted belief decomposes into a wealth-share term, $-d_{ss}\hat{\omega}_t^A$ and a time-varying disagreement term $-\delta_w d_t$. From the viewpoint of an econometrician outside the model who holds objective beliefs, both act as mispricing forces. Take the first. Under the normalization $d_{ss} > 0$, group-A

arbitrageurs are the optimists about Foreign bonds: they expect higher excess returns on Foreign bonds than group-B does. A rise in their wealth share, $\hat{\omega}_t^A > 0$, makes the optimists more dominant, so their views carry more weight in today's equilibrium prices. The Home currency depreciates by more than fundamentals alone would warrant, and expected excess returns fall. Put differently, the predominance of optimists compresses excess returns and so pushes carry returns down. The effect of time-varying disagreement d_t depends on the steady-state wealth imbalance. When $\omega^A > \omega^B$, a widening belief gap raises the wealth-weighted forecast. This direct effect vanishes when the two groups have equal wealth shares. The remaining terms capture bond supply relative to arbitrageur wealth, as in segmented-asset-market models, such as Itskhoki and Mukhin (2021); Kekre and Lenel (2024).

Wealth–share dynamics One of the key components of the current model is the dynamics of the wealth share. They are characterized by the following equation:

Lemma 2. *The first-order dynamics of the wealth-share deviation of group A in the benchmark with unbiased consensus beliefs are given by*

$$\begin{aligned}\hat{\omega}_{t+1}^A = (1-\xi)\Bigg\{ & \frac{1}{\beta}\left[1 + \frac{\omega_x(\bar{L} - \delta_w d_{ss})}{\Gamma}\right]\hat{\omega}_t^A \\ & + \omega^A\omega^B\left[\frac{\omega_x}{\beta\Gamma}d_t + \frac{d_{ss}}{\Gamma}\left(\frac{\omega_x}{\beta}\hat{W}_t + \hat{R}_{t+1}^p\right)\right]\Bigg\},\end{aligned} \tag{6}$$

where $\omega_x \equiv \beta/\beta^* - 1$, $\bar{L} = \log\beta - \log\beta^*$, $d_t = UIP_t^A - UIP_t^B$ *and* $\hat{R}_{t+1}^p = \frac{1}{\beta^*}(\hat{e}_{t+1} - \hat{e}_t + \hat{r}_t^*) - \frac{1}{\beta}\hat{r}_t$.

Appendix B.4 derives this law and its symmetric analytical limit.

Equation (6) makes the role of belief heterogeneity explicit. Absent disagreement, the two groups choose the same carry position per unit of wealth. Aggregate portfolio demand then depends only on total wealth, so a representative arbitrageur suffices to determine prices and the wealth share drops out of the pricing state vector. Disagreement creates differences in positions per unit of wealth and turns the share into a state that affects prices, driven by two forces.

The first is revaluation. Under the normalization $d_{ss} > 0$, group-A arbitrageurs are the optimists about Foreign bonds: they expect higher excess returns on Foreign bonds than group-B does. In steady state their Foreign-bond exposure per unit of wealth is larger by d_{ss}/Γ, so any shock that delivers a positive carry return $\hat{R}_{t+1}^p$—whatever its source—revalues wealth from group-B toward group-A, raising $\hat{\omega}_{t+1}^A$.

The second is the drift earned on the steady-state excess return $\omega_x = \beta/\beta^* - 1$. This force multiplies not the realized return but the *gap* between the two groups' positions, and that gap moves with two further states. Aggregate arbitrageur wealth $\hat{W}_t$ scales the steady-state gap d_{ss}/Γ: a larger balance sheet supports larger carry positions, so the same per-unit return redistributes more, amplifying the drift when $\hat{W}_t > 0$ and damping it when $\hat{W}_t < 0$. Time-varying disagreement d_t widens the gap itself: holding the steady-state return fixed, a rise in d_t pushes the optimists further long Foreign and the pessimists further short, so more wealth rides on the carry. Because the steady-state excess return on Foreign bonds is positive, consistent with the data between the U.S. (Home) and other advanced economies (Foreign), this widening tilts the average carry gain toward group-A; whether it survives in general equilibrium is taken up in the next section, where the anticipated unwind of the position pulls in the opposite direction.

3 Analytical Results

This section solves the model in closed form. Three limits reduce the equilibrium to a feedback loop between the arbitrageurs' wealth share and the exchange rate: news about fundamentals revalues the optimists' wealth, and their wealth is then repriced into the exchange rate. The feedback loop has a fixed point, and that fixed point is a multiplier on every fundamental surprise. The multiplier is bounded above, and the bound is set by the goods market rather than by anything financial. That same bound returns twice more: once as the limiting Fama coefficient, and once as the threshold that decides the sign of the Backus–Smith correlation. Persistent and volatile exchange rates, a negative Fama coefficient, and a negative Backus–Smith correlation all follow without an exogenous UIP shock.

3.1 Calibration towards analytical results

To isolate the model's core mechanism I impose three limits on the parameter space. The first follows Itskhoki and Mukhin (2021) and is standard in the disconnect literature (Itskhoki and Mukhin 2021; Kekre and Lenel 2024; Fukui et al. 2025): I let the financial sector grow large relative to the real economy, $\frac{b^*}{W} \to 0$. Using the equivalent Foreign-bond representation in Appendix B.3, the bond-supply terms then drop out of the modified UIP condition, which becomes

$$\mathbb{E}_t\left[\Delta \hat{e}_{t+1}\right] + \hat{r}_t^* - \hat{r}_t = -d_{ss}\hat{\omega}_t^A - \delta_w d_t. \tag{7}$$

The right side is minus the UIP wedge. The optimists' wealth share and the disagreement flow do not enter expected currency returns separately; they enter as one object.

The second limit sends the steady-state discount factor differential to zero, $\beta/\beta^* \to 1$. With equal entrant endowments and $a \equiv (1-\xi)/\beta < 1$, the steady-state wealth formula in Appendix A.2 gives $W^A = W^B$, so $\omega^A = \omega^B = 1/2$ and $\delta_w = 0$. Positive finite steady-state wealth requires $a < 1$. The wedge collapses to $d_{ss}\hat{\omega}_t^A$, and the wealth-share dynamics reduce to

$$\hat{\omega}_{t+1}^A = (1-\xi)\left(\frac{1}{\beta}\hat{\omega}_t^A + \omega^A\omega^B\frac{d_{ss}}{\beta\Gamma}\left(\hat{e}_{t+1} - \hat{e}_t + \hat{r}_t^* - \hat{r}_t\right)\right).$$

The realized log carry return, $q_{t+1}^p \equiv \beta\hat{R}_{t+1}^p = \hat{e}_{t+1} - \hat{e}_t + \hat{r}_t^* - \hat{r}_t$, decomposes into

$$\begin{aligned} q_{t+1}^p &= q_{t+1}^p - \mathbb{E}_t q_{t+1}^p + \mathbb{E}_t\Delta\hat{e}_{t+1} + \hat{r}_t^* - \hat{r}_t \\ &= \hat{e}_{t+1} - \mathbb{E}_t\hat{e}_{t+1} - d_{ss}\hat{\omega}_t^A, \end{aligned}$$

where the second line uses the reduced UIP condition (7). Substituting gives the law of motion

$$\hat{\omega}_{t+1}^A = \lambda\hat{\omega}_t^A + b_w\varepsilon_{t+1}^e, \tag{8}$$

in which $\lambda = a - b_w d_{ss} = a\left(1 - \frac{d_{ss}^2}{4\Gamma}\right)$ is the persistence of the wealth share, $b_w = \frac{ad_{ss}}{4\Gamma}$ is its revaluation exposure to news, and $\varepsilon_{t+1}^e \equiv \hat{e}_{t+1} - \mathbb{E}_t\hat{e}_{t+1}$ is the equilibrium exchange rate innovation. Here Γ is the effective risk-cost parameter, so a larger value reduces positions per unit of wealth, ξ their exit rate, and ω^A, ω^B the steady-state wealth shares of the two groups.

Two sign conventions carry through the rest of the paper. A rise in $\hat{e}_t$ is a Home depreciation. Group- A is optimistic about foreign-currency returns and is long the Foreign bond, so a surprise depreciation revalues wealth toward the optimists and $b_w > 0$. A positive wedge holds the exchange rate above its fundamental level, and implies expected appreciation as the wedge decays.

This limit also removes total arbitrageur wealth from the state vector, which Lemma 2 carries. The reason is the one that made δ_w vanish. Away from the limit the steady-state carry trade earns a nonzero baseline return, so the two groups earn different returns per unit of wealth; scaling both groups up—a movement in *common* wealth—scales the transfer between them, and common wealth redistributes at first order. At $\frac{\beta}{\beta^*} \to 1$ the baseline carry return is zero, so common wealth leaves relative wealth untouched.

The third limit adopts Remark 1: fundamental realizations follow AR(1) processes, $d_t^x = d_{ss}^x$, and the centered exchange-rate disagreement is $d_t = 0$. The constant d_{ss} still determines steady-state portfolio differences and the wealth-revaluation feedback. These restrictions are maintained throughout this section and its proofs.

3.2 The wealth–disagreement multiplier

Equations (8) and (7) form a feedback loop. Realized exchange rate news revalues the wealth share through b_w, and the wealth share is repriced into the exchange rate through the wedge. Deriving the loop's fixed point takes one intermediate step: the smallest block of equilibrium conditions that pins down the exchange rate. Goods market clearing implies

$$\hat{c}_t - \hat{c}_t^* = \phi_y \left(\hat{y}_t - \hat{y}_t^*\right) - \chi \phi_e \hat{e}_t, \tag{9}$$

where $\phi_y = \frac{1}{1-2\gamma}$, $\phi_e = \frac{4\theta\gamma(1-\gamma)}{(1-2\gamma)^2}$, and $\chi = 1 - \frac{\xi_v}{\theta_v - 1}$ is the degree of incomplete pass-through. Combining the relative household Euler equation, $\hat{r}_t - \hat{r}_t^* = \sigma\left(\mathbb{E}_t \Delta\hat{c}_{t+1} - \mathbb{E}_t \Delta\hat{c}_{t+1}^*\right) - \left(\hat{\beta}_t - \hat{\beta}_t^*\right)$, with goods market clearing yields

$$\hat{r}_t - \hat{r}_t^* = -\left(\hat{\beta}_t - \hat{\beta}_t^*\right) - \sigma\phi_y \left[(1-\rho_y)\hat{y}_t - (1-\rho_{y^*})\hat{y}_t^*\right] - S\mathbb{E}_t \Delta\hat{e}_{t+1}, \tag{10}$$

where $S \equiv \sigma\chi\phi_e > 0$.

The composite S measures how strongly an expected depreciation shifts consumption across time, and it will turn out to govern every bound in this section. Equation (9) supplies the mechanism. A real depreciation switches Home expenditure toward Home goods; with output fixed and home-biased consumption, Home relative consumption has to fall for the market to clear. An expected depreciation is therefore expected negative relative consumption growth, which the Euler equation prices into a lower Home interest differential today, scaled by risk aversion σ. Hence the form $S = \sigma\chi\phi_e$: expenditure switching supplies $\chi\phi_e$, and intertemporal substitution supplies σ. Using the modified UIP condition (7) to eliminate expected depreciation delivers the reduced system.

The equilibrium collapses to three equations in the exchange rate, the interest differential, and the optimists' wealth share.

Proposition 1. *Under Remark 1, the financial limits above, stationary fundamental processes, and $0 < \lambda < 1$, the reduced equilibrium with $d_t = 0$ is characterized by:*

i. Interest rate differential: $\hat{r}_t - \hat{r}_t^* = \frac{1}{1+S}\hat{\iota}_t^f + \frac{S}{1+S} d_{ss}\hat{\omega}_t^A$, where $\hat{\iota}_t^f = -\left(\hat{\beta}_t - \hat{\beta}_t^*\right) - \sigma\phi_y \left[(1-\rho_y)\hat{y}_t - (1-\rho_{y^*})\hat{y}_t^*\right]$.

ii. Modified UIP: $\mathbb{E}_t \Delta\hat{e}_{t+1} + \hat{r}_t^* - \hat{r}_t = -d_{ss}\hat{\omega}_t^A$.

iii. Wealth share dynamics: $\hat{\omega}_{t+1}^A = \lambda\hat{\omega}_t^A + b_w \varepsilon_{t+1}^e$.

The system is block recursive: these three equations determine exchange rates, interest differentials, and wealth shares given the driving forces, and consumption and goods prices follow

from that solution. Read in order, they say three things. Part (i) breaks the interest differential as a sufficient statistic for fundamentals: a fraction $\frac{S}{1+S}$ of the wedge $d_{ss}\hat{\omega}_t^A$ is priced into observed interest rates, so an econometrician reading interest rates alone attributes belief-driven movements to fundamentals. Part (ii) prices the same wedge into expected currency returns. Part (iii) is the one that matters for what follows, because it is not an exogenous shock process: the innovation ε_{t+1}^e that drives the wealth share is itself determined by the other two equations. A stationary solution requires $\lambda \in (0,1)$, which the forward solution below uses.

Solving parts (i) and (ii) forward under stationarity maps the wealth share into the level of the exchange rate,

$$\hat{e}_t = -\frac{1}{1+S}\mathbb{E}_t \sum_{j=0}^{\infty} \hat{\imath}_{t+j}^f + \frac{d_{ss}}{(1+S)(1-\lambda)}\hat{\omega}_t^A, \tag{11}$$

where the frictionless fundamental component is

$$f_t \equiv -\frac{1}{1+S}\mathbb{E}_t \sum_{j=0}^{\infty} \hat{\imath}_{t+j}^f = \frac{1}{1+S}\left[\frac{\hat{\beta}_t}{1-\rho_\beta} - \frac{\hat{\beta}_t^*}{1-\rho_{\beta^*}} + \sigma\phi_y(\hat{y}_t - \hat{y}_t^*)\right].$$

The coefficient $\kappa_\omega \equiv d_{ss}/[(1+S)(1-\lambda)]$ is the price of one unit of wealth share. The wedge is d_{ss} per period, its expected duration is $1/(1-\lambda)$, and goods-market adjustment attenuates its exchange-rate effect by $1/(1+S)$.

Taking innovations in (11), direct fundamental news is

$$\varepsilon_{t+1}^{e,f} \equiv f_{t+1} - \mathbb{E}_t f_{t+1} = \frac{1}{1+S}\left[\frac{v_{t+1}^{\beta}}{1-\rho_\beta} - \frac{v_{t+1}^{\beta^*}}{1-\rho_{\beta^*}} + \sigma\phi_y(v_{t+1}^{y} - v_{t+1}^{y^*})\right].$$

By part (iii), the exchange-rate surprise revalues the wealth share by $b_w\varepsilon_{t+1}^e$. The same revaluation enters the exchange rate through κ_ω, giving the innovation fixed point $\varepsilon_{t+1}^e = \varepsilon_{t+1}^{e,f} + \kappa_\omega b_w \varepsilon_{t+1}^e$.

Solving it shows that disagreement multiplies every fundamental exchange rate surprise by a factor greater than one, and that the factor has a ceiling.

Corollary 1. *In the stationary benchmark of Proposition 1, with $d_{ss} > 0$, conditional on macro fundamental shocks $\hat{x}_t \in \left\{\hat{\beta}_t, \hat{\beta}_t^*, \hat{y}_t, \hat{y}_t^*\right\}$, the equilibrium exchange rate innovation ε_{t+1}^e and its counterpart in an economy without the wealth share wedge, $\varepsilon_{t+1}^{e,f}$, satisfy*

$$\varepsilon_{t+1}^e = \Xi \varepsilon_{t+1}^{e,f}, \qquad \Xi \equiv \frac{1}{1-g},$$

with

$$g = \frac{b_w d_{ss}}{(1+S)(1-\lambda)}.$$

Moreover,

$$1 < \Xi < \Xi_{\max} \equiv 1 + \frac{1}{S}.$$

The multiplier Ξ is strictly increasing in d_{ss}, and strictly decreasing in $\Gamma, \xi, \beta, \sigma, \chi, \theta$, and γ.

The proof is in Appendix B.5. The multiplier reflects a feedback between exchange rates and the distribution of wealth. A surprise Home depreciation enriches arbitrageurs who are optimistic about Foreign-currency returns and therefore hold larger Foreign positions. Their additional wealth gives their beliefs greater weight in equilibrium pricing, amplifying the initial depreciation. The strength of this feedback is

$$g = \underbrace{b_w}_{\text{wealth moved by news}} \quad \underbrace{\frac{d_{ss}}{1+S}}_{\text{pricing effect of that wealth}} \quad \underbrace{\frac{1}{1-\lambda}}_{\text{duration of that influence}}.$$

The first factor, b_w, measures how much an exchange-rate surprise changes the optimists' wealth share. The second measures how strongly that redistribution affects pricing: greater disagreement increases its effect, while consumption and interest-rate adjustment attenuate it through $1/(1+S)$. The third captures persistence. Because wealth shares decay at rate λ, prices capitalize the entire expected sequence of future belief wedges, $1 + \lambda + \lambda^2 + \cdots = 1/(1-\lambda)$. Each additional round of the feedback contributes a fraction g of the preceding round, giving

$$\Xi = 1 + g + g^2 + \cdots = \frac{1}{1-g}.$$

The comparative statics follow the same mechanism. Everything that raises the gain of the feedback raises Ξ, and the gain is the product of two terms: how much wealth a given surprise moves, b_w, and how much price a given unit of wealth commands, $\frac{d_{ss}}{(1+S)(1-\lambda)}$. Steady-state disagreement d_{ss} enters both, which is why it is the one parameter that raises Ξ. The effective risk-cost parameter Γ and the exit rate ξ cut the first term: a higher risk cost makes a given belief gap translate into a smaller position per unit of wealth, and faster

turnover replaces incumbents with entrants whose fixed endowments dilute accumulated wealth differences. The goods-market parameters σ, χ, θ, and γ cut the second, all through S.

The ceiling comes from the goods market, not from the financial block. Only S appears in $\Xi_{\max} = 1 + 1/S$, and the reason is a bound on how far consumption can be pushed. A depreciation lowers Home relative consumption through goods market clearing (9). Given expected future consumption, that is lower expected consumption growth, which the relative Euler equation (10) turns into a lower Home interest differential. A lower Home rate induces Home households to consume more today, which pushes back against the initial fall. Restoring balance requires an expected Home appreciation, and that expected appreciation caps how far the currency can depreciate on impact. The division of labor is clean: financial parameters place Ξ inside $(1, \Xi_{\max})$, and the goods market fixes the endpoint.

This impact amplification does not depend on the volatility or persistence of the exogenous shocks, whose values are the most contested inputs in this literature. The multiplier is built from standard preference and trade parameters, the arbitrageurs' wealth persistence, and steady-state disagreement. Both of the latter two can be measured outside the exchange rate data. Section 4 disciplines wealth persistence with intermediary net worth (He et al. (2017)), because the exit rate affects both aggregate wealth and wealth-share dynamics, and it disciplines d_{ss} with dispersion in the Blue Chip Financial Forecasts exchange rate survey.

Figures 1 and 2 plot the impulse responses to a positive Home discount-factor shock and a positive Home endowment shock, respectively[3]. Following the discount-factor shock, Home households become more willing to save, causing Home consumption to fall relative to Foreign consumption. The resulting increase in Home net exports is supported by a depreciation of the Home currency. Because optimists are long Foreign-currency bonds, this depreciation generates a valuation gain and transfers wealth from pessimists to optimists. The redistribution strengthens optimists' demand for Foreign assets and thereby amplifies the initial exchange-rate response. It also produces an endogenous reversal: optimists' increased demand for Foreign bonds lowers the Foreign interest rate relative to the economy without disagreement, while the initial depreciation leaves the Foreign currency temporarily overvalued. Expected Foreign-currency excess returns therefore turn negative, gradually eroding optimists' wealth share and returning the economy toward its steady state. The response to a positive Home endowment shock operates through the same wealth-revaluation mechanism. In this case, however, greater Home supply raises Home consumption while

[3]With the current parameterization from Section 4.2, the multiplier is $\Xi = 1.826$, with its upper-bound $\Xi_{\max} = 1.9474$. This multiplier implies about 70% of the impact exchange rate variance is attributable to the wealth revaluation amplification.

depreciating the Home currency, generating the negative comovement between relative consumption and the exchange rate associated with the Backus–Smith anomaly. Thus, disagreement amplifies exchange-rate adjustment without requiring a commensurate increase in relative-consumption volatility.

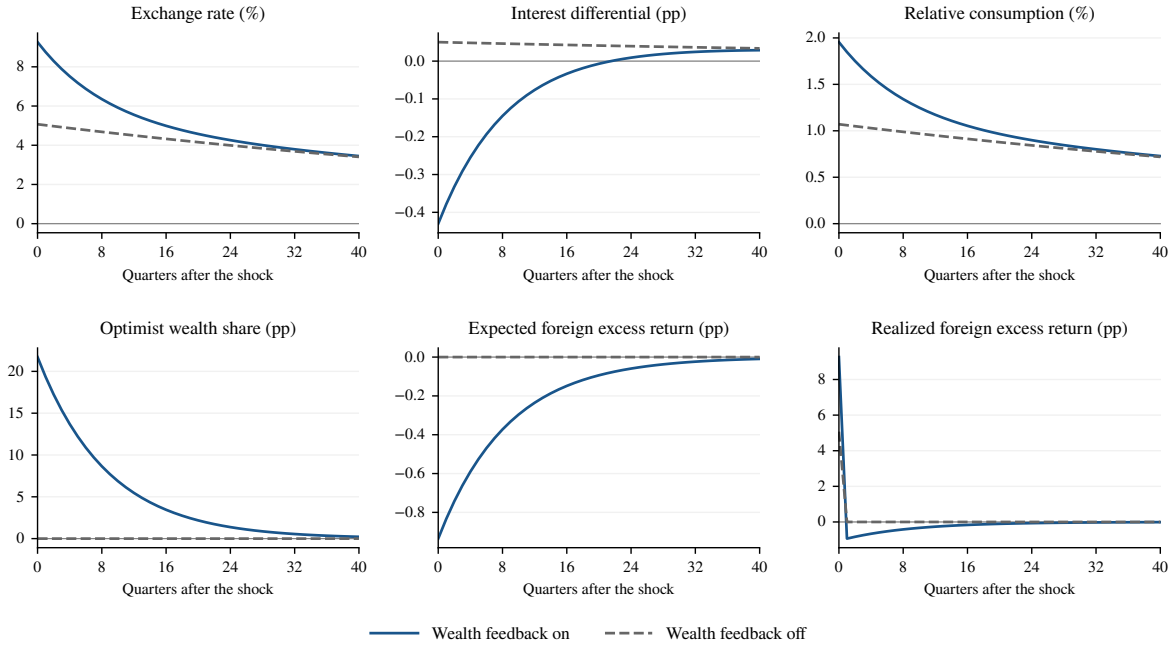


Figure 1: Reduced-equilibrium demand shock: wealth-revaluation feedback on versus off

Notes: Impulse responses in the no-NFA, no-transitory-noise reduced equilibrium to a one-standard-deviation Home demand (discount-factor) shock. The solid line retains the wealth-revaluation feedback; the dashed line shuts down only that feedback while holding fundamentals and the shock size fixed. Hence the ratio of the impact exchange-rate responses is exactly Ξ. Exchange rates and consumption are log deviations multiplied by 100; the remaining panels report percentage-point changes. Interest and consumption differentials are Foreign minus Home.

3.3 FX disconnect moments

The four moments at the center of the disconnect literature all follow from the multiplier: the near-random-walk behavior of the exchange rate, the small volatility of interest differentials relative to currency movements, the negative Fama coefficient, and the negative Backus–Smith correlation. One assumption on the driving forces makes the algebra transparent.

Assumption 1. *Within the information benchmark of Remark 1, the persistent components of the exogenous driving forces share a common persistence* $\rho_x \in (0, 1)$.

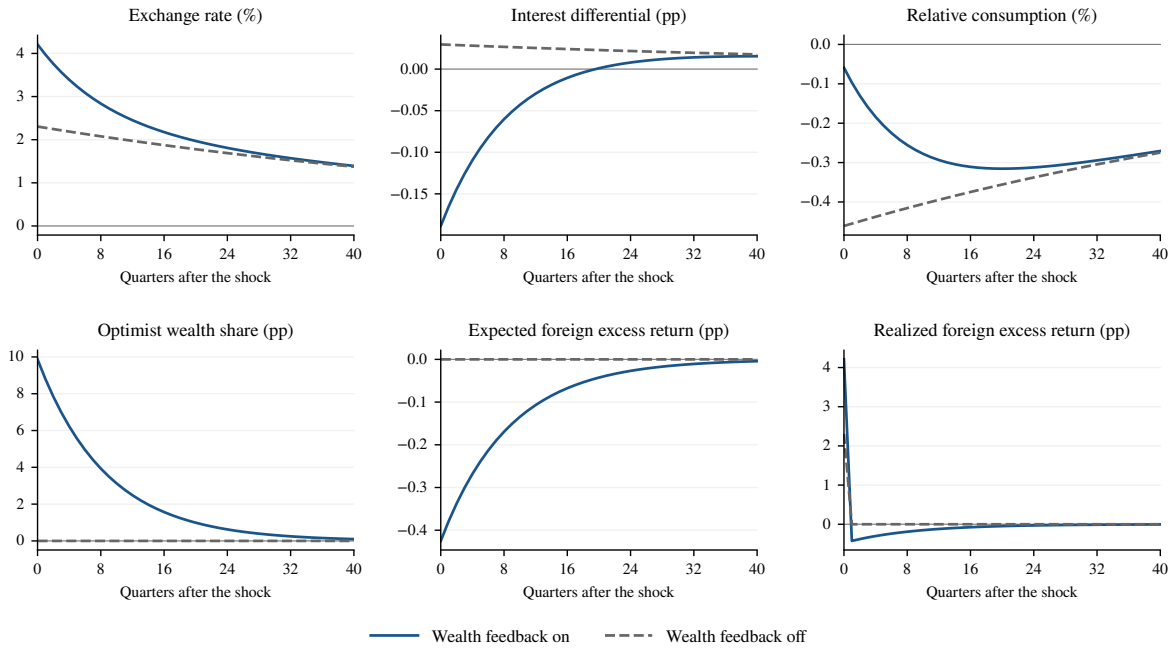


Figure 2: Reduced-equilibrium supply shock: wealth-revaluation feedback on versus off

Notes: Impulse responses in the no-NFA, no-transitory-noise reduced equilibrium to a one-standard-deviation Home endowment shock. The solid line retains the wealth-revaluation feedback; the dashed line shuts down only that feedback while holding fundamentals and the shock size fixed. Hence the ratio of the impact exchange-rate responses is exactly Ξ. Exchange rates and consumption are log deviations multiplied by 100; the remaining panels report percentage-point changes. Interest and consumption differentials are Foreign minus Home.

Assumption 1 is a simplifying restriction, not an essential one. It buys a single fundamental state in place of one per shock, so that the exchange rate is driven by one persistence parameter rather than a vector of them. Write $f_t \equiv -\frac{1}{1+S}\mathbb{E}_t \sum_{j=0}^{\infty} \hat{\iota}^f_{t+j}$ for the frictionless fundamental component of (11). Under the assumption it follows $f_t = \rho_x f_{t-1} + \nu_t$, where ν_t is fundamental news, and the fundamental interest differential is $\hat{\imath}^f_t \equiv \hat{\iota}^f_t/(1+S) = \mathbb{E}_t \Delta f_{t+1} = -(1-\rho_x) f_t$. Persistent fundamentals thus generate small interest differentials: as $\rho_x \to 1$, the level of f_t becomes large while its expected change becomes small.

The disconnect moments emerge together in one limit, in which fundamentals and the arbitrageurs' wealth share both become nearly permanent.

Proposition 2. *In the reduced equilibrium of Proposition 1, consider a sequence with* $0 < \rho_x, \lambda < 1$*, holding* $S > 0$ *and* $\Xi \in (1, \Xi_{\max})$ *fixed. As* $\rho_x \lambda \to 1$*,*

1. the exchange rate approaches a random walk;
2. the volatility of $\hat{r}_t - \hat{r}^*_t$ relative to that of $\Delta \hat{e}_t$ approaches zero; and
3. the Fama coefficient β_F turns negative provided ρ_x approaches one faster than λ, and the associated R^2_F approaches zero along every nondegenerate joint path.

The proof is in Appendix B.6. The limit isolates persistence: fundamentals and wealth shares become nearly permanent at any fixed amplification factor $1 < \Xi < \Xi_{\max}$. The appendix shows how financial primitives support this limit within the constant-disagreement benchmark of Remark 1.[4]

Part 1 is the Meese–Rogoff fact, and the model does not deliver it by removing structure. The exchange rate is the sum of two persistent pieces, the fundamental f_t and the priced wealth share, both driven by the same news. Amplification puts Ξ units of the surprise into the price on impact but only $\Xi - 1$ units into the wealth state, and that stored piece unwinds at rate λ. The unwind is predictable, so the depreciation rate is negatively autocorrelated at any interior point—the overshooting signature. As $\lambda \to 1$ the unwind becomes arbitrarily slow, the predictable component shrinks relative to the impact component, and the autocorrelation vanishes at every fixed horizon. The exchange rate looks like a random walk not because nothing is happening, but because the mean reversion has been stretched beyond the sample.

Part 2 is the disconnect between currency movements and interest rates, and it follows from an asymmetry in how the wedge reaches the two prices. By part (i) of Proposition 1,

[4]For example, hold β, Γ, and S fixed and let $d_{ss} \to 0$ and $\xi \to 1 - \beta$, keeping $a d^2_{ss}/[4\Gamma(1-a)]$ fixed and positive, where $a = (1-\xi)/\beta$. Then $\lambda \to 1$ while Ξ is unchanged: weaker disagreement reduces the feedback per period, while slower net capital replacement prolongs its influence. Disagreement remains constant over time within each economy; its level changes across the sequence.

the interest differential prices the wedge only through the flow it generates this period, which is proportional to the unwind rate $1-\lambda$. By (11), the exchange rate prices the entire stock. Slowing the unwind therefore shrinks the interest rate signal while leaving the currency signal bounded below by $\Xi \geq 1$ times fundamental news. Both the fundamental and wealth components of the interest differential vanish relative to exchange-rate innovations; their relative speeds determine the rate of convergence. Assumption 1 is doing real work here. A slow unwind alone would not suffice if fundamentals still moved interest rates; it is the common ρ_x that sends the fundamental differential $\hat{\imath}_t^f = -(1-\rho_x)f_t$ to zero at the same time.

Part 3 is the Fama puzzle. Holding the wealth component fixed, fundamental interest-rate variation forecasts depreciation at slope $+1$. Holding fundamentals fixed, the wealth component forecasts appreciation at slope $-1/S = 1 - \Xi_{\max}$. The two components covary, so their finite-parameter regression coefficient depends on that covariance as well as their variances. Under the rate condition in Proposition 2, $1-\rho_x = o(1-\lambda)$, the fundamental variance and covariance terms vanish relative to the wealth component. The Fama coefficient is then eventually negative and converges to $-1/S$.

The limiting Fama coefficient is one minus the multiplier ceiling. Both depend on $S = \sigma\chi\phi_e$: it prices the unwind of the wealth wedge into the interest differential and attenuates the wedge's exchange-rate effect. The exact coefficient at an interior parameterization retains the covariance terms shown in Appendix B.6. Along the same joint-persistence sequence, $R_F^2 \to 0$ because predictable depreciation vanishes relative to its innovation. The model therefore delivers a negative coefficient together with little explained variation.

The Backus–Smith correlation needs one more limit, because it depends on how close the loop runs to its ceiling.

Proposition 3. *In the reduced equilibrium of Proposition 1, with orthogonal demand and supply innovations, consider stationary parameterizations with fixed $S > 0$ and $0 < \rho_x, \lambda < 1$. As $\rho_x\lambda \to 1$ and $\Xi \to \Xi_{\max}$,*

$$\operatorname{Corr}(\Delta(\hat{c}_t - \hat{c}_t^*), \Delta\hat{e}_t) \longrightarrow -\sqrt{1-s_y},$$

where $s_y = \operatorname{Var}(\nu_{t+1}^y)/\sigma_\nu^2 \in (0,1)$ is the supply share of direct fundamental-news variance.

In this limit the correlation is negative for any positive demand-news share and approaches minus one as the supply share tends to zero. The mechanism is dilution rather than insulation. Relative consumption is $\phi_y(\hat{y}_t - \hat{y}_t^*) - \chi\phi_e\hat{e}_t$ by (9), so it contains two pieces: an income term that only supply shocks move, and an expenditure-switching term proportional to the exchange rate. The loop multiplies the exchange rate, and with it the

expenditure-switching term, by Ξ. It leaves the income term alone. Amplification therefore does not change the expenditure-switching channel's share of consumption movements—that piece scales with the exchange rate in numerator and denominator alike—but it shrinks the income term's share by $1/\Xi$. Since the income term is the only source of positive comovement between relative consumption and the exchange rate, amplification dilutes precisely the piece that would make the correlation positive.

Away from the limit the sign condition is sharp and easy to read: the impact correlation is negative if and only if $s_y < \Xi/\Xi_{\max}$, the fraction of its ceiling at which the loop runs. Without disagreement the multiplier is one and the threshold is only $1/\Xi_{\max} = S/(1+S)$. Amplification raises this threshold and expands the range of shock mixes with negative impact comovement. This bound rules out a negative impact correlation under pure supply news. It does not rule out a negative unconditional correlation: wealth-share adjustment following past shocks can reverse the sign, as Appendix C.2 shows. Section 4 assesses which force dominates at the reported calibration.

The same completed square governs relative volatility. The ratio of exchange rate volatility to the volatility of the relative consumption growth differential rises with Ξ and saturates at $(\sigma/S)/\sqrt{1-s_y}$, the reciprocal of the Backus–Smith limit up to the factor σ/S. Two features of that expression are worth separating, because they are often run together. The factor $\sigma/S = 1/(\chi\phi_e)$ is large because home bias and incomplete pass-through keep exchange rate movements from reaching consumption one for one; this sets the level of the ratio and has nothing to do with disagreement. The dependence on Ξ is the dilution effect above, and it is proportional to s_y: when all news is demand news the ratio does not respond to amplification at all. Amplification raises measured disconnect only to the extent that there is unamplified income variation for it to crowd out.

Taken together the three results describe one economy rather than three. A slow-moving wealth state makes the exchange rate persistent and its predictable component small, which produces both the random walk and the vanishing R^2_F. The goods-market ceiling $\Xi_{\max}$ pins down the limiting Fama coefficient and, when amplification approaches the ceiling, the Backus–Smith correlation. Neither the ceiling nor the wedge requires an exogenous UIP shock: both are properties of a market in which arbitrageurs disagree and their relative wealth is a state variable. What the analytics cannot say is whether the implied magnitudes are the observed ones, and whether the required demand share of exchange rate news is plausible. Section 4 takes up both questions, disciplining d_{ss} with survey dispersion and λ with intermediary net worth, and using the FIRE and no-disagreement comparisons to distinguish fit from mechanism.

4 Quantitative Analysis

This section asks whether survey-measured disagreement allows otherwise standard macroeconomic shocks to reproduce the central features of exchange rates jointly: a near-random-walk level, high volatility relative to consumption and output, a negative Backus–Smith correlation, and a negative Fama coefficient. The answer is yes, and the no-disagreement counterfactual is stark. Within this model class, the disagreement channel is indispensable for delivering the full set of moments together; even after the conventional shock block is re-estimated, the no-disagreement economy remains too stable and returns to a positive Fama coefficient. Persistent fundamentals determine how long exchange-rate movements last, while disagreement changes how strongly those fundamentals move prices relative to quantities. I discipline the shock processes with macroeconomic and yield moments, the belief block with Blue Chip survey expectations, and intermediary wealth with broker-dealer equity relative to GDP. I then use model ablations, variance decompositions, and a historical shock inversion to separate the origin of exchange-rate news from its propagation. Roughly 80% of short-horizon exchange-rate variation originates in conventional supply and demand shocks. This decomposition classifies the origin of the news, not its propagation. Fundamental shocks provide the news; disagreement determines whether that news becomes a large currency movement by changing positions and relative wealth. The reduced-form UIP shock is secondary, and the core joint fit survives without it.

4.1 Quantitative specification

The analytical results use a first-order approximation to isolate disagreement and endogenous wealth. The quantitative model adds three ingredients, each for a specific empirical purpose. Average bias lets consensus expectations adjust sluggishly and generates predictable survey forecast errors. Nonlinear products between disagreement, positions, and realized returns make conditional exchange-rate volatility depend on the state of beliefs. A parsimonious reduced-form UIP shock captures movements in risk-bearing capacity that the model does not generate internally. The ablations below show a clear division of labor. Average bias helps with the horizon profile of forecast errors, the nonlinear terms generate state-dependent volatility, and the UIP shock adds a secondary source of exchange-rate variation. The joint unconditional FX–macro fit, however, is already present in the parsimonious disagreement economy. Removing the extensions leaves that fit largely intact; removing disagreement destroys it.

Average bias process So far I have imposed no assumptions on the nature of the average bias process, $\bar{\mu}_t$. In the context of international macroeconomics and finance, Candian and De Leo (2025) show that the predictability reversal documented in Engel (2016) is largely driven by subjective forecast errors, and they propose a model of misperceived persistence of the exogenous driving forces that is consistent with the reversal pattern. The current model is flexible enough to accommodate such a bias process. In particular, parameterizing

$$\bar{\mu}_t^x = \left(\frac{\kappa_x}{G_x} - 1\right)\hat{x}_t + \frac{(1-\kappa_x)\tilde{\rho}_x - \rho_x(1-G_x)}{G_x}\bar{\tau}_{t-1|t-1}^x,$$

where κ_x and $\tilde{\rho}_x$ denote the information gain and the persistence under misperception, respectively, reproduces the consensus posterior dynamics of Candian and De Leo (2025). When subjective forecasts use the same perceived persistence $\tilde{\rho}_x$, the conditional-mean forecasts also coincide at every horizon. Appendix D.1 proves these statements and derives the implied forecast-error dynamics. This information structure adds two parameters without adding new primitive shocks. The two parameters have natural counterparts in standard expectations regressions. The first, from Coibion and Gorodnichenko (2015), regresses forecast errors on forecast revisions and measures the sluggishness of expectations to newly available information, κ_x. The second, from Kohlhas and Walther (2021), regresses forecast errors on the current realization of the variable itself. For a range of macroeconomic variables, this regression typically yields overreaction to the current realization, suggesting that agents over-extrapolate currently available information. It therefore disciplines perceived persistence. Because both coefficients are sensitive to the sample and forecast horizon, I use the predictability-reversal profile as the primary calibration target and treat these regressions as external diagnostics. I refer the reader to Candian and De Leo (2025) for details.

Time-varying disagreement and financial transmission Disagreement changes how news moves exchange rates through two channels. When $d_{ss} + d_t$ is larger, the groups' currency positions move further apart, so a given return redistributes more wealth. The same redistribution then has a larger effect on the market's wealth-weighted forecast. To retain both channels as beliefs change, the quantitative model keeps the products $d_t\widehat{\omega}_t^A$ in bond pricing and $d_t\widehat{R}_{t+1}^p$ in wealth accumulation. It holds perceived risk fixed and uses the baseline linear forecasting rules and first-order coefficients elsewhere. Appendix E states these assumptions and the solution method explicitly. The interactions make exchange-rate volatility depend on the state of beliefs, while the ablations show that the main unconditional moments change little.

Reduced form UIP shock A number of recent studies, such as Itskhoki and Mukhin (2021); Kekre and Lenel (2024), rely on some form of exogenous UIP shock. To study the role of the wealth–disagreement wedge relative to such a shock, I also introduce an additional, reduced-form AR(1) UIP shock, ψ_t, which enters the right-hand side of the modified UIP condition (5). To discipline the process empirically, I follow the interpretation of Kekre and Lenel (2024), under which the shock captures the time-varying nature of arbitrageurs' risk-bearing capacity, the popular notion of risk-on and risk-off episodes. In this paper I use the excess bond premium by Gilchrist and Zakrajšek (2012) as a proxy for risk-bearing capacity, but the persistence of the process is almost unchanged across a number of risk measures such as the VIX. Another interpretation of the UIP shock is a currency demand shock, as in Dao et al. (2025). Dao et al. (2025) use changes in CFTC dealer positions as a proxy for currency demand shocks, but dealer positions are far less persistent than the various risk measures. The persistent excess-bond-premium specification deliberately gives the reduced-form risk channel a favorable opportunity to matter. The counterfactuals below nevertheless show that it remains a secondary source of exchange-rate variation.

4.2 Calibration

The calibration combines external restrictions, direct moment matching, and a simulated-method-of-moments criterion. I fix a small set of conventional preference and technology parameters from the literature. Long yields and macroeconomic quantities discipline the fundamental shock processes; survey dispersion disciplines disagreement; and broker-dealer equity disciplines intermediary wealth. Following the empirical strategy of Kekre and Lenel (2024), I also use exchange-rate scale and persistence when selecting the quantitative specification. I therefore do not treat every entry in the results table as an out-of-sample test. The main evidence for the mechanism comes from a sharper restriction and the ablations: the disagreement parameters are fixed by surveys and intermediary data before I ask whether an economy without disagreement can reproduce the same joint pattern. It cannot, even when I re-estimate the conventional shock block.

I set the degree of openness $\gamma = 0.05$ and household risk aversion $\sigma = 5$, both following Chahrour et al. (2024); Candian and De Leo (2025). The trade elasticity is $\theta = 1.5$, a common value in this literature (Itskhoki and Mukhin 2021; Candian and De Leo 2025). The pass-through of the exchange rate to relative consumer prices, $\chi = 1 - \xi_v/(\theta_v - 1)$, is set to 0.60, following Kekre and Lenel (2024, 2026b). That value sits between the long-run pass-through of 0.51 for U.S. imports and 0.71 on average for other G10 imports, reported by Burstein and Gopinath (2014). The reduced-form UIP shock persistence is $\rho_\psi = 0.85$,

again following Kekre and Lenel (2024). This comes from the persistence of the excess bond premium; using the VIX instead gives almost the same number. The two belief groups are equally populated, $\eta_A = \eta_B = 0.5$.

Table **??** reports the internally calibrated parameters and their targets. I describe the shock processes first, then the arbitrageurs' parameters, and last the disagreement block. Data moments are computed over 1991Q2–2020Q1. Home is the U.S. and Foreign is a set of advanced economies matched to the Blue Chip Financial Forecasts Fed AFE dollar forecasts: Australia, Canada, Switzerland, the Euro Area, the United Kingdom, Japan, and Sweden. All macro variables are aggregated with the trade weights between those countries and the U.S., matching the convention of the reported Fed AFE dollar exchange rates.

Interest rate moments discipline the discount factor processes. The strategy follows Kekre and Lenel (2024), who show that discount factor shocks can be proxied by long-term interest rates. The model has a steady-state interest rate differential, which the steady-state discount factors pin down. The U.S. discount factor $\beta = 0.9980$ matches an annualized U.S. real interest rate of 0.804%, the 0.2% quarterly real rate deflated by realized inflation. The average real interest differential over the sample, 0.0502% quarterly, then pins down $\beta^* = 0.9975$. The U.S. discount factor is the higher of the two, so the U.S. steady-state real rate is below the Foreign rate and the dollar is the funding currency, consistent with Kekre and Lenel (2026a). The persistence $\rho_\beta = 0.9861$ targets the AR(1) of the 10-year home-minus-foreign expectations-hypothesis yield differential, σ_β targets the standard deviation of changes in the annualized differential, and the cross-country correlation $\text{Corr}(\hat{\beta}, \hat{\beta}^*) = 0.219$ targets the comovement of changes in U.S. and advanced-foreign-economy 10-year expectations-hypothesis yields.

That last target departs from Kekre and Lenel (2024), who calibrate the correlation to match the volatility of long yields. I target the empirical counterpart of the correlation directly instead. The resulting yield volatility is close to the data either way, so little rides on the choice. The calibrated values are close to those in Kekre and Lenel (2024), with persistence slightly higher here because the country set differs.

The endowment processes are calibrated in parallel, though the construction of output and the choice of comovement target both need a word. Output in the model is the per capita value of final goods in units of consumption. For its empirical counterpart I follow Kekre and Lenel (2024) and define output per capita as nominal GDP less investment and government spending, scaled by the consumption deflator and population. The persistence and scale of the endowment process are disciplined by the persistence of relative output and the volatility of relative output growth. I choose the cross-country correlation of endowment innovations within a joint criterion that also includes the cross-country correlation of consumption growth

and the volatility of relative consumption compared with relative output. The consumption-correlation discipline keeps the model from generating a counterfactually negative cross-country consumption correlation, which the calibrated models of Kekre and Lenel (2024) and Candian and De Leo (2025) both produce. The estimated endowment persistence is again somewhat higher than in Kekre and Lenel (2024), and again the country set explains it: output per capita in advanced economies is more persistent than in emerging markets.

U.S. intermediaries' equity-to-GDP ratio disciplines the arbitrageurs' parameters. He et al. (2017) find that the primary dealer counterparties of the New York Federal Reserve are the marginal investors in many risky asset classes, currencies among them. In that spirit I discipline intermediary wealth with the equity capital of U.S. broker-dealers relative to GDP[5]. This measure is the natural empirical counterpart to intermediary net worth in the model, since it tracks the risk-bearing capacity of the institutions most active in international asset markets. Its average level disciplines steady-state net worth W, and its persistence disciplines the arbitrageurs' exit rate ξ[6]. The fit is informative but not exact: the model produces a mean equity-to-GDP ratio of 0.0777 and a finite-sample persistence of 0.97, compared with 0.0726 and 0.92 in the data. The persistence miss partly reflects the equilibrium lower bound on the exit rate. The Appendix reports sensitivity to both intermediary parameters.

The intermediation friction Γ is estimated jointly with exchange rate volatility among the targets, because it governs how far arbitrageurs will move capital between the two countries. When Γ is small, arbitrageurs adjust positions aggressively on their subjective expected excess returns, which is what the modified UIP condition prices and what exchange rate volatility reveals. Stationarity of the wealth share, $\lambda = (1-\xi)\left(\frac{1}{\beta} - \frac{\omega^A \omega^B d_{ss}^2}{\beta^* \Gamma}\right) \in (0,1)$, places a lower bound on Γ. Because this lower bound moves with the steady state, the scale-free object is more informative than the raw value. The calibration gives $\Gamma/\Gamma_{\min} = 2.79$, comfortably inside the stationary region. The resulting finite-sample wealth persistence of 0.97 is near the persistent end of estimates used in the intermediary literature.

The new layer of the calibration disciplines the disagreement block with survey data. The source is the *Blue Chip Financial Forecasts* (BCFF), a monthly survey of professional forecasters published by Wolters Kluwer. The BCFF collects forecasts from roughly 45–50

[5]He et al. (2017) define the intermediary capital ratio as primary dealers' aggregate market equity divided by market equity plus book debt. It therefore measures net worth per dollar of intermediary assets, whereas W_t in the model is the level of intermediary net worth relative to aggregate output. I convert the ratio into the model's units by multiplying it by broker-dealer assets relative to quarterly GDP from the Financial Accounts.

[6]Kekre et al. (2024); Kekre and Lenel (2026a) pin down these two parameters from the conditional dynamics of their models in response to monetary policy shocks. The present model, like Kekre and Lenel (2024), has no monetary layer, which makes a direct proxy for W_t the natural choice here. The proxy is imperfect, so I report the sensitivity of the model's performance to both parameters.

institutions each month, including major banks and broker-dealers, and it reports forecasts institution by institution rather than only the consensus. That is what makes it usable here. Forecasters predict the "major currencies dollar index," a trade-weighted broad dollar measure, one quarter ahead, which maps onto the broad exchange rate of the two-country model. I measure disagreement as the spread between the top-decile and bottom-decile expected one-quarter-ahead log dollar changes, computed under a common current exchange rate. It corresponds directly to the model's gap between the two groups' expected exchange rate changes.

Three survey moments pin down three parameters. Steady-state disagreement $d_{ss} = 0.0448$ targets the sample mean of the spread, 0.0440, or roughly 4.4 percentage points per quarter. The Kalman gain $G = 0.3182$ targets its AR(1) of 0.648. The interpretation-shock volatility $\sigma_\mu = 0.001408$ targets its innovation standard deviation of 0.0056. The persistence and innovation volatility are matched almost exactly, while the mean is 0.0448 in the model and 0.0440 in the data. The survey therefore fixes the size, persistence, and volatility of the new channel rather than leaving them free. In the estimation all fundamentals share the common gain G and interpretation-shock scale σ_μ, and d_{ss} and d_t denote the implied one-quarter-ahead exchange rate disagreement. The resulting steady state is asymmetric: the optimists' cumulated carry gains imply a steady-state wealth share $\omega^A = 0.68$, so $\delta_w = (\omega^A - \omega^B)/2 > 0$.

I also calibrate the average belief bias, following Candian and De Leo (2025), who trace the observed predictability reversal to subjective forecast errors. In their model agents overestimate the persistence of fundamentals and update with suboptimal information gains. I parameterize those same two objects — perceived persistence and information gains — to generate a plausible predictability-reversal profile. The forecast regressions in Coibion and Gorodnichenko (2015) and Kohlhas and Walther (2021) provide external diagnostics rather than exact targets. The resulting parameters are $\kappa_y = 0.259$ and $\widetilde{\rho}_y = 0.9995$ for output, and $\kappa_\beta = 0.969$ and $\widetilde{\rho}_\beta = 0.9924$ for discount-factor shocks. At the one-quarter horizon, the Coibion–Gorodnichenko regression implies an effective gain for exchange-rate expectations of 0.66 in the model, compared with 0.78 in the data. The Kohlhas–Walther regression gives 0.11 in the model and 0.08 in the data; by that measure exchange rate expectations show no overreaction. These estimates are noisy across samples and horizons, so I use them to assess plausibility rather than to claim exact survey fit. Average bias improves the horizon profile of expected returns, but the results below show that it is not what generates the standard unconditional disconnect moments.

Table 1: Quantitative parameters and empirical disciplines

Parameter	Value	Target moment	Data	Model
W	0.07768	mean (intermed. equity/y)	0.0726	0.07768
ξ	0.009734	ρ_{-1} (intermed. equity$_t/y_t$)	0.92	0.97
Γ	0.0041	$\sigma(\Delta\hat{e}_t)$	3.33%	4.02%
d_{ss}	0.04480	$E_t^A[\Delta\hat{e}_{t+1}] - E_t^B[\Delta\hat{e}_{t+1}]$	0.044	0.045
Common belief gain G	0.31820	$\rho_{-1}\left(E_t^A[\Delta\hat{e}_{t+1}] - E_t^B[\Delta\hat{e}_{t+1}]\right)$	0.648	0.648
$\sigma_{\mu,\beta}, \sigma_{\mu,y}$	0.001408	innovation SD of one-quarter disagreement	0.56%	0.56%
ρ_β	0.98607	$\rho_{-1}\left(\hat{r}_t^{(40)} - \hat{r}_t^{*(40)}\right)$	0.94	0.96
σ_β	0.0007167	$\sigma\left(\Delta(\hat{r}_t^{(40)} - \hat{r}_t^{*(40)})\right)$	0.273%	0.256%
Corr$(\varepsilon^\beta, \varepsilon^{\beta*})$	0.21866	corr $\left(\Delta\hat{r}_t^{(40)}, \Delta\hat{r}_t^{*(40)}\right)$	0.81	0.80
ρ_y	0.98669	$\rho_{-1}(\hat{y}_t - \hat{y}_t^*)$	0.95	0.94
σ_y	0.009155	$\sigma(\Delta(\hat{y}_t - \hat{y}_t^*))$	0.77%	0.72%
Corr$(\varepsilon^y, \varepsilon^{y*})$	0.69128	corr $(\Delta\hat{c}_t, \Delta\hat{c}_t^*)$	0.63	0.71
$(\kappa_y, \widetilde{\rho}_y)$; $(\kappa_\beta, \widetilde{\rho}_\beta)$	$(0.259, 0.9995)$; $(0.969, 0.9924)$	predictability reversal; CG and KW diagnostics		
ρ_ψ	0.85	$\rho_{-1}\left(\hat{\psi}_t\right)$	0.80	0.82
σ_ψ	0.01376	$\sigma(\hat{e}_t)$	10.62%	10.62%

Notes: Model moments are averages over 1,000 independent simulations of 116 quarters after a 1,000-quarter burn-in. The UIP-shock persistence is set from the excess bond premium. The average-bias parameters target a plausible predictability-reversal profile, with CG and KW regressions reported as external diagnostics; the remaining rows report direct or joint moment disciplines.

4.3 Exchange Rate Moments

Table 2 compares four economies. The full model contains average bias, both nonlinear cross terms, and the reduced-form UIP-risk shock. The separately calibrated FIRE benchmark retains disagreement and endogenous wealth but removes average bias and UIP risk. Its consensus forecast is rational on average even though beliefs remain dispersed across arbitrageurs. The two no-disagreement economies set $d_{ss} = 0$ and $\varepsilon_{\mu,t} = 0$. The fixed version holds the FIRE parameters unchanged; the recalibrated version re-estimates the conventional shock block, giving the model its best chance to recover the moments without disagreement. The key overall takeaway is that an exogenous UIP shock is not necessary or important, and in some cases it is even harmful, especially when it is externally disciplined using a risk proxy.

Exchange rate volatility and persistence All four models match highly persistent exchange rate dynamics, as they inherit persistent fundamental states and therefore generate persistent exchange rate levels. This is still notable as the persistence of exchange rate is not targeted. Disagreement is not needed for that fact alone. Its role is to magnify the price response through the wealth distribution, while home bias and incomplete pass-through mute the response of consumption and output. In fact, without disagreement, models undershoot the exchange rate volatility, relative to the data. Although the exchange rate volatility level is targeted, when it is jointly matched together with other moments, these levels are the best these models get because shock processes are disciplined by external proxies.

This point becomes clearer if we look at the relative volatilities. Starting from exchange rate volatility relative to consumption volatility, the full model and the FIRE variant both generate high "excess volatility". They somewhat undershoot but both models imply that real exchange rate volatilities are four to five times greater than consumption volatilities, which is consistent with the data and the conventional wisdom. However, if we drop the wealth amplification channel, models generate too little exchange rate volatility compared to consumption. In fact, no-disagreement economies imply that exchange rate volatilities are only 1.5 to 3 times larger than consumption volatilities, much less than the data. This result implies that the amplification channel through the multiplier, Ξ, actually generates a plausible magnitude of exchange rate volatility amplifications. Most notably, this success does not come from adding a reduced form UIP shock as is evident from the FIRE variant's success. The full model, which has a reduced form UIP shock, does as good as the FIRE variant, which does not have a reduced form UIP shock, suggesting that the endogenous amplification mechanism works quantitatively. The exchange-rate-to-output volatility row has the same pattern across models. It tends to be somewhat smaller because consumption

smoothing generates smaller volatility compared to output.

The volatility of the short rate differential relative to exchange rate changes varies considerably across specifications. The FIRE benchmark comes closest to the data, generating a ratio of 0.166 compared with 0.110 empirically, even though this moment is not directly targeted. This result provides another measure of the disagreement multiplier. Disagreement amplifies exchange rate movements much more than short rate differentials, thereby lowering $\sigma(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})/\sigma(\Delta\hat{e}_t)$. When disagreement is removed, exchange rate volatility falls disproportionately and the ratio rises to 0.247, or 0.309 after recalibration.

The full model instead generates a ratio of 0.513, reflecting excessive short rate volatility rather than insufficient exchange rate volatility. Average belief bias is the main proximate source. Unlike disagreement, which washes out when beliefs are averaged holding wealth shares fixed, consensus bias survives aggregation and enters the aggregate Euler equations as a forecast error wedge. It therefore induces additional movements in consumption growth and short-term interest rates. Consistent with this interpretation, removing average bias lowers the ratio from 0.513 to 0.374, while having little effect on the relative volatility of long yields. The reduced form UIP shock reinforces this effect by adding short-lived pressure to the currency market equilibrium, part of which is absorbed through the current interest rate differential. Its quantitative contribution is smaller, however: removing the UIP shock lowers the ratio only from 0.513 to 0.473. Thus, the full model's miss is concentrated at the short end of the yield curve; it still closely matches the volatility of the long rate differential.

Fama coefficient The Fama coefficient provides a sharp diagnostic for distinguishing the models. Under UIP, a regression of next-quarter depreciation, $\Delta\hat{e}_{t+1}$, on the current interest-rate differential, $\hat{r}_t - \hat{r}_t^*$, should produce a coefficient of one. The data instead produce a coefficient of -0.213: when the Foreign interest rate is higher, the Foreign currency tends to appreciate rather than depreciate. This is the classic Fama puzzle.

Both the full model and the FIRE benchmark reproduce the negative sign, with coefficients of -0.449 and -0.480, respectively. This negative coefficient result comes from different sources between the two models. For the FIRE benchmark, the result comes from disagreement rather than average bias or the reduced-form UIP shock. The endogenous wealth–disagreement wedge enters the interest rate differential and expected depreciation with opposite signs. A shift in wealth toward one belief type therefore moves the observed interest differential in one direction while pushing expected currency depreciation in the other. In particular, a high Foreign interest rate can be accompanied by further Foreign currency appreciation.

The full model's negative Fama potentially comes from three sources: the wealth–disagreement

wedge, average bias, the reduced form UIP wedge, because all of them work as UIP wedges[7]. To see the role of the wealth–disagreement wedge I remove both the UIP shock and average bias, and recomputed the Fama coefficient. Under this variant, the Fama coefficient becomes -0.576, suggesting that the major source of the UIP variation is still the wealth–disagreement wedge, even without recalibrating the whole model. This reassures the importance of the wealth–disagreement wedge as a major driver of UIP deviation.

Removing disagreement eliminates this wedge and restores the conventional UIP relationship: the two no-disagreement economies generate positive Fama coefficients of 1.023 and 0.906. Persistent fundamental shocks alone can generate persistent exchange rates, but they do not reverse currency-return predictability. The negative Fama coefficient therefore isolates the central role of the endogenous interaction between disagreement and wealth.

Table 2: Key exchange-rate moments

		Current full-model	FIRE	No-disagreement	
Moment	Data	Full model	Separate calib.	Fixed	Recalibrated
$\rho_1(\Delta\hat{e}_t)$	0.013	0.064	-0.050	-0.009	-0.003
$\rho_1(\hat{e}_t)$	0.950	0.917	0.908	0.930	0.926
$\sigma(\Delta\hat{e}_t)$	0.033	0.040	0.034	0.013	0.020
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{c}_t)$	6.43	4.46	4.11	1.58	2.95
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{y}_t)$	5.42	4.40	4.03	1.54	2.94
$\sigma(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})/\sigma(\Delta\hat{e}_t)$	0.110	0.513	0.166	0.247	0.309
$\sigma(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})/\sigma(\Delta\hat{e}_t)$	0.060	0.061	0.074	0.240	0.295
$\mathrm{Corr}(\Delta\hat{e}_t, \Delta(\hat{c}_t - \hat{c}_t^*))$	-0.169	-0.471	-0.244	0.546	-0.131
β^{Fama}	-0.213	-0.449	-0.480	1.023	0.906
R^2_{Fama}	0.0005	0.0606	0.0135	0.0431	0.0528

Notes: The full model includes the average-belief process, both nonlinear cross terms, and the reduced-form UIP shock. The FIRE benchmark is calibrated separately and removes average bias and the UIP-risk shock while retaining disagreement and the nonlinear cross terms. The fixed no-disagreement economy keeps the FIRE parameters; the recalibrated economy re-estimates its conventional shock block. Model moments are averages over 1,000 simulations of 116 quarters following a 1,000-quarter burn-in. In the volatility rows,$\Delta\hat{c}_t$ is Home consumption growth, and the output denominator is Home output growth. The joint calibration criterion uses a separate exchange-rate-to-relative-consumption volatility moment reported in the Appendix.

Backus–Smith Except for the fixed-parameter no-disagreement economy, all reported specifications reproduce the negative sign of the Backus–Smith correlation between exchange

[7]The NFA term in principle breaks the UIP condition too, but its role is quite limited. In fact, no-disagreement economies, having the NFA wedge as the only source of the UIP wedge, generate positive Fama.

rate changes and relative consumption growth. Demand shocks play a central role: the supply-only correlations remain positive at the reported calibration (Appendix Table C1).

Disagreement nevertheless strengthens the demand channel. Holding the FIRE benchmark's fundamental shock processes and other structural parameters fixed, introducing disagreement changes the Backus–Smith correlation from 0.546 to −0.244. The no-disagreement counterfactual removes both the permanent belief gap d_{ss} and transitory disagreement innovations. The wealth–disagreement channel amplifies demand's negative covariance relative to supply's positive covariance, allowing the former to dominate without changing the fundamental shock parameters. Appendix C documents this comparison and shows that the sign reversal persists when only permanent disagreement is retained.

The appendix also establishes a stronger possibility in the reduced equilibrium: with sufficiently strong feedback and suitable relative persistence, supply shocks alone can generate a negative unconditional Backus–Smith correlation. Amplification strengthens expenditure switching relative to the direct income effect. As the wealth-driven exchange rate component subsequently unwinds, relative consumption and the exchange rate move in opposite directions, creating demand-like movements following supply news. The result concerns adjustment over time; the impact comovement remains positive. Appendix C.2 gives the exact sign condition and a numerical illustration.

The role of average bias: Engel (2016) moments The preceding results attribute much of the model's FX-disconnect fit to the wealth–disagreement channel. The FIRE benchmark already reproduces the core joint patterns without average bias or a reduced-form UIP shock. Average bias has a distinct role: it helps account for the timing and cumulative sign of currency-return predictability. Subjective forecast errors therefore remain informative about exchange-rate dynamics even though they are not needed for the benchmark's main results.

Engel (2016) documents a reversal in the relationship between interest differentials and subsequent excess returns. A currency with a relatively high interest rate initially earns positive expected excess returns, but this relationship reverses at longer horizons. The later negative excess returns can more than offset the initial gains. This is a restriction on returns net of the interest differential: eventual depreciation by itself would also be consistent with UIP[8].

The full model generates this predictability reversal. Regressing future one-quarter foreign-currency excess returns on the current Home-minus-Foreign interest differential yields

[8]A closely related moment is the level-on-level or changes-on-changes correlation between exchange rates and interest rate differentials, a moment emphasized by Kekre and Lenel (2024). Appendix's full tables report these moments and the models' CI covers the data moment.

a coefficient of -1.45 at one quarter. It becomes positive at six quarters and reaches 0.22 at twelve quarters. Removing average bias while holding the remaining calibration fixed delays the sign change to eight quarters and makes the initial negative coefficients larger. Average bias thus changes the shape of the reversal; it is not necessary for every eventual sign change. Appendix D reports the horizon profiles and the corresponding cumulative-return regressions. The intuition follows the belief process in Candian and De Leo (2025). Consensus expectations initially underreact to fundamental news because the perceived information gain is below one, but agents subsequently extrapolate the news using a persistence above its true value. These beliefs enter both household Euler equations and portfolio decisions, changing interest differentials as well as exchange-rate adjustment.

4.4 What moves the exchange rate?

Fundamental shocks account for most exchange rate variation, while the additional UIP shock remains secondary at every reported horizon. Figure 3 reports a Shapley decomposition of the forecast error variance of cumulative exchange-rate changes over 1 to 40 quarters. In the full model, discount factor and output shocks account for 34 and 45 percent of the one-quarter variance, respectively. Their combined share rises from 80 percent at one quarter to 87 percent at 40 quarters, while the UIP shock's share falls from 18 to 13 percent. Although output shocks contribute more in this specification, discount-factor shocks exceed the UIP shock at every horizon and account for 63–71 percent of the non-supply contribution.

The separately calibrated FIRE benchmark shows that an independent UIP shock is not needed to reproduce the core FX-disconnect patterns. Discount factor shocks are the largest source of variation in this economy, accounting for 62–67 percent across the reported horizons. This is consistent with the predominant role of persistent demand shocks in Kekre and Lenel (2024). Their model nevertheless assigns currency-intermediation shocks an important role in short-horizon UIP deviations. Here, the FIRE benchmark generates a negative Fama coefficient through the endogenous wealth–disagreement wedge, without an additional UIP shock.

The small contribution of direct belief innovations reinforces the distinction between the source of news and its transmission. Signal-noise and disagreement innovations together account for less than 3 percent of the variance in either specification, even though they affect the UIP condition. Their small share does not imply that disagreement or the endogenous UIP wedge is unimportant. Fundamental news changes portfolios and redistributes wealth across investors with different beliefs, moving the UIP wedge even without new disagreement shocks. The robust result is therefore that conventional macroeconomic shocks generate

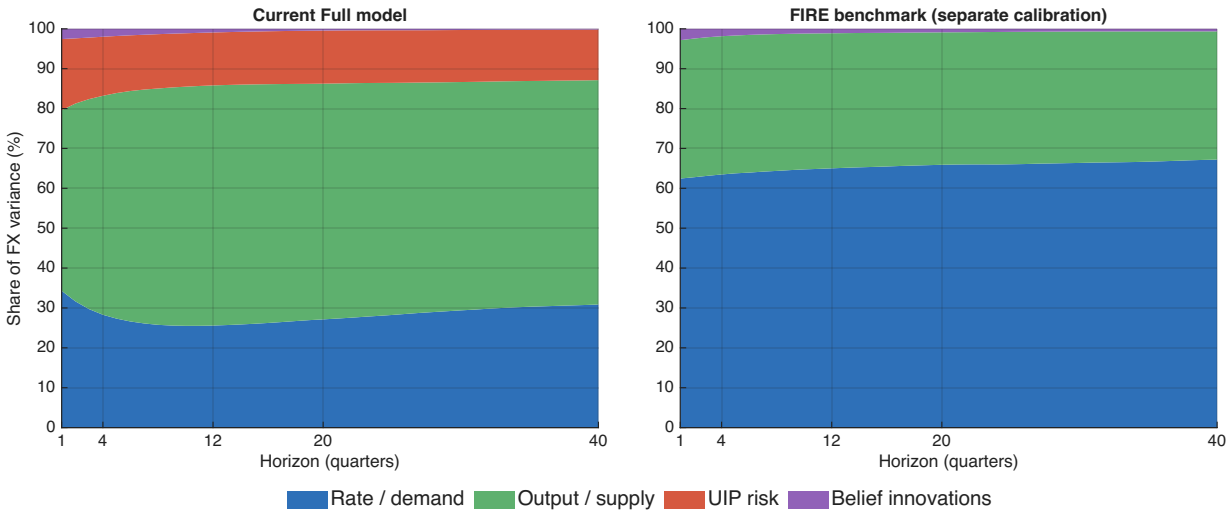


Figure 3: Exchange-rate variance decomposition across horizons

Notes: The bars report Shapley shares of the forecast-error variance of cumulative exchange-rate changes at horizons from 1 to 40 quarters. The upper panel is the current full model; the lower panel is the separately calibrated FIRE benchmark without average bias or the additional UIP shock. "Belief innovations" combines the signal and disagreement innovations shown separately in the simulation output.

most exchange-rate variation through the wealth–disagreement channel; the precise division between demand and supply depends on the specification.

4.5 Historical shock inversion

The baseline model reproduces the scale of historical dollar movements and captures part of their low-frequency comovement with the data. Over 1999Q1-2019Q4, shocks inferred from output, long-term yields, and survey disagreement generate approximately the observed exchange-rate level volatility. The representative-agent economy generates only 40 percent as much volatility as the data. This exercise gives the same message as the simulations: conventional fundamentals supply the news, and the wealth-disagreement mechanism determines how strongly that news moves the exchange rate.

Specifically, I invert the first-order model quarter by quarter, retaining the calibrated parameters and data transformations. Two country demand shocks, two output shocks, and a beta-disagreement shock match Home and Foreign long yields, detrended log output per capita, and Blue Chip disagreement. Signal-noise innovations and the additional UIP shock are zero in the baseline. The representative-agent model sets disagreement to zero, recomputes its steady state, and re-infers its four fundamental shocks from the same macroeconomic observations. Neither realized exchange rates nor survey forecast errors enter shock identification. The plotted nominal paths condition on observed relative CPI; Appendix F.3

gives the measurement and initialization details.

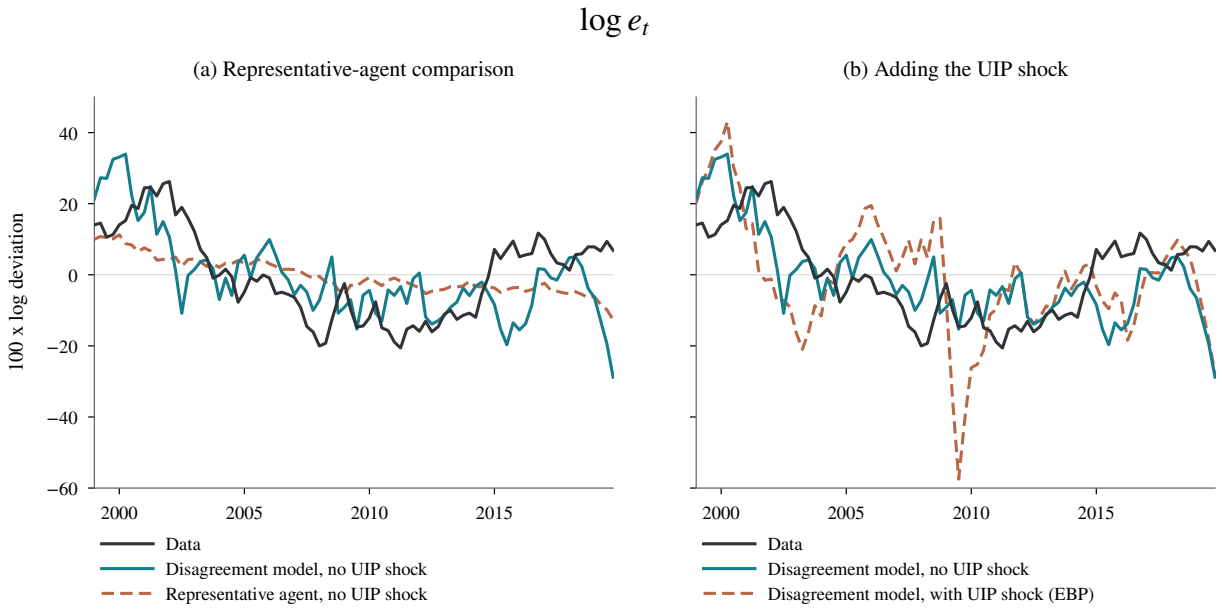


Figure 4: Historical shock inversion, 1999Q1-2019Q4

Figure 4(a) illustrates amplification. The disagreement model's level standard deviation is 1.00 times the data's, compared with 0.40 in the representative-agent economy. Their level correlations are 0.50 and 0.43. Thus the disagreement model recovers the observed scale of movements with a modestly improved level comovement. The message here is that the substantial amount of volatility is generated by the wealth-disagreement channel. The central gain is the amount of exchange-rate volatility generated by the same macroeconomic history.

The model achieves this result while attributing a substantial part of exchange-rate variation to output. A decomposition of the inferred history assigns about 43 percent of the model's real exchange-rate level variance to output shocks. Direct disagreement innovations contribute very little. This attribution concerns the origin of news. Output news changes the exchange rate, revalues the opposing positions of investors with different beliefs, and shifts wealth toward the investors who benefit from the movement. Their beliefs then receive greater weight in prices, magnifying the initial response. The amplification multiplier Ξ in Corollary 1 summarizes this feedback in the analytical model. Its quantitative importance is consistent with a large output-shock contribution: disagreement amplifies fundamental news even when no new disagreement shock arrives.

Figure 4(b) adds the EBP-implied UIP shock at its calibrated scale and re-infers the remaining shocks. The level correlation falls from 0.50 to 0.32, and level RMSE rises from 12.4

to 17.3 log percentage points. This externally disciplined risk proxy therefore weakens the baseline's level fit. The historical comparison supports the role of endogenous amplification in generating large currency movements from ordinary macroeconomic shocks. Appendix F.3 reports the longer sample and the demand-plus-beta-disagreement restriction; Appendix F.3.4 retains the complementary recursion exercise, which conditions on realized survey news.

5 Empirical Supporting Evidence

The empirical evidence links survey beliefs to exchange rates and intermediaries' positions. In the first exercise, I examine the mechanism's core assumption: arbitrageurs' positions are driven by subjective beliefs and survey data reflect them. I find that revisions in consensus forecasts explain dealer-position changes even after controlling for spot-rate movements. Additionally, I use the conditional-volatility prediction as an external validation exercise. These exercises draw on Bloomberg forecasts, a separate survey from the Blue Chip Financial Forecasts used to discipline the model.

5.1 Data

Bloomberg FX forecasts Bloomberg provides individual exchange-rate level forecasts for G10 currencies from 2006 onward. The exchange-rate regressions use AUD, CAD, CHF, EUR, GBP, JPY, NOK, NZD, and SEK against the U.S. dollar. The dealer-position and volatility exercises use the six currencies with matched CFTC data: AUD, CAD, CHF, EUR, GBP, and JPY. I define $e_{i,t}$ as the log U.S.-dollar price of one unit of foreign currency, so an increase denotes a dollar depreciation.

For the exchange-rate regressions, disagreement at each forecast horizon is the cross-forecaster standard deviation of forecast levels divided by the corresponding consensus forecast. This relative dispersion is invariant to the units in which a currency is quoted. I summarize its term structure using the first two principal components of log relative dispersion at 90-, 180-, 270-, and 360-day horizons, estimated separately for each currency. The regressions use monthly changes in these components. The archived monthly panel yields 172 to 178 complete observations per currency, from May 2006 through February 2021; NOK and SEK begin in November 2006.

CFTC dealer positions The CFTC Traders in Financial Futures report provides weekly dealer net positions in currency futures. I aggregate them to monthly frequency and divide by the 12-month moving average of total open interest. The dynamic dealer-position regres-

sions match this measure to Bloomberg 90-day consensus forecasts and contain 221 monthly observations per currency.

Realized volatility For the volatility exercise, realized volatility is the square root of the sum of squared daily log exchange-rate changes within each calendar quarter. Quarterly disagreement averages the monthly log of 90-day forecast dispersion relative to the month's mean spot rate. This exercise uses the longer survey and spot-rate sample through 2024 and measures volatility in subsequent quarters; Section 5.3 gives the sample requirements and horizon-specific observation counts.

5.2 Survey forecasts explain dealer positions

If arbitrageurs' positions reflect their beliefs, revisions of measured beliefs should move positions. We test this directly with the dynamic specification

$$\Delta f^D_{i,t} = \alpha_i + \alpha_t + \rho\,\Delta f^D_{i,t-1} + \theta_0\,\Delta \widehat{e}^F_{i,t} + \theta_1\,\Delta \widehat{e}^F_{i,t-1} + \gamma_0\,\Delta e_{i,t} + \gamma_1\,\Delta e_{i,t-1} + \varepsilon_{i,t}, \tag{12}$$

where $\Delta \widehat{e}^F_{i,t}$ is the monthly change in the Bloomberg 90-day consensus forecast for currency i and $\Delta e_{i,t}$ the spot change, with currency and month fixed effects and Driscoll–Kraay standard errors with six lags. All variables are standardized.

Table 3: Survey forecasts explain dealer positions

	Standardized coefficient
Lagged dealer-position change Δf^D_{t-1}	0.103^{***} (0.039)
Current forecast change $\Delta \widehat{e}^F_t$	0.119^{***} (0.042)
Lagged forecast change $\Delta \widehat{e}^F_{t-1}$	0.128^{***} (0.029)
Current spot change Δe_t	-0.556^{***} (0.088)
Lagged spot change Δe_{t-1}	-0.040 (0.052)
Observations	1,326
Within R^2	0.327
p-value: forecast-change block	< 0.001

Notes: f^D is the dealer net position divided by 12-month moving-average open interest. Monthly Bloomberg 90-day consensus forecasts and CFTC dealer positions; panel of six currencies (AUD, CAD, CHF, EUR, GBP, JPY). Standardized coefficients; currency and month fixed effects; Driscoll–Kraay standard errors with six lags. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 reports the pooled results. Forecast changes are associated with position changes both immediately and with a delay: the cumulative effect is $\theta_0 + \theta_1 = 0.247$ ($p < 0.001$). Dealers simultaneously lean against contemporaneous spot movements (coefficient -0.556),

consistent with the model's portfolio rule: holding expected future exchange rates, interest rates, and wealth fixed, a higher current Foreign-currency price lowers expected excess returns and desired Foreign-bond exposure.

Table 4: The forecast channel is broad across currencies

Currency	Forecast change $\theta_0 + \theta_1$	Spot change γ_0	Lagged ρ	R^2
AUD	0.324^{***} (0.110)	-0.581^{***} (0.126)	0.111^{**} (0.045)	0.364
CAD	0.302^{***} (0.085)	-0.578^{***} (0.099)	0.114 (0.074)	0.426
CHF	0.330^{***} (0.065)	-0.624^{***} (0.110)	-0.006 (0.054)	0.437
EUR	0.312^{***} (0.067)	-0.551^{***} (0.076)	0.061 (0.069)	0.334
GBP	0.216^{***} (0.067)	-0.645^{***} (0.105)	0.196^{***} (0.063)	0.389
JPY	0.187^{**} (0.094)	-0.686^{***} (0.085)	0.140^{*} (0.075)	0.451
Observations per currency		221		

Notes: Same dynamic specification as the pooled regression, estimated currency by currency. Standardized coefficients; Newey–West standard errors with six lags. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4 shows that the channel is broad: the cumulative forecast-change effect is positive and significant in every one of the six currencies, while dealer inventories everywhere combine the belief-driven component with a contrarian response to spot movements.

5.3 Disagreement and conditional exchange-rate volatility

The model also predicts that disagreement affects conditional exchange-rate volatility. In the calibrated nonlinear model, a positive disagreement innovation raises the conditional volatility of subsequent exchange-rate changes by increasing intermediaries' exposure to currency returns. This prediction provides an additional way to assess the wealth–disagreement mechanism. Matching the unconditional FX-disconnect moments does not by itself require predictable variation in volatility, and the conditional-risk evidence below is not a calibration target. I therefore use it as an external validation exercise.

I test whether disagreement predicts future realized volatility for AUD, CAD, CHF, EUR, GBP, and JPY against the U.S. dollar. Monthly disagreement is the cross-forecaster standard deviation of 90-day-ahead forecast levels divided by the same month's mean spot rate, requiring at least eight forecasters. The quarterly measure $D_{i,t}$ averages the log of this relative dispersion over available survey months. Realized volatility $RV_{i,t}$ is the square root of the sum of squared daily log exchange-rate changes within the quarter, requiring valid observations for all three months. Outcomes are measured in entirely subsequent calendar quarters, so missing survey months do not compress the forecast horizon.

For horizons $h = 1, \ldots, 12$ quarters, I estimate the panel local projection

$$\log RV_{i,t+h} = \alpha_i^h + \tau_t^h + b_h D_{i,t} + \Gamma_h' X_{i,t} + u_{i,t+h}.$$

The controls $X_{i,t}$ include current log realized volatility, two quarterly lags of both log realized volatility and disagreement, current and lagged changes in log spot rates, and the log number of forecasters. Currency effects α_i^h absorb persistent differences across currencies, while quarter effects τ_t^h absorb common movements. I scale the coefficient by the residual standard deviation of disagreement after conditioning on these variables, $s_D = 0.143$, held fixed across horizons. The reported response is $100[\exp(b_h s_D) - 1]$. Inference allows contemporaneous dependence across currencies and serial dependence over calendar time.

Figure 5 compares the model response (left) with the quarterly local projections (right). In the model, a one-standard-deviation disagreement shock raises next-quarter conditional volatility by 9.05 percent; the response declines to 2.57 percent at four quarters. In the data, unusually high disagreement predicts higher subsequent realized volatility. A one-residual-standard-deviation increase in disagreement predicts 1.83 percent higher volatility in the next quarter, with a 95 percent pointwise interval of [0.17, 3.51]. This estimate uses 426 currency-quarter observations from 2006Q4 through 2024Q2. Point estimates remain positive over the twelve-quarter horizon, but the later responses are imprecisely estimated. The figure reports both 68 and 95 percent pointwise intervals; it does not establish a precisely estimated response at every horizon.

The evidence is clearer when risk is measured over the entire following year. Replacing the outcome by the log of $RV_{i,t}^{\text{year}} = \left(\sum_{j=1}^{4} RV_{i,t+j}^2\right)^{1/2}$ gives an estimated increase of 2.01 percent, with a 95 percent interval of [0.92, 3.10]. This regression uses 408 observations and predictor quarters 2006Q4–2023Q3. The estimate is 1.84 percent using the interquartile range of forecasts and 1.97 percent when windows overlapping the global financial crisis or the onset of COVID-19 are excluded. The next-year effect also remains positive when each currency is omitted in turn.

These results support the model's qualitative prediction that disagreement contains information about subsequent currency risk beyond current volatility. Because the two panels use different shock normalizations, the comparison does not establish a quantitative match of their magnitudes. Quarter effects make this a comparison across currencies within the same quarter; replacing them with VIX controls yields an imprecise estimate. The disagreement innovation is a regression residual, not an externally identified shock, so the evidence is predictive rather than causal. Its role is to provide external support for the mechanism's conditional-volatility implication, complementing the model's fit to the core FX-disconnect moments.

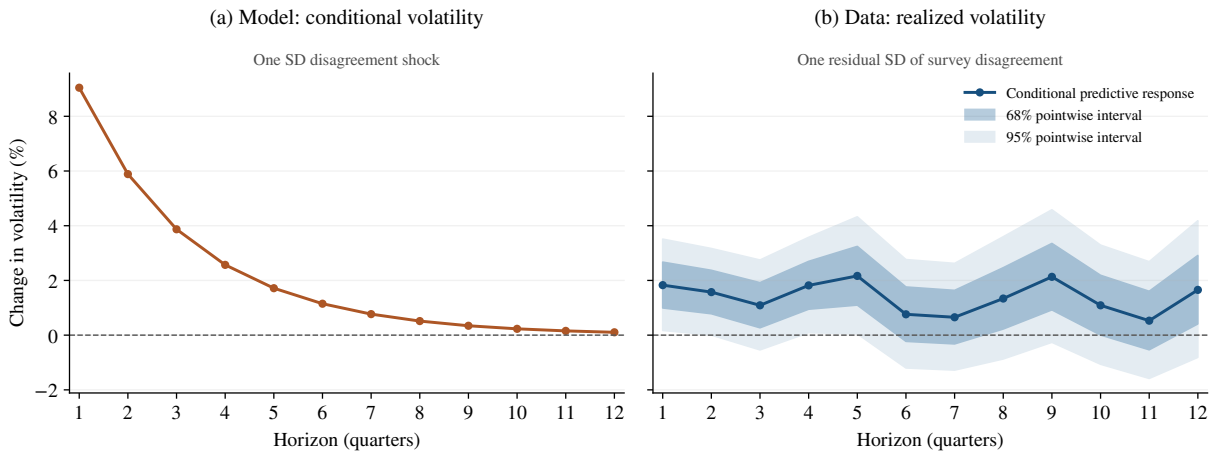


Figure 5: Disagreement and subsequent volatility: model and data

Notes: Left: generalized response of the mean conditional standard deviation of exchange-rate changes to a one-standard-deviation disagreement shock in the calibrated full model, expressed relative to the baseline mean. Right: panel local projections for six currencies against USD, scaled by one residual standard deviation of survey disagreement, with currency and quarter effects and the controls described in the text. Shock normalizations differ across panels. Dark and light shading denote 68 and 95 percent pointwise intervals. Empirical standard errors use Bartlett HAC with four quarterly lags on scores aggregated by calendar quarter and t critical values with the number of observed quarters minus one degrees of freedom. The next-year regression uses eight HAC lags. Horizon one is the next quarter in both panels.

6 Conclusion

This paper shows that disagreement among arbitrageurs can generate exchange rate disconnect through endogenous wealth redistribution. Heterogeneous beliefs and wealth dynamics create a wealth–disagreement wedge: a time-varying deviation from UIP that reflects who holds wealth when beliefs differ. This wedge provides a channel through which standard macroeconomic shocks generate large exchange rate movements, distinct from conventional risk compensation and the effects of external imbalances.

With disagreement disciplined by survey data, the calibrated model jointly reproduces high relative exchange rate volatility, negative Fama and Backus–Smith coefficients, and near-random-walk exchange rates. The mechanism also operates in a separately calibrated benchmark in which beliefs satisfy FIRE on average. At fixed parameters, removing disagreement reduces quarterly exchange rate volatility by 62 percent and reverses both the Fama and Backus–Smith signs; re-estimating conventional shocks does not restore the joint fit. Conventional rate and output shocks account for approximately 80 percent of short-horizon exchange rate variation. Disagreement matters primarily through the propagation of these shocks: currency positions redistribute wealth, changing whose expectations influence equilibrium prices.

Historical and empirical exercises provide complementary evidence. Recovering shocks

from output, long yields, and survey disagreement reproduces the observed scale of dollar level movements, while the representative-agent economy generates only 40 percent of that volatility. CFTC dealer positions respond to survey forecast revisions even after controlling for spot movements, and disagreement predicts subsequent realized currency volatility. These relationships support the model's links between beliefs, positions, and currency risk.

The broader implication is that an apparent disconnect from macroeconomic fundamentals can arise because their effects depend on the history of trading gains and losses within financial markets. The distribution of wealth across investors with different beliefs changes the exchange rate response to fundamental news, making that relationship difficult to recover from aggregate observables alone. This perspective complements work emphasizing currency demand and financial intermediation in exchange rate determination, including Dao, Gourinchas and Itskhoki (2025). A direct test of the wealth–disagreement mechanism requires investor-level data linking expectations, currency positions, wealth, and risk-bearing capacity. Such data would allow researchers to examine whether trading gains increase the pricing influence of investors whose beliefs differ from the consensus.

References

Alessandria, George, "Consumer search, price dispersion, and international relative price fluctuations," *International Economic Review*, 2009, *50* (3), 803–829.

Bacchetta, Philippe and Eric Van Wincoop, "Can information heterogeneity explain the exchange rate determination puzzle?," *American Economic Review*, 2006, *96* (3), 552–576.

_ and _, "Puzzling exchange rate dynamics and delayed portfolio adjustment," *Journal of International Economics*, 2021, *131*, 103460.

Borovička, Jaroslav, Lars Peter Hansen, and Thomas J Sargent, "Exploring recursive utility," Working paper 2023.

Burstein, Ariel and Gita Gopinath, "International prices and exchange rates," in "Handbook of International Economics," Vol. 4, Elsevier, 2014, pp. 391–451.

Caballero, Ricardo J and Alp Simsek, "Monetary policy with opinionated markets," *American Economic Review*, 2022, *112* (7), 2353–2392.

Campbell, John Y and Luis M Viceira, "Consumption and portfolio decisions when expected returns are time varying," *The Quarterly Journal of Economics*, 1999, *114* (2), 433–495.

Candian, Giacomo and Pierre De Leo, "Imperfect Exchange Rate Expectations," *The Review of Economics and Statistics*, 09 2025, *107* (5), 1406–1423.

Chahrour, Ryan, Vito Cormun, Pierre De Leo, Pablo A Guerrón-Quintana, and Rosen Valchev, "Exchange rate disconnect revisited," Technical Report, National Bureau of Economic Research 2024.

Chang, Chun, Zheng Liu, and Mark M. Spiegel, "Capital controls and optimal Chinese monetary policy," *Journal of Monetary Economics*, 2015, *74*, 1–15.

Coibion, Olivier and Yuriy Gorodnichenko, "Information rigidity and the expectations formation process: A simple framework and new facts," *American Economic Review*, 2015, *105* (8), 2644–2678.

Dao, Mai C, Pierre-Olivier Gourinchas, and Oleg Itskhoki, "Breaking Parity: Equilibrium Exchange Rates and Currency Premia," Technical Report, National Bureau of Economic Research 2025.

Du, Qingyuan, Stefano Eusepi, and Bruce Preston, "Non-rational beliefs in an open economy," *Review of Economic Dynamics*, 2021, *41*, 174–204.

Engel, Charles, "Exchange rates, interest rates, and the risk premium," *American Economic Review*, 2016, *106* (2), 436–474.

_ **and Steve Pak Yeung Wu**, "Exchange Rate Models are Better than You Think, and Why They Didn't Work in the Old Days," Technical Report, National Bureau of Economic Research 2024.

Evans, Martin DD and Richard K Lyons, "Order flow and exchange rate dynamics," *Journal of political economy*, 2002, *110* (1), 170–180.

Fukui, Masao, Emi Nakamura, and Jón Steinsson, "The macroeconomic consequences of exchange rate depreciations," *The Quarterly Journal of Economics*, 2025, *140* (4), 3015–3065.

Gabaix, Xavier and Matteo Maggiori, "International liquidity and exchange rate dynamics," *The Quarterly Journal of Economics*, 2015, *130* (3), 1369–1420.

Gilchrist, Simon and Egon Zakrajšek, "Credit spreads and business cycle fluctuations," *American economic review*, 2012, *102* (4), 1692–1720.

Gourinchas, Pierre-Olivier and Aaron Tornell, "Exchange rate puzzles and distorted beliefs," *Journal of International Economics*, 2004, *64* (2), 303–333.

Hassan, Tarek A, Thomas M Mertens, and Tony Zhang, "Not so disconnected: Exchange rates and the capital stock," *Journal of International Economics*, 2016, *99*, S43–S57.

He, Zhiguo, Bryan Kelly, and Asaf Manela, "Intermediary asset pricing: New evidence from many asset classes," *Journal of Financial Economics*, 2017, *126* (1), 1–35.

Itskhoki, Oleg and Dmitry Mukhin, "Exchange rate disconnect in general equilibrium," *Journal of Political Economy*, 2021, *129* (8), 2183–2232.

_ and _ , "What drives the exchange rate?," *IMF Economic Review*, 2025, *73* (1), 86–117.

Jiang, Zhengyang, Arvind Krishnamurthy, Hanno Lustig, and Jialu Sun, "Convenience yields and exchange rate puzzles," Technical Report, National Bureau of Economic Research 2024.

Kekre, Rohan and Moritz Lenel, "Exchange rates, natural rates, and the price of risk," *University of Chicago, Becker Friedman Institute for Economics Working Paper*, 2024, (2024-114).

_ and _ , "A model of us monetary policy and the global financial cycle," *Working Paper*, 2026.

_ and _ , "What Do Asset Prices in April 2025 Say About Demand for the Dollar?," Working Paper 35466, National Bureau of Economic Research July 2026.

_ , _ , and Federico Mainardi, "Monetary policy, segmentation, and the term structure," Technical Report, National Bureau of Economic Research 2024.

Kohlhas, Alexandre N. and Ansgar Walther, "Asymmetric Attention," *American Economic Review*, September 2021, *111* (9), 2879–2925.

Lilley, Andrew, Matteo Maggiori, Brent Neiman, and Jesse Schreger, "Exchange rate reconnect," *Review of Economics and Statistics*, 2022, *104* (4), 845–855.

Molavi, Pooya, Alireza Tahbaz-Salehi, and Andrea Vedolin, "Model complexity, expectations, and asset prices," *Review of Economic Studies*, 2024, *91* (4), 2462–2507.

Stavrakeva, Vania and Jenny Tang, "A fundamental connection: Exchange rates and macroeconomic expectations," *Review of Economics and Statistics*, 2024, pp. 1–49.

Valchev, Rosen, "Bond convenience yields and exchange rate dynamics," *American Economic Journal: Macroeconomics*, 2020, *12* (2), 124–166.

Appendix contents

A Full first-order equilibrium conditions **50**
A.1 Notation . . . 50
A.2 Steady-state quantities . . . 51
A.3 First-order equilibrium system . . . 52
A.4 Fundamentals, filtering, and expectations . . . 55

B Omitted Proofs and Derivations **57**
B.1 Information structure and belief dynamics . . . 57
B.2 Optimal portfolio choice . . . 61
B.3 Proof of Lemma 1 . . . 62
B.4 Proof of Lemma 2 . . . 63
B.5 Proof of Proposition 1 and Corollary 1 . . . 65
B.6 Proof of Proposition 2 . . . 66
B.7 Proof of Proposition 3 . . . 70

C Disagreement and the Backus–Smith correlation **73**
C.1 The calibrated economies and the fixed-shock comparison . . . 74
C.2 How supply shocks alone can produce a negative correlation . . . 75
C.3 A numerical illustration in the reduced equilibrium . . . 77

D Average bias and currency-return predictability **77**
D.1 Equivalence to the Candian–De Leo belief process . . . 77
D.2 Predictability across horizons . . . 79

E Time-varying disagreement in the quantitative model **83**
E.1 Beliefs and portfolio demand . . . 83
E.2 The two financial interactions . . . 83
E.3 Forecasting and solution . . . 84

F Additional quantitative results **86**
F.1 Full comparison of exchange-rate moments across model variants . . . 86
F.2 Full impulse responses across model variants . . . 90
F.3 Historical inversion and recursive accounting . . . 97

G Additional Empirical Results **103**
G.1 Disagreement and exchange-rate changes . . . 103

A Full first-order equilibrium conditions

This appendix collects the full first-order system of the structural model around the possibly asymmetric steady state. It retains the Home-bond UIP formulation, pricing to market, both groups' wealth dynamics, and the information structure in which current fundamentals are observed. Coefficients are written in terms of primitives and steady-state quantities; Γ is the effective risk-cost parameter defined in the portfolio approximation.

A.1 Notation

The Home and Foreign intermediate-good endowments are y_t and y_t^*, respectively, with steady-state values y and y^*. The producer prices are p_t, p_t^*, whereas $p_{Ht}, p_{Ft}, p_{Ht}^*, p_{Ft}^*$ are consumer prices. A star denotes the Foreign counterpart; for consumption quantities and consumer prices, it identifies the country of consumption. A subscript H or F on consumption quantities and consumer prices identifies the good's origin. The exchange rate e_t is Home consumption units per Foreign consumption unit. We use equal group masses, $\eta_A = \eta_B = 1/2$, and steady-state values have no time subscript.

For positive quantities and prices, $\hat{z}_t = \log(z_t/z)$. Interest-rate hats refer to gross returns:

$$\hat{r}_t = \log \frac{1+r_t}{1+r}, \qquad \hat{r}_t^* = \log \frac{1+r_t^*}{1+r^*}.$$

Bond positions, carry positions, and wealth shares can be zero or signed, so they use the following level deviations:

$$\widetilde{b}_t = \frac{b_t - b}{y}, \qquad \widetilde{b}_t^* = \frac{b_t^* - b^*}{y^*}, \qquad \widetilde{N}_t^i = \frac{N_t^i - N^i}{W^i}, \qquad \hat{\omega}_t^i = \omega_t^i - \omega^i.$$

Thus each household bond deviation is normalized by its own country's steady-state endowment, and each carry-position deviation is normalized by its own group's steady-state wealth. The latent persistent fundamental is denoted by τ_t^x, as in the information section.

Aggregate arbitrageur wealth and the Home-valued carry position are $W_t = (W_t^A + W_t^B)/2$ and $N_t = (N_t^A + N_t^B)/2$, measured after exit and entry. The bond markets clear when

$$\frac{b_t}{1+r_t} + W_t - N_t = 0, \qquad \frac{e_t b_t^*}{1+r_t^*} + N_t = 0.$$

The aggregate wealth law implies that exiting arbitrageurs rebate $\frac{\xi}{1-\xi}(W_t - \xi W_0)$, while new

entrants receive ξW_0. Net transfers and their first-order deviations are

$$\pi_t^a = \frac{\xi}{1-\xi}(W_t - W_0), \qquad \pi_t^a - \pi^a = \frac{\xi}{1-\xi} W \hat{W}_t.$$

A.2 Steady-state quantities

Normalize $e = 1$ and choose endowments so that all four consumer prices are one. The producer prices are $p = p^* = (\theta_v - 1)/\theta_v$, rather than one, because domestic firms charge a constant markup. Household Euler equations give $1 + r = 1/\beta$ and $1 + r^* = 1/\beta^*$. In the benchmark with offsetting permanent forecasts,

$$\frac{N^A}{W^A} = \frac{\log(\beta/\beta^*) + d_{ss}/2}{\Gamma}, \qquad \frac{N^B}{W^B} = \frac{\log(\beta/\beta^*) - d_{ss}/2}{\Gamma}.$$

With the same entrant endowment in each group,

$$W^i = \frac{\xi W_0}{1 - (1-\xi)\left[\frac{1}{\beta} + \left(\frac{1}{\beta^*} - \frac{1}{\beta}\right)\frac{N^i}{W^i}\right]}, \qquad i \in \{A, B\}.$$

The position ratios in this expression are given explicitly above. Positive finite wealth requires positive denominators. Aggregate wealth, positions, and shares satisfy

$$W = \frac{W^A + W^B}{2}, \quad N = \frac{N^A + N^B}{2}, \quad \omega^i = \frac{W^i}{2W}, \quad b = -\frac{W - N}{\beta}, \quad b^* = -\frac{N}{\beta^*}.$$

Offsetting permanent forecasts do not imply equal wealth shares when the steady-state carry return is nonzero.

The remaining steady-state quantities satisfy

$$\begin{aligned}
&c_H = (1-\gamma)c, && c_F = \gamma c, && c_H^* = \gamma c^*, \quad c_F^* = (1-\gamma)c^*, \\
&y = c_H + c_H^*, && y^* = c_F + c_F^*, && \\
&c = y + (1-\beta)b + \frac{\xi}{1-\xi}(W - W_0), && c^* = y^* + (1-\beta^*)b^*. &&
\end{aligned}$$

Firm profits are already included in the last two equations: producer income plus firm profits equals sales revenue. One may normalize $y = 1$ and solve these conditions jointly for the remaining steady-state quantities. Only an independent subset is needed because the aggregate budget identity is implied by market clearing.

A.3 First-order equilibrium system

The following equations apply for $i \in \{A, B\}$. Subjective forecast terms are first-order changes around each group's own steady-state forecast. The permanent forecast difference is already separated into d_{ss} and must not be added again to $E_t^A \hat{e}_{t+1} - E_t^B \hat{e}_{t+1}$.

Household Euler equations Because current discount factors are observed under the stated information structure,

$$\sigma(\mathbb{E}_t \hat{c}_{t+1} - \hat{c}_t) = \hat{\beta}_t + \hat{r}_t,$$
$$\sigma(\mathbb{E}_t \hat{c}^*_{t+1} - \hat{c}^*_t) = \hat{\beta}^*_t + \hat{r}^*_t.$$

Filtering affects forecasts of future fundamentals. The current discount factor is not replaced by the posterior estimate of its persistent component in these equations.

Consumer demand and price indices CES demand and the two consumption-price normalizations give

$$\hat{c}_{Ht} = \hat{c}_t - \theta \hat{p}_{Ht}, \qquad \hat{c}_{Ft} = \hat{c}_t - \theta \hat{p}_{Ft},$$
$$\hat{c}^*_{Ht} = \hat{c}^*_t - \theta \hat{p}^*_{Ht}, \qquad \hat{c}^*_{Ft} = \hat{c}^*_t - \theta \hat{p}^*_{Ft},$$
$$0 = (1-\gamma)\hat{p}_{Ht} + \gamma \hat{p}_{Ft}, \qquad 0 = \gamma \hat{p}^*_{Ht} + (1-\gamma)\hat{p}^*_{Ft}.$$

Firm pricing The elasticity across domestic varieties is $\theta_v > 1$, distinct from the trade elasticity θ. The imported-variety elasticities are $\theta_{v,t} = \theta_v (e_t/e)^{\xi_v}$ and $\theta^*_{v,t} = \theta_v (e_t/e)^{-\xi_v}$. Differentiating the markup $\theta_{v,t}/(\theta_{v,t}-1)$ and the firms' optimal prices gives

$$\hat{p}_{Ht} = \hat{p}_t, \qquad \hat{p}^*_{Ft} = \hat{p}^*_t,$$
$$\hat{p}^*_{Ht} = \hat{p}_t - \left(1 - \frac{\xi_v}{\theta_v - 1}\right)\hat{e}_t, \qquad \hat{p}_{Ft} = \hat{p}^*_t + \left(1 - \frac{\xi_v}{\theta_v - 1}\right)\hat{e}_t.$$

Thus the law of one price holds when $\xi_v = 0$. The pass-through coefficient called χ elsewhere in the paper is $1 - \xi_v/(\theta_v - 1)$; it is written out here. Solving with the price-index equations gives

$$\hat{p}_{Ht} = -\frac{\gamma}{1-2\gamma}\left(1 - \frac{\xi_v}{\theta_v - 1}\right)\hat{e}_t, \qquad \hat{p}_{Ft} = \frac{1-\gamma}{1-2\gamma}\left(1 - \frac{\xi_v}{\theta_v - 1}\right)\hat{e}_t,$$
$$\hat{p}^*_{Ht} = -\frac{1-\gamma}{1-2\gamma}\left(1 - \frac{\xi_v}{\theta_v - 1}\right)\hat{e}_t, \qquad \hat{p}^*_{Ft} = \frac{\gamma}{1-2\gamma}\left(1 - \frac{\xi_v}{\theta_v - 1}\right)\hat{e}_t.$$

These solved prices can replace the pricing and index equations; they are not additional independent restrictions.

Goods-market clearing Market clearing in physical quantities is

$$y\hat{y}_t = c_H\hat{c}_{Ht} + c^*_H\hat{c}^*_{Ht}, \qquad y^*\hat{y}^*_t = c_F\hat{c}_{Ft} + c^*_F\hat{c}^*_{Ft}.$$

Equivalently, substituting demand and prices gives a system directly in aggregate consumption and the exchange rate:

$$\begin{aligned} y\hat{y}_t &= c_H\hat{c}_t + c^*_H\hat{c}^*_t + \frac{\theta[\gamma c_H + (1-\gamma)c^*_H]}{1-2\gamma}\left(1 - \frac{\xi_v}{\theta_v - 1}\right)\hat{e}_t, \\ y^*\hat{y}^*_t &= c_F\hat{c}_t + c^*_F\hat{c}^*_t - \frac{\theta[(1-\gamma)c_F + \gamma c^*_F]}{1-2\gamma}\left(1 - \frac{\xi_v}{\theta_v - 1}\right)\hat{e}_t. \end{aligned}$$

Household budgets, including firm profits Consolidating producer income and the profits of domestically owned firms gives the exact revenue identities

$$\begin{aligned} p_t y_t + \pi_t &= p_{Ht}c_{Ht} + e_t p^*_{Ht}c^*_{Ht}, \\ p^*_t y^*_t + \pi^*_t &= \frac{p_{Ft}}{e_t}c_{Ft} + p^*_{Ft}c^*_{Ft}. \end{aligned}$$

At first order, use goods clearing and the pricing equations to obtain

$$\begin{aligned} \beta y\widetilde{b}_t - \beta b\hat{r}_t - y\widetilde{b}_{t-1} &= y(\hat{y}_t + \hat{p}_{Ht}) - c\hat{c}_t + \frac{\xi_v}{\theta_v - 1}c^*_H\hat{e}_t + \frac{\xi}{1-\xi}W\hat{W}_t, \\ \beta^* y^*\widetilde{b}^*_t - \beta^* b^*\hat{r}^*_t - y^*\widetilde{b}^*_{t-1} &= y^*(\hat{y}^*_t + \hat{p}^*_{Ft}) - c^*\hat{c}^*_t - \frac{\xi_v}{\theta_v - 1}c_F\hat{e}_t. \end{aligned}$$

The terms proportional to $\xi_v/(\theta_v - 1)$ reflect export revenues when firms price differently across destinations. They vanish with full pass-through. Producer income alone would omit these profit-rebate terms. The two budgets are shown for completeness; once goods clearing, both bond markets, and the aggregate wealth law hold, only one household budget is independent.

Carry positions and group wealth The position rules are

$$\begin{aligned} \widetilde{N}^A_t &= \frac{\log(\beta/\beta^*) + d_{ss}/2}{\Gamma}\hat{W}^A_t + \frac{E^A_t\hat{e}_{t+1} - \hat{e}_t + \hat{r}^*_t - \hat{r}_t}{\Gamma}, \\ \widetilde{N}^B_t &= \frac{\log(\beta/\beta^*) - d_{ss}/2}{\Gamma}\hat{W}^B_t + \frac{E^B_t\hat{e}_{t+1} - \hat{e}_t + \hat{r}^*_t - \hat{r}_t}{\Gamma}. \end{aligned}$$

Differentiating the group wealth law with fixed entrant endowment gives

$$\hat{W}^i_{t+1} = (1-\xi)\Bigg\{ \frac{\hat{W}^i_t + \hat{r}_t}{\beta} + \left(\frac{1}{\beta^*} - \frac{1}{\beta}\right)\widetilde{N}^i_t + \frac{N^i}{W^i}\left[\frac{\hat{e}_{t+1} - \hat{e}_t + \hat{r}^*_t}{\beta^*} - \frac{\hat{r}_t}{\beta}\right]\Bigg\}.$$

The steady-state ratios N^i/W^i are given above; they are not additional parameters. This expression keeps the level deviation of the arithmetic carry return explicit, including both gross-return factors.

Wealth aggregation and Home-bond UIP The first-order aggregation identities are

$$\hat{W}_t = \omega^A \hat{W}^A_t + \omega^B \hat{W}^B_t, \qquad \hat{\omega}^A_t = \omega^A\omega^B(\hat{W}^A_t - \hat{W}^B_t), \qquad \hat{\omega}^B_t = -\hat{\omega}^A_t.$$

Substituting the portfolio rules into Home-bond clearing gives the full UIP condition with its coefficients expanded:

$$\omega^A E^A_t \hat{e}_{t+1} + \omega^B E^B_t \hat{e}_{t+1} - \hat{e}_t + \hat{r}^*_t - \hat{r}_t = -d_{ss}\hat{\omega}^A_t + \frac{\beta\Gamma}{W}\left[y\widetilde{b}_t - b(\hat{W}_t + \hat{r}_t)\right]. \tag{13}$$

In the unbiased-consensus benchmark, the weighted forecast on the left equals $\mathbb{E}_t\hat{e}_{t+1} + (\omega^A - \omega^B)(E^A_t\hat{e}_{t+1} - E^B_t\hat{e}_{t+1})/2$, recovering Lemma 1 without introducing absorption coefficients. The group wealth equations and aggregation identities already imply Lemma 2; its share law is not an additional equilibrium equation.

Foreign bonds and accounting closure Foreign-bond clearing recovers the Foreign household's bond position:

$$y^*\widetilde{b}^*_t = -\frac{1}{\beta^*}\left[W\sum_i \omega^i \widetilde{N}^i_t + N(\hat{r}^*_t - \hat{e}_t)\right].$$

For comparison, Home-bond clearing before substituting the position rules is

$$\beta y\widetilde{b}_t - \beta b\hat{r}_t + W\left(\hat{W}_t - \sum_i \omega^i \widetilde{N}^i_t\right) = 0.$$

Use either this Home-clearing equation or the UIP equation, not both as independent restrictions. The exact consolidated financial identity behind the household-budget redundancy is

$$\frac{b_t}{1+r_t} + \frac{e_t b_t^*}{1+r_t^*} - b_{t-1} - e_t b_{t-1}^* = \frac{\xi}{1-\xi}(W_t - W_0) = \pi_t^a.$$

It follows directly from the two bond-clearing equations and the aggregate wealth law. Together with goods clearing and the firms' revenue identities, it verifies the Foreign budget whenever the Home budget holds.

A.4 Fundamentals, filtering, and expectations

For each $x \in \{\beta, \beta^*, y, y^*\}$, the observable realization and its latent persistent component satisfy

$$\hat{x}_t = \tau_t^x + \varepsilon_t^x, \qquad \tau_t^x = \rho_x \tau_{t-1}^x + v_t^x.$$

The innovations have standard deviations σ_{ε^x} and σ_{v^x}. Their cross-country correlations are those specified in the calibration. The realization $\hat{x}_t$ is observed at t; filtering concerns its decomposition, rather than an additional noisy observation of $\hat{x}_t$.

Under the maintained scalar filtering rule for each fundamental, group i's posterior $\tau_{t|t}^{xi} = E_t^i[\tau_t^x]$ evolves as

$$\tau_{t|t}^{xi} = \rho_x(1-G_x)\tau_{t-1|t-1}^{xi} + G_x(\hat{x}_t + \mu^{xi} + \hat{\mu}_t^{xi}).$$

The gain in the information section is determined by

$$G_x = \frac{\Sigma_{\tau,x}}{\Sigma_{\tau,x} + \sigma_{\varepsilon^x}^2}, \qquad \Sigma_{\tau,x} = \rho_x^2(1-G_x)\Sigma_{\tau,x} + \sigma_{v^x}^2.$$

Here $\Sigma_{\tau,x}$ is the prior variance of τ_t^x, not the posterior variance. The steady-state posterior bias of group i is $G_x\mu^{xi}/[1-\rho_x(1-G_x)]$. The interpretation differences $\hat{\mu}_t^{xA} - \hat{\mu}_t^{xB}$ are the mean-zero shocks with variance $\sigma_{\mu,x}^2$.

Define consensus and the fundamental-specific disagreement gap by $\bar{\tau}_{t|t}^x = (\tau_{t|t}^{xA} + \tau_{t|t}^{xB})/2$ and $d_t^x = \tau_{t|t}^{xA} - \tau_{t|t}^{xB}$. Averaging and differencing the individual rules gives

$$\begin{aligned}
\bar{\tau}_{t|t}^x &= \rho_x(1-G_x)\bar{\tau}_{t-1|t-1}^x + G_x\hat{x}_t + \frac{G_x}{2}(\mu_t^{xA} + \mu_t^{xB}),\\
d_t^x - d_{ss}^x &= \rho_x(1-G_x)(d_{t-1}^x - d_{ss}^x) + G_x(\hat{\mu}_t^{xA} - \hat{\mu}_t^{xB}),\\
d_{ss}^x &= \frac{G_x(\mu^{xA} - \mu^{xB})}{1-\rho_x(1-G_x)}.
\end{aligned}$$

These are consequences of the individual filtering equations, not extra independent restric-

tions. The average-interpretation term is zero in the benchmark and is retained for extensions with average bias. The gap d_t^x concerns a particular latent fundamental; it is distinct from exchange-rate forecast disagreement $d_t = D_t - d_{ss} = UIP_t^A - UIP_t^B$ and from the steady-state exchange-rate forecast gap d_{ss}.

Current posteriors and interpretations are mutually observed. In the benchmark, subjective future beliefs follow the higher-order expectation formulas in Proposition 4, derived in Appendix B.1. Both groups apply those beliefs to the same equilibrium price and wealth laws. In particular, $E_t^i \hat{e}_{t+1}$ is the conditional forecast of the endogenous exchange-rate solution, not an independent shock or a freely chosen forecast coefficient. Given the exogenous innovation laws, initial wealth, inherited bonds, and initial posteriors, the equations above and these conditional expectation rules close the system. The solution is selected to satisfy household transversality and the maintained non-explosive equilibrium restriction.

Average-bias extension The zero-average-bias benchmark sets $\bar{\mu}_t^x = (\mu_t^{xA} + \mu_t^{xB})/2 = 0$. For the extension specified in the quantitative section, substitute

$$\bar{\mu}_t^x = \left(\frac{\kappa_x}{G_x} - 1 \right) \hat{x}_t + \frac{(1-\kappa_x)\tilde{\rho}_x - \rho_x(1-G_x)}{G_x} \bar{\tau}_{t-1|t-1}^x.$$

The consensus recursion then simplifies to

$$\bar{\tau}_{t|t}^x = (1-\kappa_x)\tilde{\rho}_x \bar{\tau}_{t-1|t-1}^x + \kappa_x \hat{x}_t.$$

Here κ_x is the perceived information gain and $\tilde{\rho}_x$ the perceived persistence, as in the main text. This extension uses the existing fundamental shocks. When households share the common biased beliefs, the conditional expectations in both household Euler equations are evaluated under that same subjective law. The current observed discount factors on the right-hand sides remain $\hat{\beta}_t$ and $\hat{\beta}_t^*$.

The structural system above sets the additional reduced-form UIP shock ψ_t to zero. The quantitative pricing-wedge extension specifies $\psi_t = \rho_\psi \psi_{t-1} + \varepsilon_t^\psi$ and adds ψ_t to the right-hand side of UIP. That specification replaces the structural pricing restriction; it is not an additional restriction to impose alongside unmodified Home-bond clearing and portfolio demand. A structural interpretation of this extra shock requires specifying its counterpart in portfolio demand or outside bond demand.

B Omitted Proofs and Derivations

B.1 Information structure and belief dynamics

Subjective interpretation of public information For each driving variable, agents observe $\hat{x}_t = \tau_t^x + \varepsilon_t^x$ but not its persistent and transitory components separately. Under group i's subjective probability measure, the observation satisfies

$$\hat{x}_t =^i \tau_t^x - \mu_t^{xi} + \varepsilon_t^x. \tag{14}$$

The notation $=^i$ denotes equality under group i's perceived data-generating process. Group i thus treats $\hat{x}_t + \mu_t^{xi}$ as a noisy but unbiased observation of τ_t^x.

Each group considers the other's interpreted observation a garbled version of its own:

$$\hat{x}_t + \mu_t^{xj} =^i \tau_t^x + \varepsilon_t^x + \mu_t^{xj} - \mu_t^{xi}, \qquad j \neq i.$$

Interpretations and current posteriors are mutually observed. Conditional on its own interpretation, group i regards the other group's interpretation as uninformative about τ_t^x. Equilibrium prices therefore reveal no additional information.

Interpretations have permanent and transitory components, $\mu_t^{xi} = \mu^{xi} + \hat{\mu}_t^{xi}$. The transitory components are jointly normal and i.i.d. across periods, with

$$\hat{\mu}_t^{xA} - \hat{\mu}_t^{xB} \sim N(0, \sigma_{\mu,x}^2). \tag{15}$$

The groups have equal population shares, and the zero-average-bias benchmark imposes $\bar{\mu}_t^x = (\mu_t^{xA} + \mu_t^{xB})/2 = 0$. The following proposition collects the resulting filtering and higher-order expectation formulas.

Proposition 4. *Denote the filtered time-t persistent state by group $i \in \{A, B\}$ given time t information as $\tau_{t|t}^{xi} \equiv E_t^i[\tau_t^x]$. Let the steady state Kalman gain be $G_x = \frac{\left(\sigma_{\varepsilon^x}^2\right)^{-1}}{(\Sigma_{\tau,x})^{-1} + \left(\sigma_{\varepsilon^x}^2\right)^{-1}}$, where $\Sigma_{\tau,x}$ is the steady-state prior variance of τ_t^x, which solves $\Sigma_{\tau,x} = \rho_x^2 (1 - G_x) \Sigma_{\tau,x} + \sigma_{v^x}^2$. The individual beliefs, consensus beliefs, $\bar{\tau}_{t|t}^x \equiv \frac{1}{2}\left(\tau_{t|t}^{xA} + \tau_{t|t}^{xB}\right)$, and disagreement, $d_t^x \equiv \tau_{t|t}^{xA} - \tau_{t|t}^{xB}$, evolve according to*

$$\begin{aligned}
\tau_{t|t}^{xi} &= G_x \mu^{xi} + \varphi_x \tau_{t-1|t-1}^{xi} + G_x \left(\hat{x}_t + \hat{\mu}_t^{xi}\right), \\
\bar{\tau}_{t|t}^x &= G_x \bar{\mu}_t^x + \varphi_x \bar{\tau}_{t-1|t-1}^x + G_x \hat{x}_t, \\
d_t^x &= (1 - \varphi_x)\, d_{ss}^x + \varphi_x d_{t-1}^x + G_x (\hat{\mu}_t^{xA} - \hat{\mu}_t^{xB}),
\end{aligned}$$

respectively, where $\varphi_x = \rho_x (1 - G_x)$*, and* $d^x_{ss} = \frac{G_x}{1-\varphi_x} \left(\mu^{xA} - \mu^{xB}\right)$*. Moreover, current beliefs are mutually observed, and for* $h \geq 1$*,*

$$\begin{aligned}
E^A_t \left[\tau^{xA}_{t+h|t+h}\right] &= \rho^h_x \tau^{xA}_{t|t},\ E^B_t \left[\tau^{xB}_{t+h|t+h}\right] = \rho^h_x \tau^{xB}_{t|t}, \\
E^A_t \left[\tau^{xB}_{t+h|t+h}\right] &= \varphi^h_x \tau^{xB}_{t|t} + \left(\rho^h_x - \varphi^h_x\right) \tau^{xA}_{t|t} - \left(1 - \varphi^h_x\right) d^x_{ss}, \\
E^B_t \left[\tau^{xA}_{t+h|t+h}\right] &= \varphi^h_x \tau^{xA}_{t|t} + \left(\rho^h_x - \varphi^h_x\right) \tau^{xB}_{t|t} + \left(1 - \varphi^h_x\right) d^x_{ss}.
\end{aligned}$$

Consequently, both groups agree on the path for disagreement:

$$E^A_t \left[d^x_{t+h}\right] = E^B_t \left[d^x_{t+h}\right] = d^x_{ss} + \varphi^h_x \left(d^x_t - d^x_{ss}\right).$$

Proof

Individual belief updating Each agent i observes the current realization $\hat{x}_t$ but not its persistent and transitory components separately. Group i treats $\hat{x}_t + \mu^{xi}_t$ as a noisy observation of the latent persistent component. Combining the prior mean $\rho_x \tau^{xi}_{t-1|t-1}$ with this interpreted observation gives the steady-state Kalman update

$$\tau^{xi}_{t|t} = (1 - G_x)\rho_x \tau^{xi}_{t-1|t-1} + G_x(\hat{x}_t + \mu^{xi}_t) = \varphi_x \tau^{xi}_{t-1|t-1} + G_x(\hat{x}_t + \mu^{xi}_t),$$

where $\varphi_x \equiv \rho_x(1 - G_x)$. There is no second signal for the realized innovation: adding one would double count information already contained in the observed $\hat{x}_t$.

Consensus beliefs Taking the weighted average across agents with weights η_i:

$$\bar{\tau}^x_{t|t} = \sum_i \eta_i \tau^{xi}_{t|t} = \varphi_x \bar{\tau}^x_{t-1|t-1} + G_x \left(\hat{x}_t + \sum_i \eta_i \mu^{xi}_t\right).$$

Since $\sum_i \eta_i \mu^{xi}_t = 0$ in the zero-average-bias benchmark, this reduces to the benchmark consensus equation. With average bias, the same averaging identity retains the interpretation term shown in the main text.

Disagreement Taking the difference between agents i and j:

$$d^x_t = \varphi_x d^x_{t-1} + G_x \Delta\mu^x_t = (1 - \varphi_x) d^x_{ss} + \varphi_x d^x_{t-1} + G_x \varepsilon^x_{\mu,t}.$$

The second equality uses $\Delta\mu^x_t \equiv \mu^{xA}_t - \mu^{xB}_t = \mu^{xA} - \mu^{xB} + \varepsilon^x_{\mu,t}$ and $d^x_{ss} = G_x(\mu^{xA} - \mu^{xB})/(1 - \varphi_x)$.

Higher-order expectations For $h = 0$, agent i observes agent j's current belief directly, so $E_t^i[\tau_{t|t}^{xj}] = \tau_{t|t}^{xj}$.

For one period ahead, the subjective conditional means implied by the common transition law and the interpretation convention are

$$\begin{aligned} E_t^A[\tau_{t+1|t+1}^{xA}] &= \rho_x \tau_{t|t}^{xA}, \\ E_t^A[\tau_{t+1|t+1}^{xB}] &= \varphi_x \tau_{t|t}^{xB} + (\rho_x - \varphi_x)\tau_{t|t}^{xA} - (1 - \varphi_x) d_{ss}^x, \end{aligned}$$

with the symmetric expressions under group B. Iterating these affine recursions proves the higher-order-expectation formulas in Proposition 4. In particular,

$$\begin{aligned} E_t^A[\tau_{t+h|t+h}^{xB}] &= \varphi_x^h \tau_{t|t}^{xB} + (\rho_x^h - \varphi_x^h)\tau_{t|t}^{xA} - (1 - \varphi_x^h) d_{ss}^x, \\ E_t^A[d_{t+h}^x] &= d_{ss}^x + \varphi_x^h (d_t^x - d_{ss}^x), \end{aligned}$$

and the corresponding formulas for group B follow by exchanging the group labels and reversing the sign of the permanent gap. This also shows formally that the groups agree on the future path of disagreement even though their forecasts of the persistent component differ.

B.1.1 Private information and price learning

This subsection proves Remark 2 conditional on a candidate stationary linear price signal. For each fundamental x, retain the objective state τ_t^x and let group i use a fixed subjective long-run mean ℓ^{xi}. Its prior mean is $\tau_{t|t-1}^{xi} = \rho_x \tau_{t-1|t-1}^{xi} + (1 - \rho_x)\ell^{xi}$. Suppose the public observation is $\hat{x}_t = \tau_t^x + \varepsilon_t^x$, the private signal is $s_t^{xi} = \tau_t^x + \zeta_t^{xi}$, and a common, normalized price signal is $q_t^{x,p} = \tau_t^x + \zeta_t^{x,p}$. Within each subjective model, the signal errors are jointly Gaussian and centered conditional on the state and prior information, with a common time-invariant covariance matrix. The private-signal difference is objectively mean zero and iid over time. Signal precision and prior covariance assumptions are symmetric across groups.

Filtering and disagreement Let $K_{0,x}$, $K_{s,x}$, and $K_{p,x}$ be the common constant gains, with $k_x \equiv K_{0,x} + K_{s,x} + K_{p,x}$, $\psi_x \equiv \rho_x(1 - k_x)$, and $|\psi_x| < 1$. Bayesian updating within each subjective model gives

$$\tau_{t|t}^{xi} = \psi_x \tau_{t-1|t-1}^{xi} + (1 - k_x)(1 - \rho_x)\ell^{xi} + K_{0,x}\hat{x}_t + K_{s,x} s_t^{xi} + K_{p,x} q_t^{x,p}.$$

Subtracting cancels the public and price signals. With $\Delta\ell^x \equiv \ell^{xA} - \ell^{xB}$, the posterior gap and its stationary mean satisfy

$$\begin{aligned} d_t^x &= \psi_x d_{t-1}^x + (1-k_x)(1-\rho_x)\Delta\ell^x + K_{s,x}(\zeta_t^{xA} - \zeta_t^{xB}), \\ d_{ss}^x &= \frac{(1-k_x)(1-\rho_x)}{1-\psi_x}\Delta\ell^x, \\ d_t^x - d_{ss}^x &= \psi_x(d_{t-1}^x - d_{ss}^x) + K_{s,x}(\zeta_t^{xA} - \zeta_t^{xB}). \end{aligned}$$

The one-period fundamental forecast gap is $F_t^x = \rho_x d_t^x + (1-\rho_x)\Delta\ell^x$. Therefore

$$F_{ss}^x = \frac{1-\rho_x}{1-\psi_x}\Delta\ell^x, \qquad F_t^x - F_{ss}^x = \psi_x(F_{t-1}^x - F_{ss}^x) + \rho_x K_{s,x}(\zeta_t^{xA} - \zeta_t^{xB}).$$

Both centered gaps have the AR(1) form of the baseline, with gains determined by the alternative information structure. With zero private-signal differences, constancy also requires initialization at the stationary gap unless $\psi_x = 0$.

Wealth weighting and scope For any two one-period exchange-rate forecasts F_t^A, F_t^B, write $\bar{F}_t = (F_t^A + F_t^B)/2$, $D_t = F_t^A - F_t^B = d_{ss} + d_t$, and $\omega_t^A = 1/2 + \delta_w + \hat{\omega}_t^A$. Direct aggregation gives

$$\begin{aligned} F_t^W &= \bar{F}_t + (\delta_w + \hat{\omega}_t^A)(d_{ss} + d_t), \\ F_t^W - F_{ss}^W &= (\bar{F}_t - \bar{F}_{ss}) + d_{ss}\hat{\omega}_t^A + \delta_w d_t + \hat{\omega}_t^A d_t. \end{aligned}$$

Dropping the product of deviations yields the same first-order pricing wedge as in Lemma 1, irrespective of how the forecasts are formed.

The result is conditional, rather than a solution of a price-learning equilibrium. Price coefficients, signal covariances, and filtering gains must satisfy their joint equilibrium fixed point. Averaging the filters retains the average private signal and the price signal, so $\ell^{xA} + \ell^{xB} = 0$ alone does not make consensus equal the objective public-information filter. Nor does this argument recover arbitrary average-bias coefficients or all the higher-order beliefs in Proposition 4. The fixed ℓ^{xi} are persistent subjective model differences, not unknown means that agents eventually learn under a common prior. With a common prior and objectively correct forecasts, the law of iterated expectations gives $\mathbb{E}(F_t^A - F_t^B) = 0$; it therefore does not provide the nonzero deterministic d_{ss} used in the symmetric analytical steady state. □

B.2 Optimal portfolio choice

Fix a date, group, and arbitrageur, and suppress their indices temporarily. Let $R = 1+r_t > 0$, $w = w_t^i(\iota) > 0$, $\mu = E_t^i[R_{t+1}^x]$, and $v = \mathrm{Var}_t^i(R_{t+1}^x) > 0$. Substituting $w' = Rw + nR^x$ into the stated mean-variance objective gives

$$J(n) = Rw + n\mu - \frac{\nu n^2 v}{2Rw}, \qquad J''(n) = -\frac{\nu v}{Rw} < 0.$$

Thus the first-order condition is sufficient for the unique unconstrained optimum:

$$\mu - \frac{\nu n v}{Rw} = 0, \qquad \frac{n}{w} = \frac{R\mu}{\nu v} = \frac{E_t^i[z_{t+1}]}{\nu \,\mathrm{var}_t^i(z_{t+1})}, \quad z_{t+1} \equiv \frac{R_{t+1}^x}{R}. \tag{16}$$

The gross Home return cancels when both the mean and variance are expressed in terms of the normalized arithmetic carry return z_{t+1}.

For the log-return approximation, define

$$q_{t+1}^p \equiv \Delta \log e_{t+1} + \log \frac{1+r_t^*}{1+r_t} = \log(1 + z_{t+1}).$$

The maintained approximation replaces $z_{t+1} = \exp(q_{t+1}^p) - 1$ by q_{t+1}^p in its conditional mean and variance. This step is an additional return approximation; the exact mean-variance optimum is the preceding equation. Since interest rates are known at t, $\mathrm{var}_t^i(q_{t+1}^p) = \mathrm{var}_t^i(\Delta \log e_{t+1})$. Members of a group share conditional moments, so their optimal positions are proportional to wealth. Averaging within the group yields

$$\frac{N_t^i}{W_t^i} = \frac{M_t^i}{D_t}, \qquad M_t^i \equiv E_t^i[q_{t+1}^p], \qquad D_t \equiv \nu \,\mathrm{var}_t^i(q_{t+1}^p). \tag{17}$$

Equality here and below refers to the maintained log-return approximation. Under the common, fixed-variance normalization, $D_t = \Gamma$ at the relevant order. In steady state $M^i = L_i$ and $N^i/W^i = L_i/\Gamma$. Since the first-order deviation of M_t^i is UIP_t^i, differentiating $N_t^i = W_t^i M_t^i/\Gamma$ gives

$$\widetilde{N}_t^i \equiv \frac{N_t^i - N^i}{W^i} = \frac{L_i}{\Gamma}\hat{W}_t^i + \frac{1}{\Gamma} UIP_t^i. \tag{18}$$

The objective also has a local certainty-equivalent interpretation. For CRRA utility u, let $w' = Rw + \epsilon Y$ and define $u(CE) = E_t^i[u(w')]$. A second-order expansion at Rw, holding ν fixed, gives

$$CE = Rw + \epsilon E_t^i[Y] - \frac{\nu \epsilon^2}{2Rw} \,\mathrm{var}_t^i(Y) + o(\epsilon^2).$$

Here $-u''(Rw)/u'(Rw) = \nu/(Rw)$, including the log-utility case $\nu = 1$. We take the per-

period mean-variance objective as primitive; this expansion supplies its local CRRA interpretation. The choice takes current wealth and conditional moments as given and contains no continuation value, so intertemporal hedging demand is excluded by the myopia assumption. □

B.3 Proof of Lemma 1

Use the portfolio rule in Appendix B.2. Bond-market clearing gives

$$\frac{b_t}{1+r_t} + W_t - N_t = 0, \qquad \frac{e_t b_t^*}{1+r_t^*} + N_t = 0. \tag{19}$$

With M_t^i and D_t defined in the portfolio derivation, Home clearing implies

$$\sum_i \omega_t^i \frac{M_t^i}{D_t} = \frac{N_t}{W_t} = 1 + \frac{b_t}{(1+r_t)W_t}.$$

This proves the level condition in Lemma 1.

At first order $D_t = \Gamma$. The deviation of $\sum_i \omega_t^i M_t^i$ is

$$\sum_i \omega^i UIP_t^i + \sum_i L_i \hat{\omega}_t^i = \mathbb{E}_t \hat{e}_{t+1} - \hat{e}_t + \hat{r}_t^* - \hat{r}_t + \delta_w d_t + d_{ss}\hat{\omega}_t^A.$$

This uses unbiased consensus beliefs, $\hat{\omega}_t^B = -\hat{\omega}_t^A$, $d_{ss} = L_A - L_B$, and $\delta_w = (\omega^A - \omega^B)/2$. The product of changing wealth shares and changing disagreement is second order and drops out. With average forecast bias, replace $\mathbb{E}_t \hat{e}_{t+1}$ by the simple average of the groups' forecasts.

The constant one in the Home-clearing identity has zero deviation. Differentiating the bond ratio, with $1 + r = 1/\beta$, gives

$$\Gamma \, \mathrm{d}\left[\frac{b_t}{(1+r_t)W_t}\right] = \frac{\beta\Gamma}{W}\left[y\widetilde{b}_t - b(\hat{W}_t + \hat{r}_t)\right]. \tag{20}$$

Equating the two deviations proves the first-order Home-bond UIP condition. The factor y converts the normalized bond deviation $\widetilde{b}_t$ back to Home bond face value, whereas $\hat{r}_t$ is the log deviation of the gross Home return.

Equivalence with the Foreign-bond version Foreign clearing implies the exact identity

$$1 + \frac{b_t}{(1+r_t)W_t} = \frac{N_t}{W_t} = -\frac{e_t b_t^*}{(1+r_t^*)W_t}.$$

The factor $e_t/(1+r_t^*)$ converts Foreign bond face value to Home-valued investment. Since the two ratios differ only by the constant one, their first-order deviations agree. Using $e=1$ and $\widetilde{b}_t^* = (b_t^* - b^*)/y^*$ gives the explicit equality

$$\begin{aligned}\frac{\beta\Gamma}{W}\left[y\widetilde{b}_t - b(\hat{W}_t + \hat{r}_t)\right] \\ = -\frac{\beta^*\Gamma}{W}\left[y^*\widetilde{b}_t^* + b^*(\hat{e}_t - \hat{r}_t^* - \hat{W}_t)\right].\end{aligned}$$

Consequently the equivalent Foreign-bond UIP condition is

$$\begin{aligned}\mathbb{E}_t\hat{e}_{t+1} - \hat{e}_t + \hat{r}_t^* - \hat{r}_t = -d_{ss}\hat{\omega}_t^A - \delta_w d_t \\ - \frac{\beta^*\Gamma}{W}\left[y^*\widetilde{b}_t^* + b^*(\hat{e}_t - \hat{r}_t^* - \hat{W}_t)\right].\end{aligned}$$

The equality follows from market clearing; the Home and Foreign bond positions cannot be varied independently in equilibrium. Both derivations remain valid at zero steady-state bond holdings because bond deviations are in levels. □

B.4 Proof of Lemma 2

At the steady state, $1+r = 1/\beta$, $1+r^* = 1/\beta^*$, $R^x = \omega_x/\beta$, and $N^i/W^i = L_i/\Gamma$. Differentiate the group wealth law, holding the entrant endowment fixed, and divide by steady-state group wealth:

$$\hat{W}_{t+1}^i = (1-\xi)\left[\frac{\hat{W}_t^i + \hat{r}_t}{\beta} + \frac{\omega_x}{\beta}\widetilde{N}_t^i + \frac{L_i}{\Gamma}\hat{R}_{t+1}^p\right], \tag{21}$$

where

$$\hat{R}_{t+1}^p = \frac{\hat{e}_{t+1} - \hat{e}_t + \hat{r}_t^*}{\beta^*} - \frac{\hat{r}_t}{\beta}.$$

The hat on R^p denotes the level deviation of the arithmetic carry return. Its coefficient is the steady-state carry position per unit of wealth, L_i/Γ, with no additional ω_x factor.

Since population weights are fixed, first-order aggregation and differentiation of the wealth share give

$$\hat{W}_t = \omega^A\hat{W}_t^A + \omega^B\hat{W}_t^B, \qquad \hat{\omega}_t^A = \omega^A\omega^B(\hat{W}_t^A - \hat{W}_t^B).$$

In particular, the wealth-share hat is a level deviation. Set $\Delta W_t = \hat{W}_t^A - \hat{W}_t^B$. Then $\hat{W}_t^A = \hat{W}_t + \omega^B\Delta W_t$ and $\hat{W}_t^B = \hat{W}_t - \omega^A\Delta W_t$. Substituting these identities into the difference

of portfolio rules yields

$$\widetilde{N}_t^A - \widetilde{N}_t^B = \frac{d_{ss}}{\Gamma}\hat{W}_t + \frac{\omega^B L_A + \omega^A L_B}{\Gamma}\Delta W_t + \frac{d_t}{\Gamma}.$$

Under unbiased consensus beliefs in steady state, $L_A = \bar{L} + d_{ss}/2$ and $L_B = \bar{L} - d_{ss}/2$. The cross-weighted return is therefore

$$\omega^B L_A + \omega^A L_B = \bar{L} - \delta_w d_{ss}.$$

Subtract the group wealth equations and multiply by $\omega^A\omega^B$. The common risk-free-rate term cancels, leaving

$$\hat{\omega}_{t+1}^A = (1-\xi)\Bigg\{\frac{1}{\beta}\left[1 + \frac{\omega_x(\bar{L} - \delta_w d_{ss})}{\Gamma}\right]\hat{\omega}_t^A + \omega^A\omega^B\left[\frac{\omega_x}{\beta\Gamma}d_t + \frac{d_{ss}}{\Gamma}\left(\frac{\omega_x}{\beta}\hat{W}_t + \hat{R}_{t+1}^p\right)\right]\Bigg\}.$$

This is the general wealth-share law in the lemma. The term $-\delta_w d_{ss}$ in its persistence coefficient vanishes under symmetric steady-state wealth shares.

For the analytical limit, impose $\beta^* = \beta$, the symmetric steady-state shares $\omega^A = \omega^B = 1/2$, and the zero-NFA benchmark used for reduced UIP. Then $\omega_x = \bar{L} = \delta_w = 0$, and Remark 1 gives $d_t = 0$. Define the first-order log carry return $q_{t+1}^p \equiv \beta\hat{R}_{t+1}^p = \hat{e}_{t+1} - \hat{e}_t + \hat{r}_t^* - \hat{r}_t$. The share law becomes

$$\hat{\omega}_{t+1}^A = a\hat{\omega}_t^A + b_w q_{t+1}^p, \qquad a = \frac{1-\xi}{\beta}, \qquad b_w = \frac{(1-\xi)\omega^A\omega^B d_{ss}}{\beta\Gamma}.$$

Reduced UIP gives $\mathbb{E}_t q_{t+1}^p = -d_{ss}\hat{\omega}_t^A$. Since rates at t are known, the innovation in q_{t+1}^p is $\varepsilon_{t+1}^e = \hat{e}_{t+1} - \mathbb{E}_t\hat{e}_{t+1}$. Consequently,

$$\hat{\omega}_{t+1}^A = \lambda\hat{\omega}_t^A + b_w\varepsilon_{t+1}^e, \qquad \lambda = a - b_w d_{ss} = \frac{1-\xi}{\beta}\left(1 - \frac{\omega^A\omega^B d_{ss}^2}{\Gamma}\right).$$

These are the reduced-law coefficients in the text, evaluated at $\beta^* = \beta$. Symmetry is part of the maintained analytical benchmark; it is not inferred solely from a zero steady-state carry return. □

B.5 Proof of Proposition 1 and Corollary 1

Maintain Remark 1, zero Foreign bond-supply terms, $\beta^* = \beta$, equal entrant endowments, and $a = (1-\xi)/\beta < 1$. Then $\omega^A = \omega^B = 1/2$. The terms multiplying d_t vanish from the first-order UIP and wealth equations because $\delta_w = \omega_x = 0$. Thus the reduced financial equations follow even before verifying that the centered gap is zero.

Reduced equilibrium and forward solution Household Euler equations and goods-market clearing imply $\hat{r}_t - \hat{r}_t^* = \hat{\imath}_t^f - S\mathbb{E}_t\Delta\hat{e}_{t+1}$, where $\hat{\imath}_t^f$ is the numerator defined in Proposition 1. Combining this with reduced UIP gives

$$\hat{r}_t - \hat{r}_t^* = \frac{\hat{\imath}_t^f + Sd_{ss}\hat{\omega}_t^A}{1+S}, \qquad \mathbb{E}_t\Delta\hat{e}_{t+1} = \frac{\hat{\imath}_t^f - d_{ss}\hat{\omega}_t^A}{1+S}.$$

Appendix B.4 gives $\hat{\omega}_{t+1}^A = \lambda\hat{\omega}_t^A + b_w\varepsilon_{t+1}^e$, with $b_w = ad_{ss}/(4\Gamma)$ and $\lambda = a - b_w d_{ss}$. For $0 < \rho_x, \lambda < 1$, the stationary forward solution is

$$\hat{e}_t = f_t + \kappa_\omega\hat{\omega}_t^A, \qquad f_t = -\frac{1}{1+S}\mathbb{E}_t\sum_{j=0}^{\infty}\hat{\imath}_{t+j}^f, \qquad \kappa_\omega = \frac{d_{ss}}{(1+S)(1-\lambda)}.$$

The explicit f_t in the text follows by summing the AR(1) fundamentals. In particular, its supply component is $\sigma\phi_y(\hat{y}_t - \hat{y}_t^*)/(1+S)$, as used in the Backus--Smith proof.

Multiplier and parameter restrictions Taking innovations in the forward solution yields $\varepsilon_{t+1}^e = \varepsilon_{t+1}^{e,f} + \kappa_\omega b_w\varepsilon_{t+1}^e$. Let $h = d_{ss}^2/(4\Gamma)$, so $\lambda = a(1-h)$ and $b_w d_{ss} = ah$. For $0 < a < 1$ and $0 < \lambda < 1$,

$$g = \frac{ah}{(1+S)(1-a+ah)}, \qquad \Xi = \frac{(1+S)(1-a+ah)}{(1+S)(1-a)+Sah}, \qquad 0 < g < \frac{1}{1+S}.$$

Hence $1 < \Xi < (1+S)/S$ when $d_{ss} > 0$; at $d_{ss} = 0$, $\Xi = 1$. Writing $R = ah/(1-a)$ gives $\Xi = (1+S)(1+R)/(1+S+SR)$, which increases with R and decreases with S. Since R increases with d_{ss} and a, it decreases with Γ, ξ, and β. The goods-market comparative statics follow from $S = \sigma\chi\phi_e$ in the home-bias region $0 < \gamma < 1/2$. This proves the corollary within the stationary parameter region.

Constant exchange-rate forecast disagreement It remains to verify that the reduced solution is consistent with the information benchmark. At $G_x = 1$, $\varphi_x = 0$, and zero interpretation innovations, $\tau_{t|t}^{xi} = \hat{x}_t + \mu^{xi}$. Under group i's perceived observation equation,

$\mathbb{E}_t^i \hat{x}_{t+1} = \rho_x \tau_{t|t}^{xi} - \mu^{xi} = \rho_x \hat{x}_t + (\rho_x - 1)\mu^{xi}$. Consequently, if $f_t = \sum_x A_x \hat{x}_t$ denotes the linear fundamental component above,

$$C_f \equiv E_t^A[f_{t+1}] - E_t^B[f_{t+1}] = \sum_x A_x(\rho_x - 1)d_{ss}^x$$

is constant. The current wealth share and interest rates are observed by both groups. Taking their forecast difference in the realized share law $\hat{\omega}_{t+1}^A = a\hat{\omega}_t^A + b_w q_{t+1}^p$ therefore gives $E_t^A[\hat{\omega}_{t+1}^A] - E_t^B[\hat{\omega}_{t+1}^A] = b_w D_t$, where D_t is the total exchange-rate forecast gap. Taking the forecast difference in the affine exchange-rate solution now yields

$$D_t = C_f + \kappa_\omega b_w D_t, \qquad D_t = \Xi C_f.$$

Since $g = \kappa_\omega b_w < 1$, the total gap is constant, and its steady-state level must satisfy $d_{ss} = \Xi C_f$. Thus $d_t = D_t - d_{ss} = 0$. This is the consistency condition between the fundamental biases and the steady-state carry-return gap; they are not independent primitive parameters. It verifies Proposition 1 and Remark 1 for the maintained affine solution. Away from the joint zero-noise limit, zero disagreement innovations alone only make $d_t^x - d_{ss}^x$ decay at φ_x; exact constancy requires initialization at d_{ss}^x. □

B.6 Proof of Proposition 2

Maintain Remark 1, Proposition 1, and Assumption 1. In particular, $d_t = 0$ and d_{ss} is constant over time at each interior point of the stationary parameter sequence.

Fixed amplification is compatible with this limit. Fix $S > 0$ and $1 < \Xi < \Xi_{\max}$, and write $c_\Xi \equiv (1 + S)(1 - \Xi^{-1}) \in (0, 1)$. For each $0 < \lambda < 1$, choose

$$a = \lambda + c_\Xi(1 - \lambda), \qquad h \equiv \frac{d_{ss}^2}{4\Gamma} = \frac{c_\Xi(1 - \lambda)}{a}.$$

Then $0 < a, h < 1$, $\lambda = a(1 - h)$, $b_w d_{ss} = ah$, and the multiplier remains Ξ. For example, fix $\beta \in (0, 1)$ and $d_{ss} > 0$, and set $\xi = 1 - a\beta$ and $\Gamma = d_{ss}^2/(4h)$, with fundamental biases satisfying the consistency condition in Appendix B.5. Thus λ can approach one with amplification and the steady-state forecast gap held fixed.

Proof. Let

$$z_t \equiv \kappa_\omega \widehat{\omega}_t^A, \qquad q_t \equiv \hat{\imath}_t^f \equiv \imath_t^f/(1 + S) = \mathbb{E}_t \Delta f_{t+1} = -(1 - \rho_x) f_t, \qquad p_t \equiv (1 - \lambda) z_t.$$

Under the common-persistence assumption,

$$f_{t+1} = \rho_x f_t + \nu_{t+1}, \qquad \mathbb{E}_t \nu_{t+1} = 0, \qquad \mathrm{Var}(\nu_{t+1}) = \sigma_\nu^2.$$

The reduced wealth-share law and the exchange-rate-innovation fixed point are

$$\widehat{\omega}_{t+1}^A = \lambda \widehat{\omega}_t^A + b_w \varepsilon_{t+1}^e, \qquad \varepsilon_{t+1}^e = \Xi \nu_{t+1}.$$

Because

$$\Xi = \frac{1}{1 - \kappa_\omega b_w}, \qquad \kappa_\omega b_w \Xi = \Xi - 1,$$

the priced wealth state obeys

$$z_{t+1} = \lambda z_t + (\Xi - 1)\nu_{t+1}. \tag{22}$$

Moreover,

$$\kappa_\omega = \frac{d_{ss}}{(1+S)(1-\lambda)}.$$

Part (i) of the reduced system therefore gives

$$\widehat{\imath}_t = \widehat{\imath}_t^f + \frac{S}{1+S} d_{ss} \widehat{\omega}_t^A = q_t + S(1-\lambda) z_t = q_t + S p_t, \tag{23}$$

$$\widehat{e}_t = f_t + z_t. \tag{24}$$

Combining the last identity with the laws of motion for f_t and z_t yields

$$\Delta \widehat{e}_{t+1} = q_t - p_t + \Xi \nu_{t+1}. \tag{25}$$

Thus $q_t - p_t$ is expected depreciation and $\Xi \nu_{t+1}$ is the date-$(t+1)$ exchange-rate innovation.

For $0 < \rho_x, \lambda < 1$, the two predetermined components follow

$$q_{t+1} = \rho_x q_t - (1 - \rho_x)\nu_{t+1},$$
$$p_{t+1} = \lambda p_t + (1-\lambda)(\Xi - 1)\nu_{t+1}.$$

Their stationary second moments are therefore

$$\frac{\mathrm{Var}(q_t)}{\sigma_\nu^2} = \frac{(1-\rho_x)^2}{1-\rho_x^2} = \frac{1-\rho_x}{1+\rho_x}, \tag{26}$$

$$\frac{\mathrm{Var}(p_t)}{\sigma_\nu^2} = \frac{(\Xi-1)^2(1-\lambda)^2}{1-\lambda^2} = (\Xi-1)^2\frac{1-\lambda}{1+\lambda}, \tag{27}$$

$$\frac{\mathrm{Cov}(q_t,p_t)}{\sigma_\nu^2} = -\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}. \tag{28}$$

Because (q_t, p_t) are predetermined relative to ν_{t+1}, equation (25) and (26)–(28) give

$$\begin{aligned}\frac{\mathrm{Var}(\Delta\widehat{e}_{t+1})}{\sigma_\nu^2} &= \Xi^2 + \frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} \\ &\quad + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}.\end{aligned} \tag{29}$$

The exact first autocovariance is

$$\begin{aligned}&\frac{\mathrm{Cov}(\Delta\widehat{e}_{t+1},\Delta\widehat{e}_t)}{\sigma_\nu^2} \\ &\quad = -\frac{1-\rho_x}{1+\rho_x} - (\Xi-1)^2\frac{1-\lambda}{1+\lambda} - (\Xi-1)\frac{(1-\rho_x)^2+(1-\lambda)^2}{1-\rho_x\lambda}.\end{aligned} \tag{30}$$

Consequently,

$$\mathrm{AC}_1(\Delta\widehat{e}) = -\frac{\frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + (\Xi-1)\frac{(1-\rho_x)^2+(1-\lambda)^2}{1-\rho_x\lambda}}{\Xi^2 + \frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}}. \tag{31}$$

As $\rho_x\lambda \to 1$, both $\rho_x \to 1$ and $\lambda \to 1$. Every term in the numerator of (31) then converges to zero, whereas its denominator converges to $\Xi^2 > 0$. More strongly,

$$\begin{aligned}\frac{\mathrm{Var}\left(\mathbb{E}_t\Delta\widehat{e}_{t+1}\right)}{\sigma_\nu^2} &= \frac{\mathrm{Var}(q_t-p_t)}{\sigma_\nu^2} \\ &= \frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda} \longrightarrow 0.\end{aligned}$$

Hence $(\Delta\widehat{e}_{t+1} - \Xi\nu_{t+1})/\sigma_\nu \to 0$ in mean square. After normalization by the fundamental-news standard deviation, exchange-rate changes therefore approach iid innovations at every fixed horizon, which is the precise sense in which the exchange-rate level approaches a random walk. This proves part 1.

Next, equation (23) gives

$$\frac{\mathrm{Var}(\widehat{r}_t - \widehat{r}_t^*)}{\sigma_\nu^2} = \frac{1-\rho_x}{1+\rho_x} + S^2(\Xi-1)^2\frac{1-\lambda}{1+\lambda} - 2S\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}. \tag{32}$$

Combining (32) with (29) produces the exact ratio

$$\frac{\mathrm{sd}(\widehat{r}_t - \widehat{r}_t^*)}{\mathrm{sd}(\Delta\widehat{e}_{t+1})} = \left[\frac{\frac{1-\rho_x}{1+\rho_x} + S^2(\Xi-1)^2\frac{1-\lambda}{1+\lambda} - 2S\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}}{\Xi^2 + \frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}}\right]^{1/2}. \tag{33}$$

Its numerator converges to zero and its denominator to Ξ^2 under joint persistence. This proves part 2.

For part 3, define the population Fama regression by

$$\Delta\widehat{e}_{t+1} = \alpha + \beta_F(\widehat{r}_t - \widehat{r}_t^*) + u_{t+1}.$$

Equations (23) and (25) imply

$$\frac{\mathrm{Cov}(\Delta\widehat{e}_{t+1}, \widehat{r}_t - \widehat{r}_t^*)}{\sigma_\nu^2} = \frac{1-\rho_x}{1+\rho_x} - (S-1)\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda} - S(\Xi-1)^2\frac{1-\lambda}{1+\lambda}. \tag{34}$$

Dividing (34) by (32) yields the exact coefficient

$$\boxed{\begin{aligned}\beta_F = &\left[\frac{1-\rho_x}{1+\rho_x} - (S-1)\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda} - S(\Xi-1)^2\frac{1-\lambda}{1+\lambda}\right] \\ &\times \left[\frac{1-\rho_x}{1+\rho_x} + S^2(\Xi-1)^2\frac{1-\lambda}{1+\lambda} - 2S\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}\right]^{-1}.\end{aligned}} \tag{35}$$

The exact coefficient of determination is

$$R_F^2 = \left[\frac{1-\rho_x}{1+\rho_x} - (S-1)\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda} - S(\Xi-1)^2\frac{1-\lambda}{1+\lambda}\right]^2 \times \left[\frac{1-\rho_x}{1+\rho_x} + S^2(\Xi-1)^2\frac{1-\lambda}{1+\lambda} - 2S\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}\right]^{-1} \times \left[\Xi^2 + \frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}\right]^{-1}. \tag{36}$$

Suppose $1-\rho_x = o(1-\lambda)$; this is the statement that ρ_x approaches one faster than λ. Then

$$\frac{1-\rho_x}{1+\rho_x} = o(1-\lambda), \qquad \frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda} = o(1-\lambda).$$

The terms proportional to $(\Xi-1)^2(1-\lambda)/(1+\lambda)$ therefore dominate both the numerator and denominator of (35), giving

$$\beta_F \longrightarrow -\frac{1}{S} < 0.$$

Thus β_F is eventually negative at any fixed amplification factor $\Xi > 1$.

Finally, because $\widehat{r}_t - \widehat{r}_t^*$ is date-t measurable, the variance explained by the Fama regression cannot exceed the variance of expected depreciation. Hence

$$R_F^2 \le \frac{\mathrm{Var}(\mathbb{E}_t \Delta\widehat{e}_{t+1})}{\mathrm{Var}(\Delta\widehat{e}_{t+1})} = \frac{\frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}}{\Xi^2 + \frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}} \longrightarrow 0.$$

Therefore $R_F^2 \to 0$ along every joint-persistence path for which the Fama regressor is non-degenerate. At the exact endpoint where its variance is zero, the regression itself is undefined. □

B.7 Proof of Proposition 3

Maintain the same constant-disagreement benchmark as in Appendix B.6. Here $S > 0$ is fixed, $0 < \rho_x, \lambda < 1$, and the sequence satisfies $\rho_x\lambda \to 1$ and $\Xi \to \Xi_{\max}$. The primitive construction and exact moment identities in that appendix apply at every interior parameterization. Direct news below is $\nu_{t+1} = \varepsilon_{t+1}^{e,f}$, the innovation in the normalized fundamental component f_t.

Proof. Decompose direct fundamental news as

$$\nu_{t+1} = \nu^y_{t+1} + \nu^{\perp}_{t+1}, \qquad \mathrm{Cov}(\nu^y_{t+1}, \nu^{\perp}_{t+1}) = 0,$$

where

$$s_y \equiv \frac{\mathrm{Var}(\nu^y_{t+1})}{\sigma^2_\nu}, \qquad 0 < s_y < 1.$$

Let f^y_t and $q^y_t = -(1-\rho_x) f^y_t$ denote the supply components of f_t and q_t. The supply block satisfies

$$f^y_t = \frac{\sigma \phi_y}{1+S}(\widehat{y}_t - \widehat{y}^*_t).$$

Since $f^y_{t+1} = \rho_x f^y_t + \nu^y_{t+1}$, it follows that

$$\phi_y \Delta(\widehat{y} - \widehat{y}^*)_{t+1} = \frac{S}{\sigma}\Xi_{\max}(q^y_t + \nu^y_{t+1}), \qquad \Xi_{\max} = \frac{1+S}{S}. \tag{37}$$

Goods-market clearing then gives

$$\begin{aligned}\Delta(\widehat{c} - \widehat{c}^*)_{t+1} &= \phi_y \Delta(\widehat{y} - \widehat{y}^*)_{t+1} - \frac{S}{\sigma}\Delta\widehat{e}_{t+1} \\ &= \frac{S}{\sigma}\left[\Xi_{\max}(q^y_t + \nu^y_{t+1}) - q_t + p_t - \Xi\nu_{t+1}\right]. \end{aligned} \tag{38}$$

Common persistence and innovation orthogonality imply

$$\begin{aligned}\frac{\mathrm{Var}(q^y_t)}{\sigma^2_\nu} &= s_y \frac{1-\rho_x}{1+\rho_x}, \\ \frac{\mathrm{Cov}(q^y_t, p_t)}{\sigma^2_\nu} &= -s_y \frac{(\Xi - 1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}.\end{aligned}$$

Using these identities, equations (25) and (38) give the following three exact moments:

$$\begin{aligned}
&\frac{\operatorname{Var}(\Delta\widehat{e}_{t+1})}{\sigma_\nu^2} \\
&\quad = \Xi^2 + \frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}, \qquad (39)\\
&\frac{\sigma}{S\sigma_\nu^2}\operatorname{Cov}(\Delta(\widehat{c}-\widehat{c}^*)_{t+1}, \Delta\widehat{e}_{t+1}) \\
&\quad = \Xi_{\max}s_y\left[\Xi + \frac{1-\rho_x}{1+\rho_x} + \frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}\right] \\
&\qquad - \left[\Xi^2 + \frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}\right], \qquad (40)\\
&\frac{\sigma^2}{S^2\sigma_\nu^2}\operatorname{Var}(\Delta(\widehat{c}-\widehat{c}^*)_{t+1}) \\
&\quad = \Xi^2 + \frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda} \\
&\qquad + \Xi_{\max}^2 s_y\left(1 + \frac{1-\rho_x}{1+\rho_x}\right) \\
&\qquad - 2\Xi_{\max}s_y\left[\Xi + \frac{1-\rho_x}{1+\rho_x} + \frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}\right]. \qquad (41)
\end{aligned}$$

Therefore the exact Backus–Smith correlation is

$$\begin{aligned}
&\operatorname{Corr}(\Delta(\widehat{c}-\widehat{c}^*)_{t+1}, \Delta\widehat{e}_{t+1}) \\
&= \left\{\Xi_{\max}s_y\left[\Xi + \frac{1-\rho_x}{1+\rho_x} + \frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}\right]\right. \\
&\qquad \left. - \left[\Xi^2 + \frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}\right]\right\} \\
&\quad \times \left\{\left[\Xi^2 + \frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}\right]\right. \\
&\qquad \times \left[\Xi^2 + \frac{1-\rho_x}{1+\rho_x} + (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + 2\frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}\right. \\
&\qquad\quad + \Xi_{\max}^2 s_y\left(1 + \frac{1-\rho_x}{1+\rho_x}\right) \\
&\qquad\quad \left.\left. - 2\Xi_{\max}s_y\left[\Xi + \frac{1-\rho_x}{1+\rho_x} + \frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda}\right]\right]\right\}^{-1/2}. \qquad (42)
\end{aligned}$$

Along the joint-persistence sequence,

$$\frac{1-\rho_x}{1+\rho_x} \to 0, \qquad (\Xi-1)^2\frac{1-\lambda}{1+\lambda} \to 0, \qquad \frac{(\Xi-1)(1-\rho_x)(1-\lambda)}{1-\rho_x\lambda} \to 0.$$

Before imposing the multiplier ceiling, the three moments above converge to

$$\begin{aligned}
\frac{\mathrm{Var}(\Delta\widehat{e})}{\sigma_\nu^2} &\longrightarrow \Xi^2, \\
\frac{\sigma}{S\sigma_\nu^2}\,\mathrm{Cov}(\Delta(\widehat{c}-\widehat{c}^*),\Delta\widehat{e}) &\longrightarrow \Xi\Xi_{\max}s_y - \Xi^2, \\
\frac{\sigma^2}{S^2\sigma_\nu^2}\,\mathrm{Var}(\Delta(\widehat{c}-\widehat{c}^*)) &\longrightarrow \Xi^2 - 2\Xi\Xi_{\max}s_y + \Xi_{\max}^2 s_y.
\end{aligned}$$

Hence

$$\mathrm{Corr}(\Delta(\widehat{c}-\widehat{c}^*),\Delta\widehat{e}) \longrightarrow \frac{\Xi_{\max}s_y - \Xi}{\sqrt{\Xi^2 - 2\Xi\Xi_{\max}s_y + \Xi_{\max}^2 s_y}}. \tag{43}$$

Finally set $\Xi \to \Xi_{\max}$. For fixed $0 < s_y < 1$, the numerator of (43) converges to $-\Xi_{\max}(1-s_y)$, while its denominator converges to $\Xi_{\max}\sqrt{1-s_y}$. Therefore

$$\mathrm{Corr}(\Delta(\widehat{c}-\widehat{c}^*),\Delta\widehat{e}) \longrightarrow -\sqrt{1-s_y},$$

as claimed. □

C Disagreement and the Backus–Smith correlation

This appendix separates two results. At the reported calibration, disagreement strengthens the negative covariance generated by demand shocks enough to reverse the Backus–Smith correlation in the FIRE benchmark. In the reduced equilibrium, sufficiently strong feedback can also generate a negative unconditional correlation with supply shocks alone, through the subsequent adjustment of the wealth share. The second result is a possibility result, not a property of the reported calibration. Throughout, the statistic is

$$\mathrm{Corr}(\Delta\hat{e}_t, \Delta(\hat{c}_t - \hat{c}_t^*)),$$

where an increase in e_t is a Home depreciation.

C.1 The calibrated economies and the fixed-shock comparison

Table C1 compares the reported correlations with their supply-conditional counterparts. The supply block retains output fundamentals and their associated signal and interpretation innovations; demand and UIP innovations are set to zero. Every supply-conditional correlation is positive. Thus these calibrated economies do not explain the negative sign using the supply block alone.

Table C1: Backus–Smith correlations in the reported economies

Specification	All shocks	Supply block only
Full model	−0.471	0.739
No UIP risk	−0.320	0.739
No average bias	−0.327	0.980
No average bias or UIP risk	−0.071	0.980
FIRE benchmark	−0.244	0.982
No disagreement, fixed parameters	0.546	0.998
No disagreement, re-estimated shocks	−0.131	0.997

Entries average 1,000 samples of 116 quarters after a 1,000-quarter burn-in, using the same innovations across specifications. Parameters are those of the main results table. Removing UIP risk leaves a supply-only experiment unchanged, so the corresponding entries coincide.

To isolate disagreement, Table C2 compares each economy with its own no-disagreement counterpart. Within each pair, preferences, technology, aggregate steady-state arbitrageur wealth, the absolute risk-cost parameter Γ, fundamental shock processes, signal noise, average beliefs, and the UIP process are held fixed. The counterfactual sets $d_{ss} = 0$ and the transitory disagreement innovations to zero. Steady-state objects are recomputed; as in the main counterfactual, the entrant endowment adjusts to preserve aggregate steady-state wealth. No parameters are re-estimated.

Table C2: The effect of disagreement at fixed shock parameters

Parameterization	Without disagreement	With disagreement	Change
FIRE benchmark	0.546	−0.244	−0.790
Full parameters, no UIP risk	0.657	−0.320	−0.977
Full parameters, with UIP risk	−0.599	−0.471	0.128

Entries use the same simulation protocol as Table C1. The full-parameter comparisons use the full model's own no-disagreement counterpart, which differs from the FIRE counterfactual in the main results table.

In FIRE, disagreement raises demand-driven exchange-rate volatility by a factor of 3.13, compared with 2.05 for supply-driven volatility. The magnitude of the isolated demand

covariance rises by a factor of 9.83, whereas the positive supply covariance rises by only 1.41. Demand-only Backus–Smith is already -1 in both economies: with output fixed, goods-market clearing makes relative consumption growth a negative multiple of depreciation. The effect therefore concerns the relative strength of demand and supply in the mixed-shock economy.

A further experiment turns off transitory disagreement in both economies and retains the permanent belief gap in one. The resulting linear system permits an exact population covariance decomposition across the independent demand and supply blocks. In units of 10^{-4}, the demand covariance changes from -0.163 without disagreement to -1.509 with disagreement; the supply covariance changes from 0.690 to 0.976. Their sum changes from 0.527 to -0.533, and the population correlation falls from 0.548 to -0.212. Permanent disagreement alone therefore suffices for the sign reversal at unchanged fundamental shock parameters. This additive decomposition is separate from the preceding simulations, which retain the wealth–disagreement cross terms.

The last row of Table C2 also identifies the scope of the result. With UIP risk active, disagreement reduces UIP-driven exchange-rate volatility from 3.682% to 1.678%. Because UIP-only Backus–Smith is also -1, this reduces an additional source of negative covariance. The full model's overall correlation then becomes less negative despite stronger demand amplification. Disagreement need not lower the correlation for every shock composition.

C.2 How supply shocks alone can produce a negative correlation

Consider the reduced equilibrium with a constant belief gap and supply news only, so $s_y = 1$. Goods-market clearing gives

$$\Delta(\hat{c} - \hat{c}^*)_{t+1} = \phi_y \Delta(\hat{y} - \hat{y}^*)_{t+1} - \frac{S}{\sigma} \Delta \hat{e}_{t+1}.$$

The first term is the direct income effect; the second is expenditure switching induced by exchange-rate movements. On impact, their covariance with the exchange-rate innovation is

$$\mathrm{Cov}_t(\Delta(\hat{c} - \hat{c}^*)_{t+1}, \Delta \hat{e}_{t+1}) = \frac{S\sigma_\nu^2}{\sigma} \Xi(\Xi_{\max} - \Xi) > 0.$$

Thus amplification cannot reverse the impact comovement under the multiplier ceiling. The unconditional correlation also incorporates adjustments following past shocks.

Setting $s_y = 1$ in equation (40) gives the exact unconditional covariance:

$$\begin{aligned}
&\frac{\sigma}{S\sigma_\nu^2}\operatorname{Cov}(\Delta(\hat{c}-\hat{c}^*),\Delta\hat{e}) \\
&\quad = \Xi(\Xi_{\max}-\Xi) + (\Xi_{\max}-1)\frac{1-\rho_y}{1+\rho_y} \\
&\qquad - (\Xi-1)^2\frac{1-\lambda}{1+\lambda} + (\Xi_{\max}-2)\frac{(\Xi-1)(1-\rho_y)(1-\lambda)}{1-\rho_y\lambda}.
\end{aligned} \tag{44}$$

Here $\rho_x = \rho_y$ because supply is the only driving force. The correlation is negative if and only if the right-hand side is negative. Its sign depends on both amplification and the relative persistence of supply and wealth, not on Ξ alone.

To show that this region is nonempty, recall the restrictions imposed by the symmetric reduced equilibrium:

$$\lambda = a\left(1-\frac{d_{ss}^2}{4\Gamma}\right), \qquad \Xi = \frac{(1+S)(1-\lambda)}{1-a+S(1-\lambda)}, \qquad a = \frac{1-\xi}{\beta} < 1.$$

Fix $S > 0$ and $0 < \lambda < 1$, and take $a \uparrow 1$ and $\rho_y \uparrow 1$, with $d_{ss}^2/(4\Gamma) = 1 - \lambda/a$. This respects the coefficient restrictions at every interior point and implies $\Xi \uparrow \Xi_{\max}$. Equation (44) then converges to

$$-(\Xi_{\max}-1)^2\frac{1-\lambda}{1+\lambda} < 0.$$

By continuity, stationary economies sufficiently close to this limit have negative supply-only Backus–Smith correlations. This is a different limit from Proposition 3: supply becomes highly persistent while wealth-share persistence remains strictly below one.

The adjustment mechanism is visible in the impulse responses. Following a unit supply innovation $\nu_t = 1$ from steady state, the responses at horizon $j \geq 0$ are

$$\hat{e}_{t+j} = \rho_y^j + (\Xi-1)\lambda^j, \qquad \frac{\sigma}{S}(\hat{c}-\hat{c}^*)_{t+j} = (\Xi_{\max}-1)\rho_y^j - (\Xi-1)\lambda^j.$$

When wealth adjusts faster than supply, the amplified exchange-rate component unwinds while the income effect remains. This unwinding moves the exchange rate and relative consumption in opposite directions. If feedback is strong enough, that negative covariance outweighs the positive impact covariance. Expenditure switching thus creates demand-like movements following supply news, without an additional exogenous demand shock.

C.3 A numerical illustration in the reduced equilibrium

Table C3 varies only the permanent belief gap within a separate numerical illustration. It holds fixed $\beta = \beta^* = 0.995$, $\xi = 0.01495$, $\Gamma = 0.001$, $\sigma = 5$, $\chi = 0.6$, $\theta = 1.5$, $\gamma = 0.05$, and $\rho_y = 0.95$, with symmetric wealth shares. These imply $a = 0.99$, $S = 1.0556$, and $\Xi_{\max} = 1.9474$. Only fundamental supply innovations are active; their variance is unchanged and cancels from the correlation. There are no demand, signal-noise, transitory disagreement, or UIP innovations.

Table C3: Supply-only Backus–Smith under stronger feedback

d_{ss}	λ	Ξ	Impact correlation	Unconditional correlation
0.000	0.990	1.000	1.000	1.000
0.015	0.934	1.702	1.000	0.994
0.030	0.767	1.871	1.000	0.140
0.040	0.594	1.903	1.000	−0.130
0.050	0.371	1.918	1.000	−0.265
0.060	0.099	1.927	1.000	−0.378

Exact population moments of the reduced equilibrium, calculated from equations (39)–(41) with $s_y = 1$ and independently verified using impulse-response sums. Every row satisfies $0 < \lambda < 1$ and $1 \leq \Xi < \Xi_{\max}$. This illustration does not replace the full model's calibration.

Increasing d_{ss} strengthens amplification and accelerates wealth-share adjustment. The unconditional correlation changes sign between $d_{ss} = 0.03$ and 0.04, while the impact correlation remains positive throughout. The example establishes that endogenous expenditure switching can overturn supply-driven comovement over time. Table C1 shows that the reported calibration instead reproduces the negative sign through the demand channel.

D Average bias and currency-return predictability

D.1 Equivalence to the Candian–De Leo belief process

This subsection establishes the equivalence of consensus posterior means and conditional-mean forecasts to the belief process in Candian and De Leo (2025). Fix a fundamental x and write $m_t^x \equiv \bar{\tau}_{t|t}^x$ for the simple average posterior in this paper. Averaging the two groups' filtering rules gives

$$m_t^x = \varphi_x m_{t-1}^x + G_x(\hat{x}_t + \bar{\mu}_t^x), \qquad \varphi_x \equiv \rho_x(1 - G_x). \tag{45}$$

In the Candian–De Leo perceived model, the observed fundamental is the sum of a

persistent component z_t^x and an unpredictable transitory component ν_t^x:

$$\hat{x}_t = z_t^x + \nu_t^x, \qquad z_t^x = \tilde{\rho}_x z_{t-1}^x + u_t^x.$$

With steady-state Kalman gain κ_x, the posterior mean $m_t^{x,\mathrm{CDL}}$ and forecasts satisfy

$$m_t^{x,\mathrm{CDL}} = \tilde{\rho}_x m_{t-1}^{x,\mathrm{CDL}} + \kappa_x\big(\hat{x}_t - \tilde{\rho}_x m_{t-1}^{x,\mathrm{CDL}}\big) = \delta_x m_{t-1}^{x,\mathrm{CDL}} + \kappa_x \hat{x}_t, \tag{46}$$

$$E_t^{\mathrm{CDL}} \hat{x}_{t+h} = \tilde{\rho}_x^h m_t^{x,\mathrm{CDL}}, \qquad h \geq 1, \qquad \delta_x \equiv (1-\kappa_x)\tilde{\rho}_x. \tag{47}$$

Proposition 5. *Suppose $G_x \in (0,1]$, $\kappa_x \in (0,1]$, and the two representations start from the same consensus posterior. The average interpretation rule*

$$\bar{\mu}_t^x = \left(\frac{\kappa_x}{G_x} - 1\right)\hat{x}_t + \frac{(1-\kappa_x)\tilde{\rho}_x - \rho_x(1-G_x)}{G_x} m_{t-1}^x \tag{48}$$

uniquely implements the Candian–De Leo consensus posterior for every observation history. If consensus forecasts also use the perceived persistence, $\bar{E}_t \hat{x}_{t+h} = \tilde{\rho}_x^h m_t^x$, the two representations generate identical conditional-mean forecasts and realized forecast errors at every horizon.

Proof. Substituting (48) into (45) yields

$$\begin{aligned} m_t^x &= \varphi_x m_{t-1}^x + G_x \hat{x}_t + (\kappa_x - G_x)\hat{x}_t + (\delta_x - \varphi_x) m_{t-1}^x \\ &= \delta_x m_{t-1}^x + \kappa_x \hat{x}_t. \end{aligned}$$

This is exactly (46). Equality of the initial posteriors therefore implies $m_t^x = m_t^{x,\mathrm{CDL}}$ at every date by induction, for any common sequence of observations. Conversely, equating the desired posterior update with (45) requires

$$G_x \bar{\mu}_t^x = (\kappa_x - G_x)\hat{x}_t + (\delta_x - \varphi_x) m_{t-1}^x.$$

Because $G_x > 0$, this uniquely determines the average interpretation rule along every such history. Applying the same perceived transition to the equal posterior means gives $\bar{E}_t \hat{x}_{t+h} = E_t^{\mathrm{CDL}} \hat{x}_{t+h}$ for all $h \geq 1$. Subtracting these equal forecasts from the same subsequent realization establishes equality of forecast errors. If $|\delta_x| < 1$, any initial posterior discrepancy instead decays geometrically at rate δ_x. □

The corresponding forecast-error dynamics can also be seen directly. Let $f_t^x \equiv \bar{E}_t \hat{x}_{t+1} =$

$\tilde{\rho}_x m_t^x$ and $q_t^x \equiv \hat{x}_t - f_{t-1}^x$. The posterior recursion implies

$$f_t^x = \delta_x f_{t-1}^x + \kappa_x \tilde{\rho}_x \hat{x}_t,$$
$$q_t^x - \delta_x q_{t-1}^x = \hat{x}_t - \tilde{\rho}_x \hat{x}_{t-1}.$$

These identities hold for any observation process. In the special case in which the true observed fundamental follows $\hat{x}_t = \rho_x \hat{x}_{t-1} + v_t^x$, applying $1 - \rho_x L$, where L is the lag operator, gives the Candian–De Leo ARMA$(2,1)$ representation:

$$q_t^x = (\rho_x + \delta_x) q_{t-1}^x - \rho_x \delta_x q_{t-2}^x + v_t^x - \tilde{\rho}_x v_{t-1}^x. \tag{49}$$

The construction leaves a separate role for disagreement. For any prescribed interpretation difference $\Delta\mu_t^x \equiv \mu_t^{xA} - \mu_t^{xB}$, set

$$\mu_t^{xA} = \bar{\mu}_t^x + \tfrac{1}{2}\Delta\mu_t^x, \qquad \mu_t^{xB} = \bar{\mu}_t^x - \tfrac{1}{2}\Delta\mu_t^x.$$

Their average is (48), while their difference remains $\Delta\mu_t^x$. The common component cancels from the posterior-gap recursion:

$$d_t^x = \varphi_x d_{t-1}^x + G_x \Delta\mu_t^x, \qquad d_t^x \equiv \tau_{t|t}^{xA} - \tau_{t|t}^{xB}.$$

Thus the equivalence concerns consensus posterior means and forecasts, conditional on the same observations; it neither eliminates heterogeneous beliefs nor asserts equivalence of the two general-equilibrium models. The average interpretation wedge is a function of current observations and the inherited consensus posterior, so it introduces no new primitive shock. Matching posterior updating alone would not suffice for forecast equivalence if forecasts continued to use ρ_x in place of $\tilde{\rho}_x$.

D.2 Predictability across horizons

This appendix compares the current full model with three ablations: removing the UIP shock, removing average bias, and removing both. All other parameters are held at the full-model calibration, and both wealth–disagreement cross terms are retained in the simulations. These comparisons differ from the separately calibrated FIRE benchmark. Each regression coefficient is averaged over 1,000 samples of 116 quarters following a 1,000-quarter burn-in, using common random innovations across specifications.

Let $i_t \equiv \hat{r}_t - \hat{r}_t^*$ denote the quarterly Home-minus-Foreign interest differential and define the foreign-currency excess return by $rx_{t+1} \equiv \Delta\hat{e}_{t+1} - i_t$. For each horizon, I estimate

$$rx_{t+h} = a_h + b_h i_t + u_{t+h}, \qquad h \geq 1,$$

$$\sum_{j=1}^{H} rx_{t+j} = A_H + B_H i_t + v_{t,H}, \qquad H \geq 1.$$

The first regression measures the predictability of a one-quarter return at horizon h; the second measures the predictability of returns accumulated over H quarters. With this sign convention, a negative coefficient means that a high Home interest rate predicts Home currency outperformance. Each cumulative regression uses the same set of starting dates for all returns entering its dependent variable. It therefore need not equal the sum of the reported horizon-specific coefficients, which use different numbers of observations. These are predictive regressions, distinct from the structural shock responses in Figure 7.

Figure 6 shows that the full-model one-quarter coefficient first becomes positive at six quarters, compared with eight quarters without average bias and 28 quarters without either average bias or the UIP shock. The full-model cumulative coefficient first becomes positive at 29 quarters; without the UIP shock it does so at 16 quarters. Both specifications without average bias retain negative cumulative coefficients throughout the first 40 quarters. Average bias therefore advances the reversal and changes its cumulative sign over this horizon. The result is a finite-horizon, finite-sample comparison; it does not establish a positive infinite-horizon cumulative coefficient.

The effect is strongest for output news, for which the calibrated information gain is 0.26. Figure 7 illustrates the mechanism. Following a positive Home endowment shock, the Home interest differential falls and the Home currency continues to depreciate for two quarters. Foreign deposits initially earn positive excess returns, but their return response becomes negative four quarters after the shock. Without average bias, the initial interest-rate and excess-return responses have the opposite signs, and the initial depreciation starts unwinding immediately. Average bias has a smaller effect on the demand-shock responses.

Average bias also improves the cumulative-return result. The coefficient from regressing foreign excess returns accumulated over the next 40 quarters on today's Home-minus-Foreign interest differential is 0.36 in the full model, compared with -2.10 when average bias is removed at the same calibration and -6.52 in the separately calibrated FIRE benchmark. Removing only the UIP shock leaves a positive coefficient of 1.19. This improvement is qualitative: the full-model coefficient remains much smaller than the data estimate of 20.65, and its simulated 5th–95th percentile range, $[-3.08, 3.99]$, includes zero. Average bias helps reproduce the cumulative sign and the timing of the reversal, while the wealth–disagreement

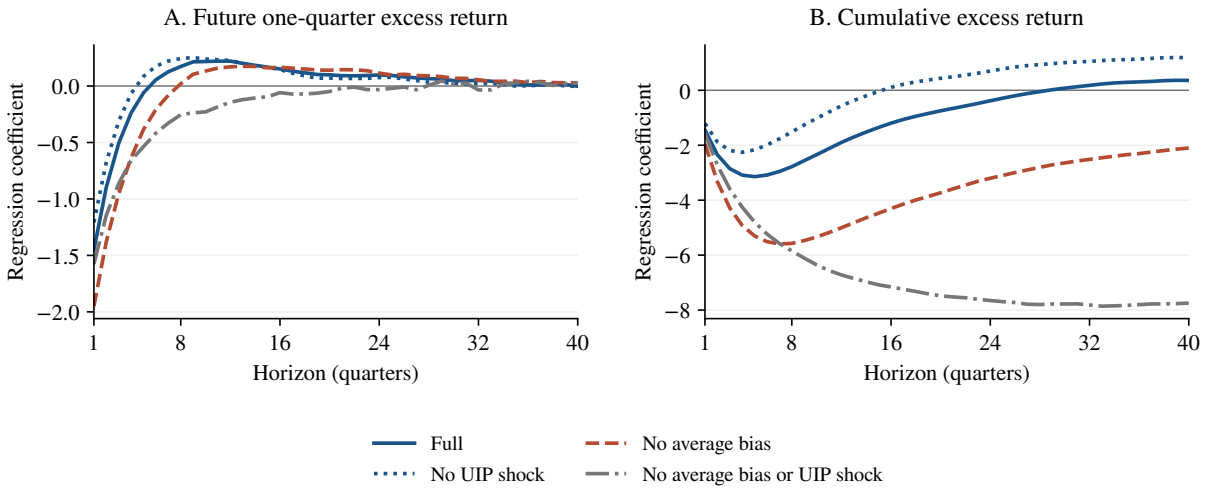


Figure 6: Average bias and the horizon profile of excess-return predictability

Notes: Mean regression coefficients across 1,000 simulations of 116 quarters. The left panel reports b_h; the right panel reports the directly estimated B_H. All ablations hold the remaining full-model parameters fixed. The figures show predictive regression coefficients, not impulse responses or confidence bands. The 40-quarter full-model coefficient is 0.36, compared with 20.65 in the data.

channel accounts for the benchmark's core FX-disconnect results.

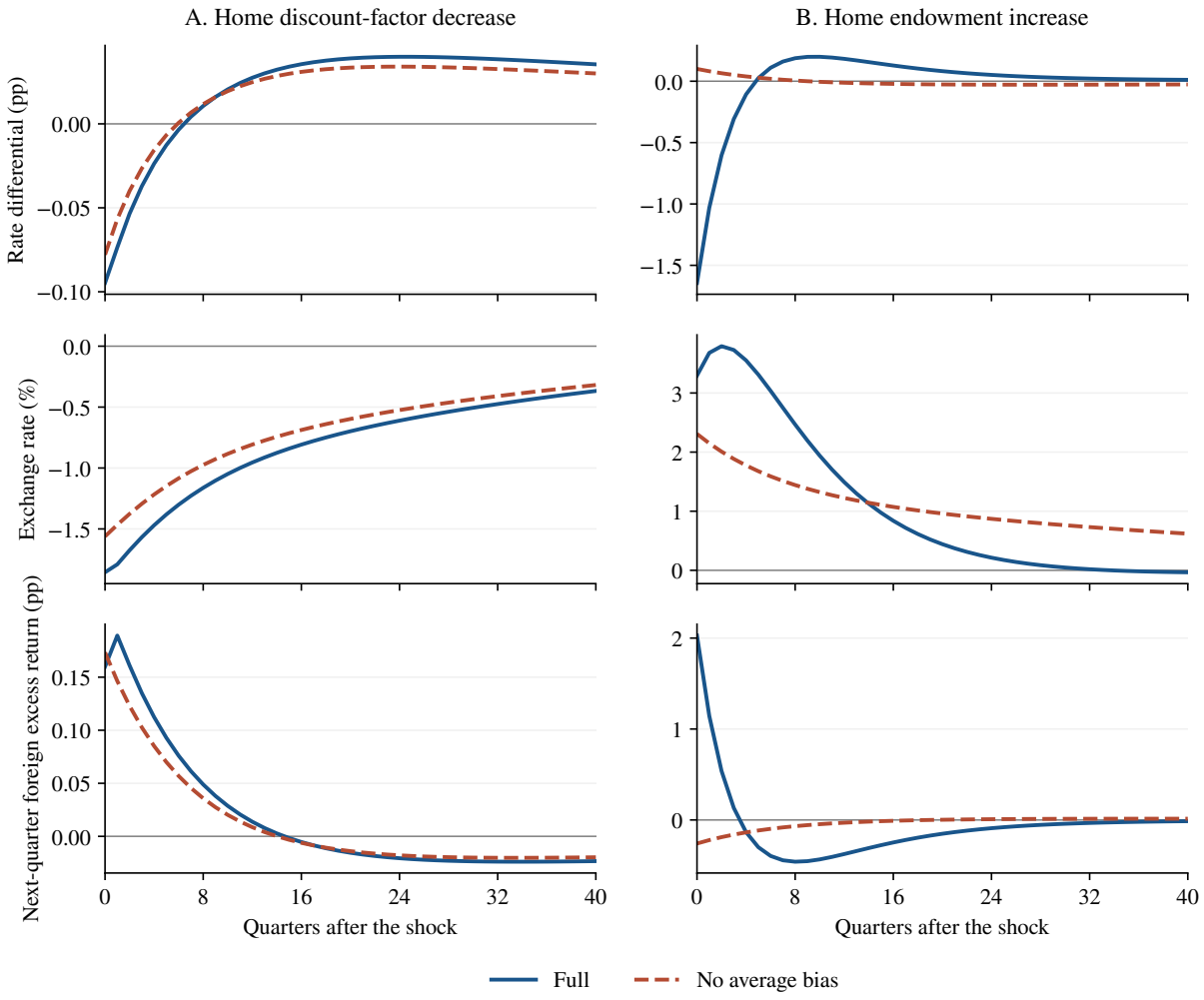


Figure 7: Average bias and the response to fundamental news

Notes: Responses to a one-standard-deviation decrease in the Home discount factor (left) and increase in the Home endowment (right), with the Foreign innovation set to zero. The paths start at the deterministic steady state and set subsequent innovations to zero. The comparison removes average bias at the full-model calibration. Rates and returns are in percentage points per quarter; an increase in the exchange rate denotes Home depreciation. At horizon h, the bottom panels report the foreign excess return earned from $t+h$ to $t+h+1$, $\hat{e}_{t+h+1} - \hat{e}_{t+h} - (\hat{r}_{t+h} - \hat{r}^*_{t+h})$.

E Time-varying disagreement in the quantitative model

The quantitative specification retains two financial interactions through which the current belief gap changes exchange-rate transmission. This appendix states the beliefs, portfolio rule, and approximation that define this specification. The interactions use existing parameters and disagreement states.

E.1 Beliefs and portfolio demand

Assumption 1: persistent types and changing beliefs. Let M_t^i denote group i's expected carry return in the maintained return normalization. The two groups' conditional means satisfy

$$M_t^A = \bar{M}_t + \frac{\mathcal{D}_t}{2}, \qquad M_t^B = \bar{M}_t - \frac{\mathcal{D}_t}{2}, \qquad \mathcal{D}_t = d_{ss} + d_t. \tag{50}$$

Here $\bar{M}_t$ is the arithmetic mean of subjective forecasts, including any average bias. The permanent difference d_{ss} reflects fixed types, while d_t is generated by the existing signal-interpretation process, with its baseline forecast loadings held fixed. The groups retain their interpretation rules as new information arrives. Thus time variation in the gap introduces no new primitive shock. Group labels remain fixed, so $\mathcal{D}_t$ is signed; with equal population weights, forecast dispersion is $|\mathcal{D}_t|/2$. The interpretation that a larger gap means greater dispersion applies on the region $\mathcal{D}_t > 0$.

Assumption 2: common, fixed perceived risk. The mean–variance rule in the maintained return normalization is

$$\frac{N_t^i}{W_t^i} = \frac{M_t^i}{\Gamma}, \qquad \frac{N_t^A}{W_t^A} - \frac{N_t^B}{W_t^B} = \frac{d_{ss} + d_t}{\Gamma}, \qquad \Gamma > 0. \tag{51}$$

Myopic demand and the common effective risk cost Γ are maintained primitives of this specification. Perceived risk remains fixed when the financial interactions are introduced. Realized conditional exchange-rate variance can vary because the transmission of primitive shocks changes with beliefs. The distinction between fixed perceived risk and changing realized volatility is part of the specified belief system.

E.2 The two financial interactions

Assumption 3: retain the disagreement interactions. The quantitative model retains the products of disagreement with the wealth share in bond pricing and with the

subsequent carry return in wealth accumulation. Other equilibrium coefficients and wealth-share normalizations retain their first-order forms.

First, aggregate portfolio demand weights each group's forecast by its current wealth. With $\widehat{\omega}_t^A = \omega_t^A - \omega^A$ and $\delta_w = \omega^A - 1/2$, the disagreement component, net of its steady-state value, is

$$\begin{aligned}\Psi_t^d &= \left(\delta_w + \widehat{\omega}_t^A\right)(d_{ss} + d_t) - \delta_w d_{ss} \\ &= d_{ss}\widehat{\omega}_t^A + \delta_w d_t + d_t\widehat{\omega}_t^A.\end{aligned} \tag{52}$$

This identity follows by substituting (50) into wealth-weighted forecasts. In the modified UIP condition, the disagreement contribution is $-\Psi_t^d$; the capacity terms retain their existing first-order form. The additional product therefore makes the pricing effect of a wealth transfer equal to the current gap:

$$\frac{\partial \Psi_t^d}{\partial \widehat{\omega}_t^A} = d_{ss} + d_t. \tag{53}$$

Second, wealth accumulation multiplies the position chosen at t by the return realized at $t+1$. In the quantitative model's coordinates, write the first-order simple-carry deviation as

$$\widehat{R}_{t+1}^p = \frac{\widehat{e}_{t+1} - \widehat{e}_t + \widehat{r}_t^*}{\beta^*} - \frac{\widehat{r}_t}{\beta}. \tag{54}$$

The changing relative exposure in (51) gives the following additions to the existing group log-wealth equations:

$$q_{t+1}^A = \frac{1-\xi}{2\Gamma} d_t \widehat{R}_{t+1}^p, \qquad q_{t+1}^B = -\frac{1-\xi}{2\Gamma} d_t \widehat{R}_{t+1}^p. \tag{55}$$

Since the retained share coordinate is $\widehat{\omega}_t^A = \omega^A\omega^B(\widehat{W}_t^A - \widehat{W}_t^B)$, these terms add $(1-\xi)\omega^A\omega^B d_t \widehat{R}_{t+1}^p/\Gamma$ to wealth-share dynamics. Combined with the existing term proportional to d_{ss}, the return loading is proportional to $d_{ss} + d_t$. The coefficient is fixed by portfolio demand and survival. At date t, pricing uses current disagreement d_t, while realized wealth gains use the disagreement d_{t-1} prevailing when the position was opened.

E.3 Forecasting and solution

Assumption 4: affine forecasts and current financial feedback. Agents forecast using the baseline affine perceived transition. Current realized prices and wealth enter the state carried into the next period. At the forecasting stage, the additional future interaction terms are evaluated at their reference value, zero. Thus today's wealth changes affect future forecasts through the baseline transition, while additional forecast corrections for future bi-

linear interactions are excluded. This restriction specifies how expectations are approximated separately from current financial accounting.

At each date the two current interactions are solved jointly. Let y_t^0 be the baseline decision rule evaluated at the inherited state and current information, and let y_t collect the corresponding quantitative-model variables. The implemented rule is

$$y_t = y_t^0 + Hq_t, \qquad q_t = \begin{pmatrix} d_t \widehat{\omega}_t^A \\ (1-\xi) d_{t-1} \widehat{R}_t^p / (2\Gamma) \\ -(1-\xi) d_{t-1} \widehat{R}_t^p / (2\Gamma) \end{pmatrix}. \tag{56}$$

The fixed matrix H contains the baseline current responses to additional terms in the UIP and two wealth equations. The entries of q_t are determined by the displayed products and introduce no independent innovations. The solved variables are then carried into the next period's state. In the implementation, these additional inputs do not change d_t directly. Conditional on d_t and d_{t-1}, the two distinct interaction terms therefore satisfy a linear two-by-two system.

Assumption 5: local regularity. The contemporaneous feedback system is nonsingular on the neighborhood under consideration, with its inverse bounded, and the underlying wealth and share interpretation is admissible there. At the reference state, all added products and their first derivatives are zero. Hence the Jacobian of $y_t - y_t^0 - Hq_t$ with respect to y_t is the identity. The implicit function theorem gives a unique smooth local solution. Scaling state and innovation deviations by ϵ, with d_{ss} and the baseline coefficients fixed, yields

$$y_t = y_t^0 + O(\epsilon^2). \tag{57}$$

This comparison holds over a fixed finite horizon while the path remains in the stated neighborhood. The specification preserves the analytical first-order solution and adds the two selected financial interactions. Solving their current feedback jointly retains the higher powers generated by those interactions. The approximation is defined by affine forecasts and bilinear financial feedback; a complete second-order expansion of all equilibrium equations is not imposed.

These two margins make financial transmission depend on beliefs: greater disagreement raises relative exposure and the pricing effect of wealth redistribution. The local response to news can therefore vary even with constant primitive shock variances. The quantitative exercise measures this state dependence and its contribution to unconditional moments.

F Additional quantitative results

F.1 Full comparison of exchange-rate moments across model variants

Table F1 reports the full moment comparison. Tables F2–F4 report the same moments with one innovation block active at a time.

Table F1: Full comparison of exchange-rate moments

		Full-model calibration				FIRE benchmark	No disagreement	
Moment	Data	Full model	No UIP risk	No avg. bias	No bias/UIP	Separate calib.	Fixed shocks	Recalibrated shocks
A. Exchange rates and macroeconomic variables								
$\rho_1(\Delta\hat{e}_t)$	0.013	0.064	0.084	-0.043	-0.051	-0.050	-0.009	-0.003
$\rho_1(\hat{e}_t)$	0.950	0.917	0.922	0.900	0.905	0.908	0.930	0.926
$\sigma(\Delta\hat{e}_t)$	0.033	0.040	0.036	0.032	0.027	0.034	0.013	0.020
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{c}_t)$	6.43	4.46	4.11	3.58	3.09	4.11	1.58	2.95
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{y}_t)$	5.42	4.40	3.98	3.55	3.00	4.03	1.54	2.94
$\sigma(\Delta(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*}))/\sigma(\Delta\hat{e}_t)$	0.032	0.426	0.407	0.278	0.093	0.117	0.041	0.053
$\sigma(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})/\sigma(\Delta\hat{e}_t)$	0.110	0.513	0.473	0.374	0.139	0.166	0.247	0.309
$\sigma(\Delta(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*}))/\sigma(\Delta\hat{e}_t)$	0.021	0.016	0.013	0.016	0.010	0.012	0.055	0.068
$\sigma(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})/\sigma(\Delta\hat{e}_t)$	0.060	0.061	0.062	0.063	0.064	0.074	0.240	0.295
$\mathrm{Corr}(\Delta\hat{e}_t, \Delta(\hat{c}_t - \hat{c}_t^*))$	-0.169	-0.471	-0.320	-0.327	-0.071	-0.244	0.546	-0.131
B. Excess-return predictability								
β^{Fama}	-0.213	-0.449	-0.208	-0.950	-0.576	-0.480	1.023	0.906
R^2_{Fama}	0.0005	0.0606	0.0184	0.1261	0.0120	0.0135	0.0431	0.0528
$\beta^{\mathrm{Fama,return}}$	-1.213	-1.449	-1.208	-1.950	-1.576	-1.480	0.023	-0.094
$R^2_{\mathrm{Fama,return}}$	0.017	0.369	0.252	0.373	0.047	0.062	0.024	0.034
$\beta^{\mathrm{Engel,cumul}}_{40q}$	20.65	0.36	1.19	-2.10	-7.75	-6.52	-3.77	-5.72
90% interval	[6.24, 35.06]	[−3.08, 3.99]	[−2.43, 4.65]	[−6.51, 2.63]	[−19.65, 2.36]	[−18.54, 3.27]	[−36.67, 26.99]	[−35.43, 21.28]
C. Changes-on-changes regressions								
$\Delta\hat{e}_t$ on $4\Delta(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})$	-0.50	-0.13	-0.33	0.56	2.08	1.62	-4.10	-2.90
90% interval	[−2.03, 1.03]	[−0.21, −0.04]	[−0.41, −0.26]	[0.43, 0.68]	[0.57, 3.61]	[0.35, 3.38]	[−4.78, −3.44]	[−3.43, −2.36]
Adjusted R^2	-0.005	0.046	0.283	0.381	0.457	0.358	0.444	0.364
$\Delta\hat{e}_t$ on $4\Delta(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})$	-1.42	1.24	-6.82	2.57	-16.78	-15.03	-4.16	-3.57
90% interval	[−3.37, 0.52]	[−1.75, 4.08]	[−10.70, −2.84]	[−0.28, 5.28]	[−20.04, −12.75]	[−18.02, −11.13]	[−4.45, −3.85]	[−3.72, −3.41]
Adjusted R^2	0.005	0.010	0.124	0.030	0.436	0.487	0.825	0.928
D. Level regressions								
$\hat{e}_t$ on $4(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})$	-3.47	-0.22	-0.45	0.34	0.90	0.55	-1.79	-1.23
90% interval	[−6.81, −0.12]	[−0.55, 0.07]	[−0.80, −0.13]	[−0.22, 0.83]	[−1.39, 2.94]	[−1.48, 2.42]	[−4.06, 0.24]	[−2.60, 0.19]
Adjusted R^2	0.222	0.038	0.104	0.084	0.106	0.084	0.344	0.276
$\hat{e}_t$ on $4(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})$	-7.79	1.98	0.42	-2.75	-7.62	-6.65	-2.23	-1.70
90% interval	[−12.81, −2.77]	[−4.70, 8.96]	[−6.72, 7.90]	[−7.94, 3.44]	[−11.62, −3.77]	[−10.66, −2.80]	[−4.27, −0.24]	[−3.09, −0.21]
Adjusted R^2	0.336	0.126	0.111	0.169	0.527	0.521	0.471	0.448

Notes: The first four model columns share the full-model calibration and respectively retain both extensions, remove UIP risk, remove average bias, and remove both. All retain the two financial interaction terms. FIRE is separately calibrated and excludes average bias and UIP risk while retaining disagreement. The fixed no-disagreement economy holds the FIRE parameters fixed; the recalibrated economy replaces only the fundamental shock block. All model entries average 1,000 independent simulations of 116 quarters after a 1,000-quarter burn-in. ρ_1 denotes first-order autocorrelation; $\hat{e}_t$ rises with Home depreciation. $\Delta\hat{c}_t$ and $\Delta\hat{y}_t$ are Home consumption and output growth. $\hat{r}_t^{(1)}$ and $\hat{r}_t^{(40)}$ are quarterly-equivalent yields at one- and 40-quarter maturities; the factors of four in Panels C and D preserve the annualized regressors used in estimation. Panel D expresses both data and model in the Home-depreciation/Home-minus-Foreign convention; reversing both original data signs leaves the slopes unchanged. The Engel coefficient uses cumulative foreign-currency excess returns over the next 40 quarters and today's quarterly Home-minus-Foreign short-rate differential. Data intervals are 90% confidence intervals; model intervals are 5th–95th percentiles across simulations. Fama, Engel, changes-on-changes, and level regressions are untargeted.

Table F2: Exchange-rate moments conditional on demand innovations

Moment	Data	Full model	No UIP risk	No avg. bias	FIRE Separate calib.	No disag. (fixed)	No disag. (recal.)
A. Exchange rates and macroeconomic variables							
$\rho_1(\Delta\hat{e}_t)$	0.013	-0.023	-0.023	-0.048	-0.048	0.005	0.005
$\rho_1(\hat{e}_t)$	0.950	0.911	0.911	0.909	0.911	0.931	0.931
$\sigma(\Delta\hat{e}_t)$	0.033	0.024	0.024	0.020	0.028	0.009	0.017
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{c}_t)$	6.43	9.51	9.51	9.51	9.51	9.48	9.48
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{y}_t)$	5.42	–	–	–	–	–	–
$\sigma(\Delta(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*}))/\sigma(\Delta\hat{e}_t)$	0.032	0.104	0.104	0.106	0.132	0.051	0.051
$\sigma(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})/\sigma(\Delta\hat{e}_t)$	0.110	0.158	0.158	0.157	0.187	0.335	0.335
$\sigma(\Delta(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*}))/\sigma(\Delta\hat{e}_t)$	0.021	0.012	0.012	0.012	0.014	0.074	0.074
$\sigma(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})/\sigma(\Delta\hat{e}_t)$	0.060	0.077	0.077	0.077	0.086	0.330	0.330
$\mathrm{Corr}(\Delta\hat{e}_t, \Delta(\hat{c}_t - \hat{c}_t^*))$	-0.169	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000
B. Excess-return predictability							
β^{Fama}	-0.213	-0.388	-0.388	-0.385	-0.366	0.840	0.840
R^2_{Fama}	0.0005	0.0106	0.0106	0.0108	0.0139	0.0531	0.0531
$\beta^{\mathrm{Fama,return}}$	-1.213	-1.388	-1.388	-1.385	-1.366	-0.160	-0.160
$R^2_{\mathrm{Fama,return}}$	0.017	0.049	0.049	0.050	0.071	0.042	0.042
$\beta^{\mathrm{Engel,cumul}}_{40q}$	20.65	-7.29	-7.29	-6.53	-5.87	-7.26	-7.26
90% interval	[6.24, 35.06]	[−17.39, 1.92]	[−17.39, 1.92]	[−17.13, 3.40]	[−17.35, 3.20]	[−34.50, 20.04]	[−34.50, 20.04]
C. Changes-on-changes regressions							
$\Delta\hat{e}_t$ on $4\Delta(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})$	-0.50	1.87	1.87	1.82	1.44	-2.31	-2.31
90% interval	[−2.03, 1.03]	[0.48, 3.39]	[0.48, 3.39]	[0.45, 3.33]	[0.28, 3.15]	[−2.89, −1.73]	[−2.89, −1.73]
Adjusted R^2	-0.005	0.445	0.445	0.426	0.335	0.218	0.218
$\Delta\hat{e}_t$ on $4\Delta(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})$	-1.42	-16.95	-16.95	-16.46	-14.66	-3.32	-3.32
90% interval	[−3.37, 0.52]	[−19.74, −13.88]	[−19.74, −13.88]	[−18.98, −13.65]	[−17.24, −11.19]	[−3.40, −3.23]	[−3.40, −3.23]
Adjusted R^2	0.005	0.634	0.634	0.666	0.651	0.969	0.969
D. Level regressions							
$\hat{e}_t$ on $4(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})$	-3.47	0.76	0.76	0.54	0.32	-1.02	-1.02
90% interval	[−6.81, −0.12]	[−1.37, 2.59]	[−1.37, 2.59]	[−1.64, 2.41]	[−1.73, 2.17]	[−2.32, 0.33]	[−2.32, 0.33]
Adjusted R^2	0.222	0.097	0.097	0.082	0.072	0.228	0.228
$\hat{e}_t$ on $4(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})$	-7.79	-6.70	-6.70	-6.82	-6.04	-1.55	-1.55
90% interval	[−12.81, −2.77]	[−9.78, −3.49]	[−9.78, −3.49]	[−9.83, −3.67]	[−9.06, −2.78]	[−2.78, −0.20]	[−2.78, −0.20]
Adjusted R^2	0.336	0.560	0.560	0.585	0.558	0.439	0.439

Notes: Only demand innovations are active: discount-factor fundamentals, signal noise, and belief innovations. Parameters and decision rules remain fixed within each specification; all other innovations are zero. The data column remains unconditional. Full model and No UIP risk coincide because the UIP innovation is inactive. FIRE and the no-disagreement columns use the same separate calibrations as Table F1. A dash denotes an undefined ratio because conditional output growth has zero variance. All model entries average 1,000 independent simulations of 116 quarters after a 1,000-quarter burn-in. ρ_1 denotes first-order autocorrelation; $\hat{e}_t$ rises with Home depreciation. $\Delta\hat{c}_t$ and $\Delta\hat{y}_t$ are Home consumption and output growth. $\hat{r}_t^{(1)}$ and $\hat{r}_t^{(40)}$ are quarterly-equivalent yields at one- and 40-quarter maturities; the factors of four in Panels C and D preserve the annualized regressors used in estimation. Panel D expresses both data and model in the Home-depreciation/Home-minus-Foreign convention; reversing both original data signs leaves the slopes unchanged. The Engel coefficient uses cumulative foreign-currency excess returns over the next 40 quarters and today's quarterly Home-minus-Foreign short-rate differential. Data intervals are 90% confidence intervals; model intervals are 5th–95th percentiles across simulations. Fama, Engel, changes-on-changes, and level regressions are untargeted.

Table F3: Exchange-rate moments conditional on supply innovations

Moment	Data	Full model	No UIP risk	No avg. bias	FIRE Separate calib.	No disag. (fixed)	No disag. (recal.)
A. Exchange rates and macroeconomic variables							
$\rho_1(\Delta\hat{e}_t)$	0.013	0.165	0.165	-0.046	-0.045	-0.018	-0.026
$\rho_1(\hat{e}_t)$	0.950	0.932	0.932	0.906	0.907	0.930	0.909
$\sigma(\Delta\hat{e}_t)$	0.033	0.027	0.027	0.018	0.020	0.010	0.010
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{c}_t)$	6.43	3.16	3.16	2.11	2.55	1.18	1.50
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{y}_t)$	5.42	3.71	3.71	2.53	2.43	1.18	1.56
$\sigma(\Delta(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*}))/\sigma(\Delta\hat{e}_t)$	0.032	0.544	0.544	0.046	0.044	0.030	0.057
$\sigma(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})/\sigma(\Delta\hat{e}_t)$	0.110	0.628	0.628	0.091	0.089	0.136	0.197
$\sigma(\Delta(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*}))/\sigma(\Delta\hat{e}_t)$	0.021	0.013	0.013	0.004	0.005	0.031	0.040
$\sigma(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})/\sigma(\Delta\hat{e}_t)$	0.060	0.046	0.046	0.043	0.045	0.123	0.133
$\mathrm{Corr}(\Delta\hat{e}_t, \Delta(\hat{c}_t - \hat{c}_t^*))$	-0.169	0.739	0.739	0.980	0.982	0.998	0.997
B. Excess-return predictability							
β^{Fama}	-0.213	-0.200	-0.200	-1.078	-1.000	1.792	1.374
R^2_{Fama}	0.0005	0.0259	0.0259	0.0132	0.0119	0.0407	0.0555
$\beta^{\mathrm{Fama,return}}$	-1.213	-1.200	-1.200	-2.078	-2.000	0.792	0.374
$R^2_{\mathrm{Fama,return}}$	0.017	0.369	0.369	0.032	0.029	0.015	0.014
$\beta^{\mathrm{Engel,cumul}}_{40q}$	20.65	1.54	1.54	-15.14	-15.11	13.14	6.33
90% interval	[6.24, 35.06]	[−1.34, 4.62]	[−1.34, 4.62]	[−27.76, −0.17]	[−28.88, 1.07]	[−30.40, 50.33]	[−19.08, 25.94]
C. Changes-on-changes regressions							
$\Delta\hat{e}_t$ on $4\Delta(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})$	-0.50	-0.38	-0.38	5.26	5.44	-8.21	-4.36
90% interval	[−2.03, 1.03]	[−0.39, −0.38]	[−0.39, −0.38]	[5.00, 5.44]	[4.76, 5.83]	[−8.32, −8.04]	[−4.40, −4.30]
Adjusted R^2	-0.005	0.690	0.690	0.939	0.906	0.988	0.990
$\Delta\hat{e}_t$ on $4\Delta(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})$	-1.42	-2.05	-2.05	-43.14	-41.36	-8.06	-6.17
90% interval	[−3.37, 0.52]	[−5.90, 1.47]	[−5.90, 1.47]	[−54.34, −32.29]	[−50.40, −32.52]	[−8.12, −7.97]	[−6.21, −6.09]
Adjusted R^2	0.005	0.014	0.014	0.540	0.620	0.995	0.992
D. Level regressions							
$\hat{e}_t$ on $4(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})$	-3.47	-0.49	-0.49	2.52	2.34	-5.38	-3.01
90% interval	[−6.81, −0.12]	[−0.62, −0.40]	[−0.62, −0.40]	[−1.47, 4.91]	[−1.85, 4.99]	[−6.95, −3.69]	[−3.88, −2.07]
Adjusted R^2	0.222	0.187	0.187	0.218	0.197	0.829	0.802
$\hat{e}_t$ on $4(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})$	-7.79	13.25	13.25	-11.58	-11.43	-6.08	-4.52
90% interval	[−12.81, −2.77]	[11.30, 14.91]	[11.30, 14.91]	[−16.01, −7.30]	[−15.67, −7.04]	[−7.35, −4.56]	[−5.64, −3.25]
Adjusted R^2	0.336	0.719	0.719	0.545	0.565	0.897	0.838

Notes: Only supply innovations are active: output fundamentals, signal noise, and belief innovations. Parameters and decision rules remain fixed within each specification; all other innovations are zero. The data column remains unconditional. Full model and No UIP risk coincide because the UIP innovation is inactive. FIRE and the no-disagreement columns use the same separate calibrations as Table F1. All model entries average 1,000 independent simulations of 116 quarters after a 1,000-quarter burn-in. ρ_1 denotes first-order autocorrelation; $\hat{e}_t$ rises with Home depreciation. $\Delta\hat{c}_t$ and $\Delta\hat{y}_t$ are Home consumption and output growth. $\hat{r}_t^{(1)}$ and $\hat{r}_t^{(40)}$ are quarterly-equivalent yields at one- and 40-quarter maturities; the factors of four in Panels C and D preserve the annualized regressors used in estimation. Panel D expresses both data and model in the Home-depreciation/Home-minus-Foreign convention; reversing both original data signs leaves the slopes unchanged. The Engel coefficient uses cumulative foreign-currency excess returns over the next 40 quarters and today's quarterly Home-minus-Foreign short-rate differential. Data intervals are 90% confidence intervals; model intervals are 5th–95th percentiles across simulations. Fama, Engel, changes-on-changes, and level regressions are untargeted.

Table F4: Exchange-rate moments conditional on the UIP-risk innovation

Moment	Data	Full model	No avg. bias
A. Exchange rates and macroeconomic variables			
$\rho_1(\Delta\hat{e}_t)$	0.013	-0.021	-0.021
$\rho_1(\hat{e}_t)$	0.950	0.886	0.886
$\sigma(\Delta\hat{e}_t)$	0.033	0.017	0.017
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{c}_t)$	6.43	9.51	9.51
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{y}_t)$	5.42	–	–
$\sigma(\Delta(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*}))/\sigma(\Delta\hat{e}_t)$	0.032	0.502	0.502
$\sigma(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})/\sigma(\Delta\hat{e}_t)$	0.110	0.681	0.681
$\sigma(\Delta(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*}))/\sigma(\Delta\hat{e}_t)$	0.021	0.026	0.026
$\sigma(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})/\sigma(\Delta\hat{e}_t)$	0.060	0.061	0.061
$\mathrm{Corr}(\Delta\hat{e}_t, \Delta(\hat{c}_t - \hat{c}_t^*))$	-0.169	-1.000	-1.000
B. Excess-return predictability			
β^{Fama}	-0.213	-0.983	-0.983
R^2_{Fama}	0.0005	0.4385	0.4385
$\beta^{\mathrm{Fama,return}}$	-1.213	-1.983	-1.983
$R^2_{\mathrm{Fama,return}}$	0.017	0.762	0.762
$\beta^{\mathrm{Engel,cumul}}_{40q}$	20.65	-1.66	-1.66
90% interval	$[6.24, 35.06]$	$[-5.29, 2.19]$	$[-5.29, 2.19]$
C. Changes-on-changes regressions			
$\Delta\hat{e}_t$ on $4\Delta(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})$	-0.50	0.47	0.47
90% interval	$[-2.03, 1.03]$	$[0.46, 0.47]$	$[0.46, 0.47]$
Adjusted R^2	-0.005	0.879	0.879
$\Delta\hat{e}_t$ on $4\Delta(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})$	-1.42	9.73	9.73
90% interval	$[-3.37, 0.52]$	$[9.68, 9.78]$	$[9.68, 9.78]$
Adjusted R^2	0.005	0.999	0.999
D. Level regressions			
$\hat{e}_t$ on $4(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})$	-3.47	0.31	0.31
90% interval	$[-6.81, -0.12]$	$[0.16, 0.42]$	$[0.16, 0.42]$
Adjusted R^2	0.222	0.174	0.174
$\hat{e}_t$ on $4(\hat{r}_t^{(40)} - \hat{r}_t^{(40)*})$	-7.79	9.00	9.00
90% interval	$[-12.81, -2.77]$	$[8.47, 9.44]$	$[8.47, 9.44]$
Adjusted R^2	0.336	0.988	0.988

Notes: Only the UIP-risk innovation is active; only specifications containing this innovation are shown. Parameters and decision rules remain fixed within each specification; all other innovations are zero. The data column remains unconditional. A dash denotes an undefined ratio because conditional output growth has zero variance. All model entries average 1,000 independent simulations of 116 quarters after a 1,000-quarter burn-in. ρ_1 denotes first-order autocorrelation; $\hat{e}_t$ rises with Home depreciation. $\Delta\hat{c}_t$ and $\Delta\hat{y}_t$ are Home consumption and output growth. $\hat{r}_t^{(1)}$ and $\hat{r}_t^{(40)}$ are quarterly-equivalent yields at one- and 40-quarter maturities; the factors of four in Panels C and D preserve the annualized regressors used in estimation. Panel D expresses both data and model in the Home-depreciation/Home-minus-Foreign convention; reversing both original data signs leaves the slopes unchanged. The Engel coefficient uses cumulative foreign-currency excess returns over the next 40 quarters and today's quarterly Home-minus-Foreign short-rate differential. Data intervals are 90% confidence intervals; model intervals are 5th–95th percentiles across simulations. Fama, Engel, changes-on-changes, and level regressions are untargeted.

F.2 Full impulse responses across model variants

Each panel compares all seven specifications in Table F1. Responses start at the deterministic steady state and follow a one-standard-deviation Home discount-factor decrease or endowment increase under each calibration; the Foreign innovation and all subsequent innovations are zero. Full-model variants share a calibration, while FIRE and the no-disagreement variants follow the table. UIP-shutdown variants coincide with their counterparts for these shocks; separate markers identify them.

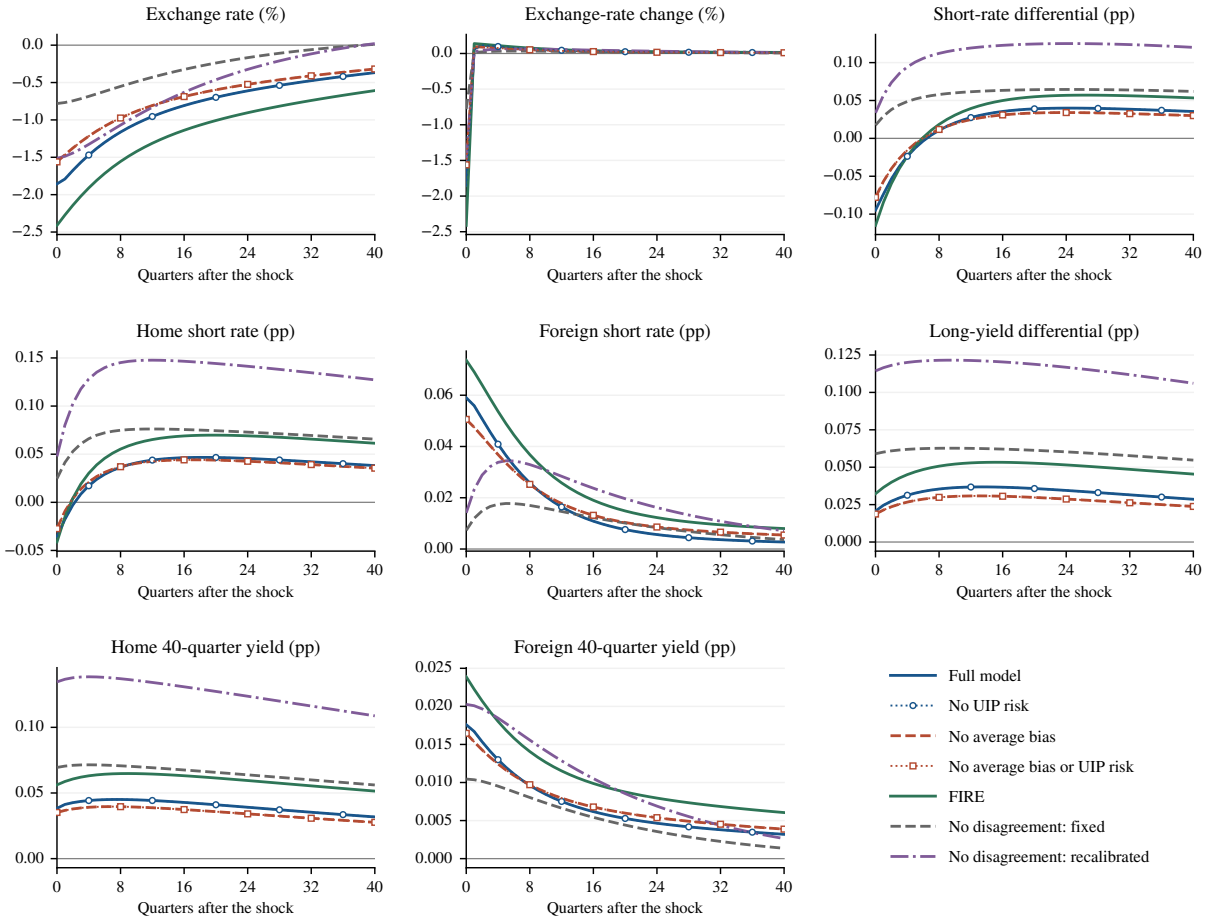


Figure F1: Home discount-factor decrease: exchange rates and yields

Notes: All seven model variants are shown over 40 quarters. The full model and No UIP risk coincide, as do No average bias and No average bias or UIP risk. An increase in the exchange rate denotes Home depreciation. Rates and yields are in percentage points per quarter; long yields have a 40-quarter maturity.

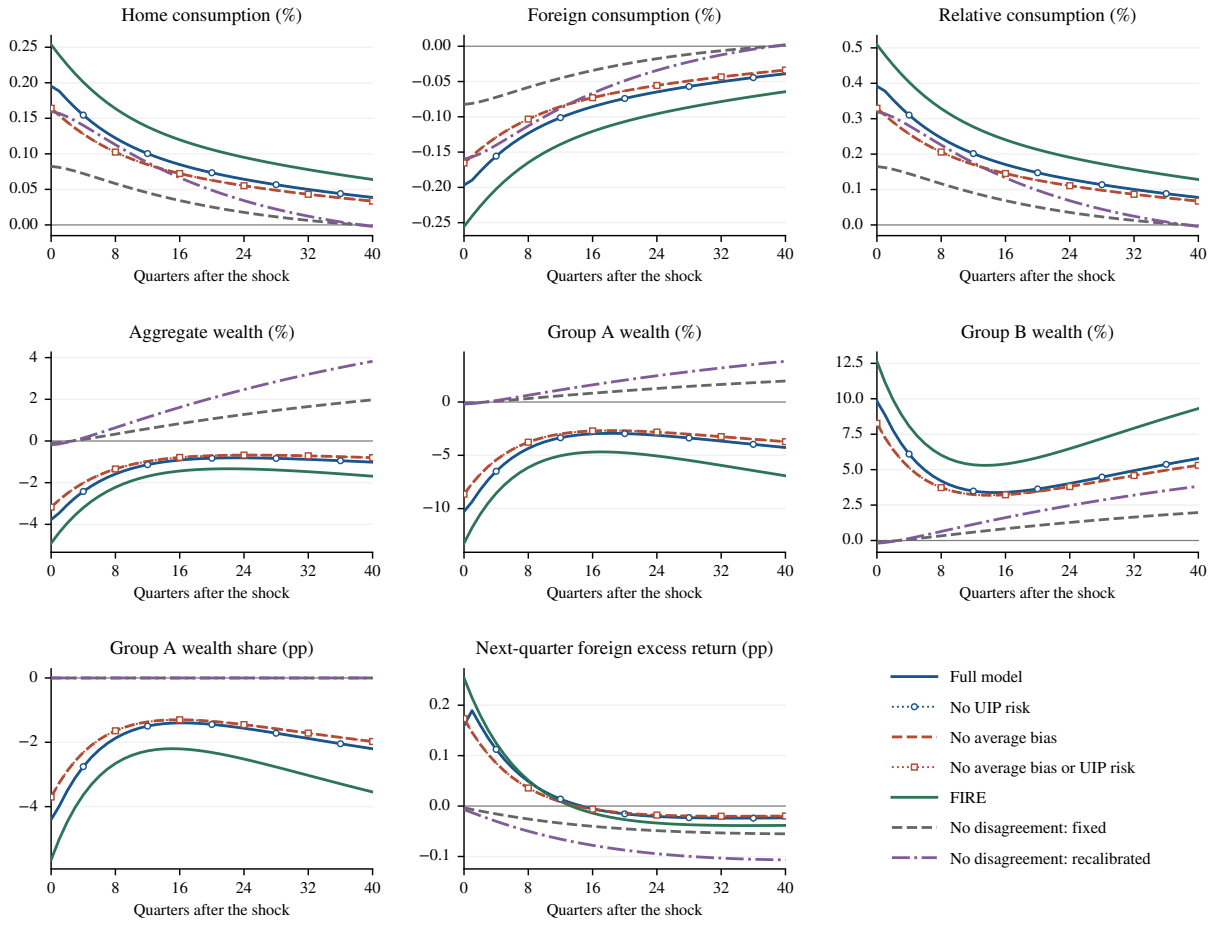


Figure F2: Home discount-factor decrease: consumption, wealth, and returns

Notes: All seven model variants are shown over 40 quarters. The full model and No UIP risk coincide, as do No average bias and No average bias or UIP risk. Consumption and wealth are percent deviations; the wealth share and next-quarter foreign excess return are in percentage points. The forward return excludes the impact valuation gain.

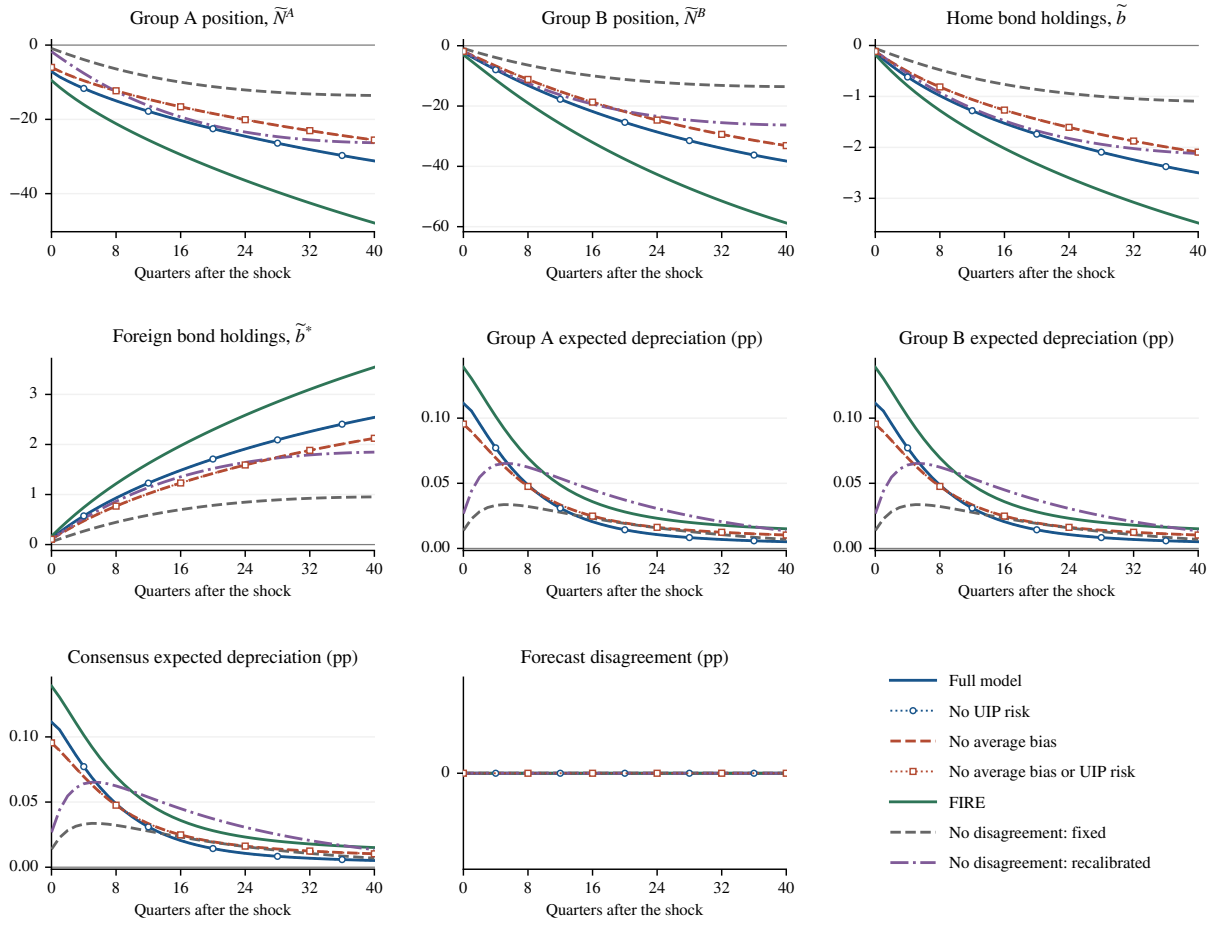


Figure F3: Home discount-factor decrease: positions and exchange-rate expectations

Notes: All seven model variants are shown over 40 quarters. The full model and No UIP risk coincide, as do No average bias and No average bias or UIP risk. Carry and bond positions are in percent of group steady-state wealth and own-country steady-state endowment, respectively. Forecasts are changes in expected next-quarter depreciation, in percentage points. The permanent belief gap is part of the steady state.

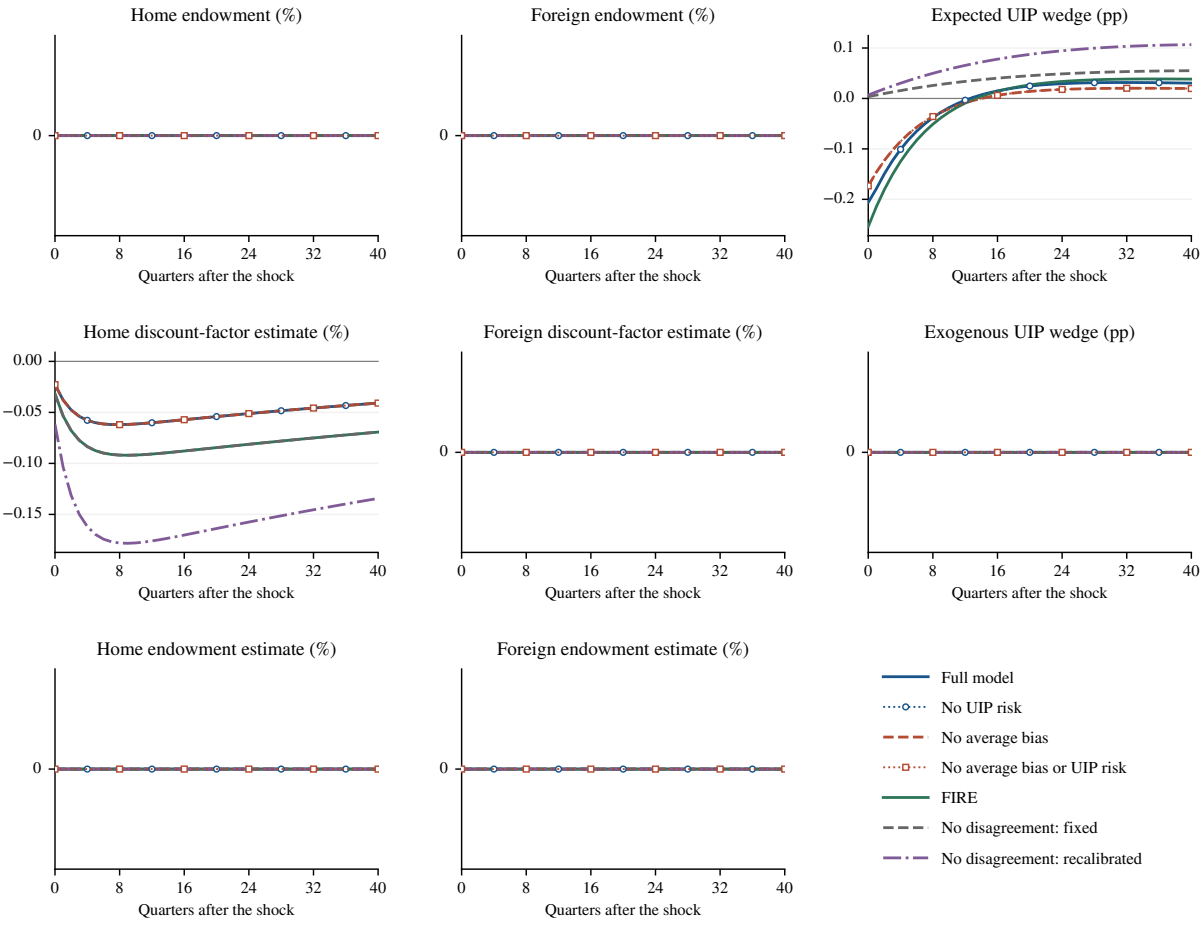


Figure F4: Home discount-factor decrease: endowments, fundamental estimates, and UIP wedges

Notes: All seven model variants are shown over 40 quarters. The full model and No UIP risk coincide, as do No average bias and No average bias or UIP risk. Fundamental estimates are posterior estimates of persistent components. The expected UIP wedge is the Home-minus-Foreign short rate less consensus expected depreciation. No Foreign endowment or UIP-risk innovation is imposed.

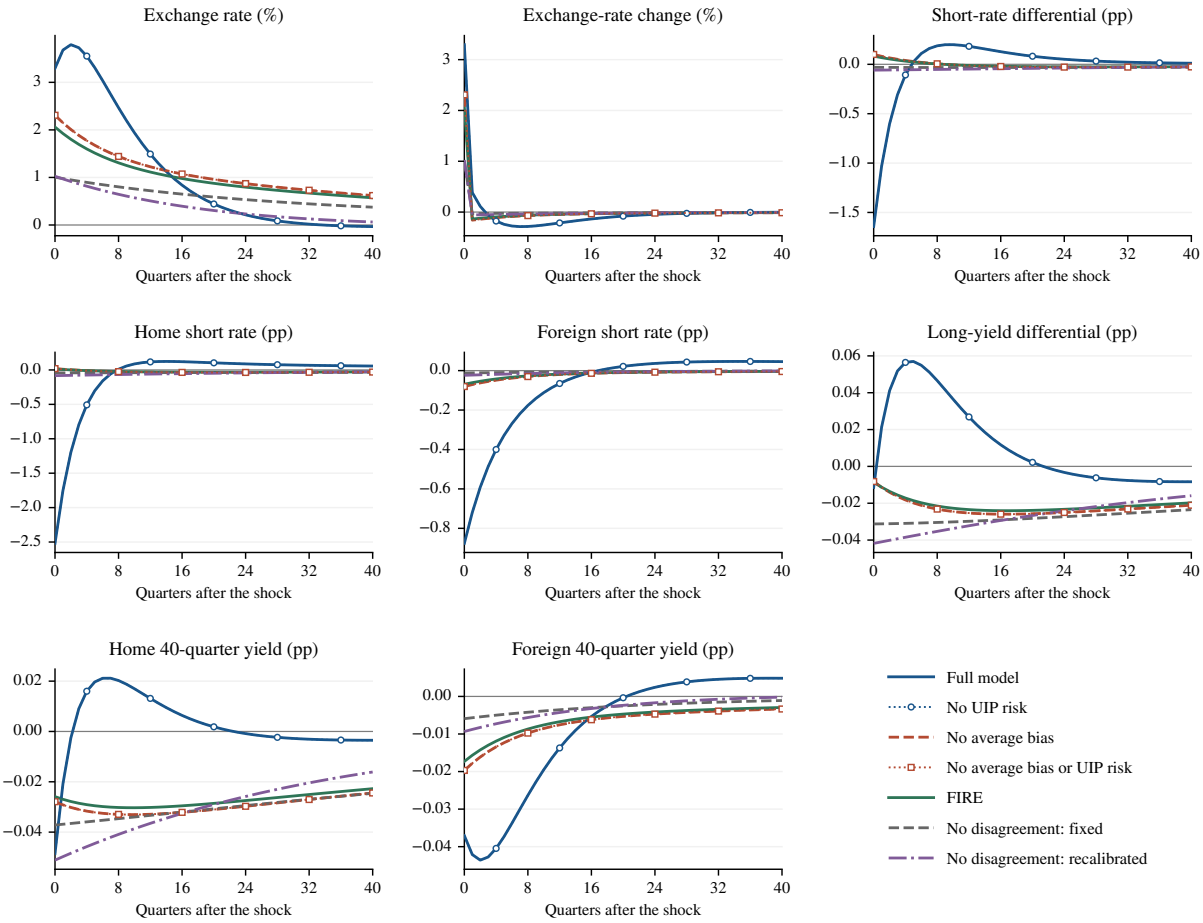


Figure F5: Home endowment increase: exchange rates and yields

Notes: All seven model variants are shown over 40 quarters. The full model and No UIP risk coincide, as do No average bias and No average bias or UIP risk. An increase in the exchange rate denotes Home depreciation. Rates and yields are in percentage points per quarter; long yields have a 40-quarter maturity.

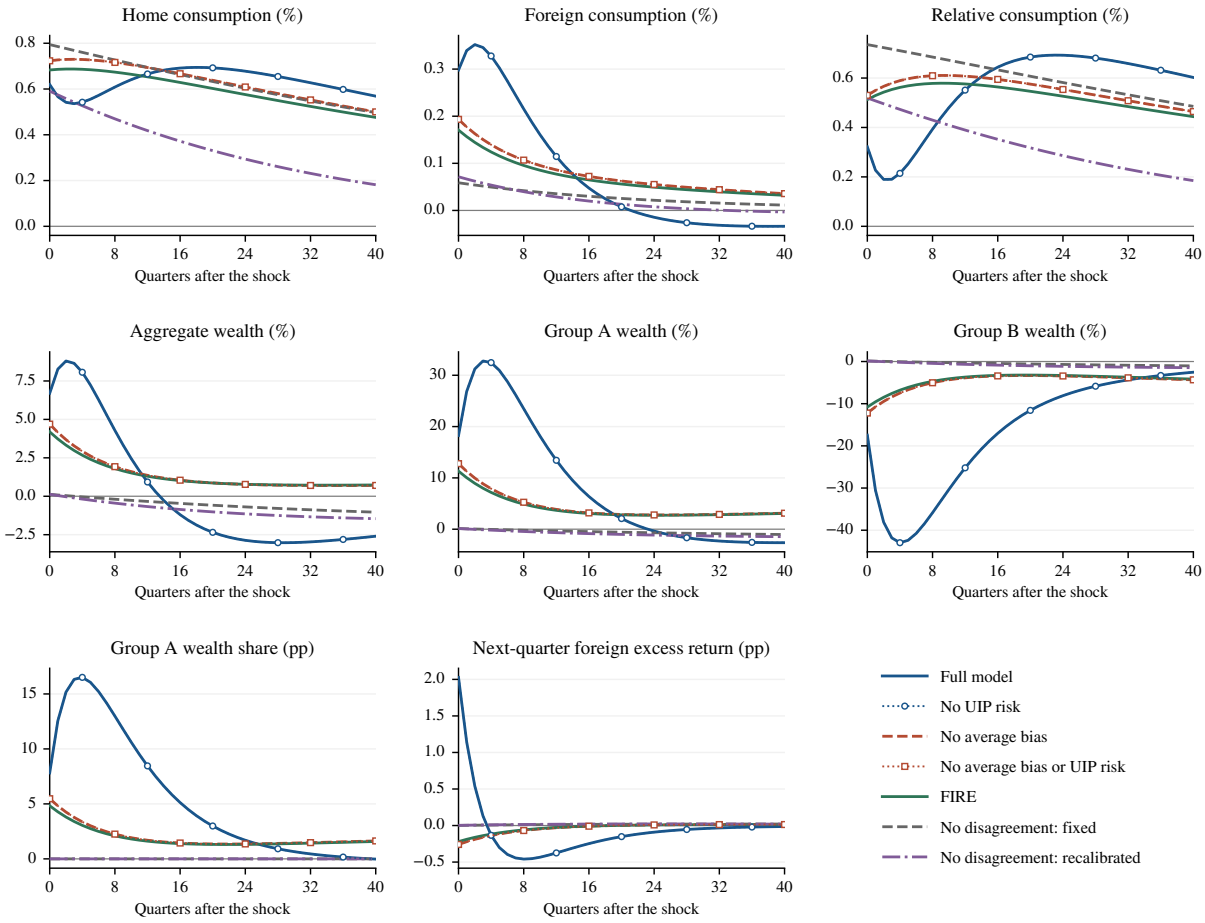


Figure F6: Home endowment increase: consumption, wealth, and returns

Notes: All seven model variants are shown over 40 quarters. The full model and No UIP risk coincide, as do No average bias and No average bias or UIP risk. Consumption and wealth are percent deviations; the wealth share and next-quarter foreign excess return are in percentage points. The forward return excludes the impact valuation gain.

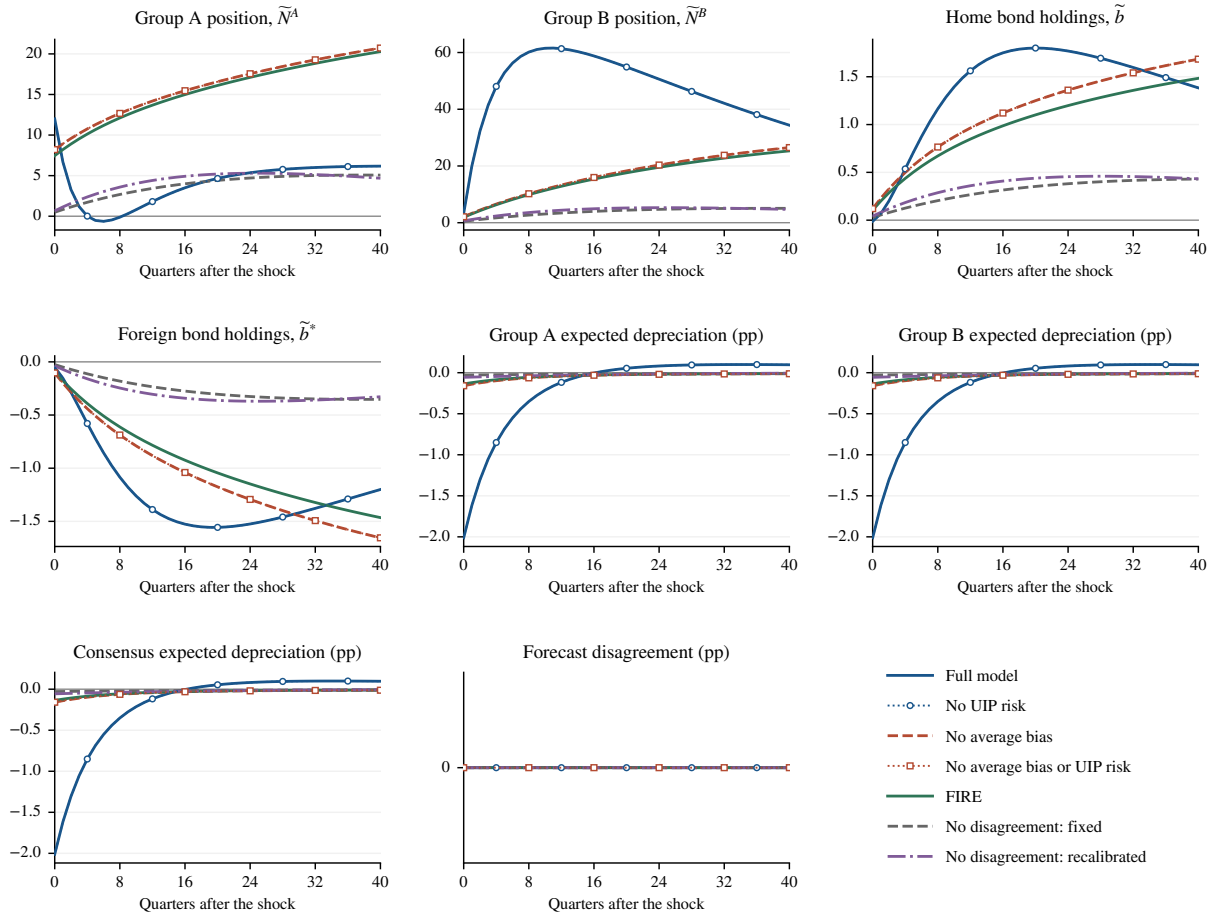


Figure F7: Home endowment increase: positions and exchange-rate expectations

Notes: All seven model variants are shown over 40 quarters. The full model and No UIP risk coincide, as do No average bias and No average bias or UIP risk. Carry and bond positions are in percent of group steady-state wealth and own-country steady-state endowment, respectively. Forecasts are changes in expected next-quarter depreciation, in percentage points. The permanent belief gap is part of the steady state.

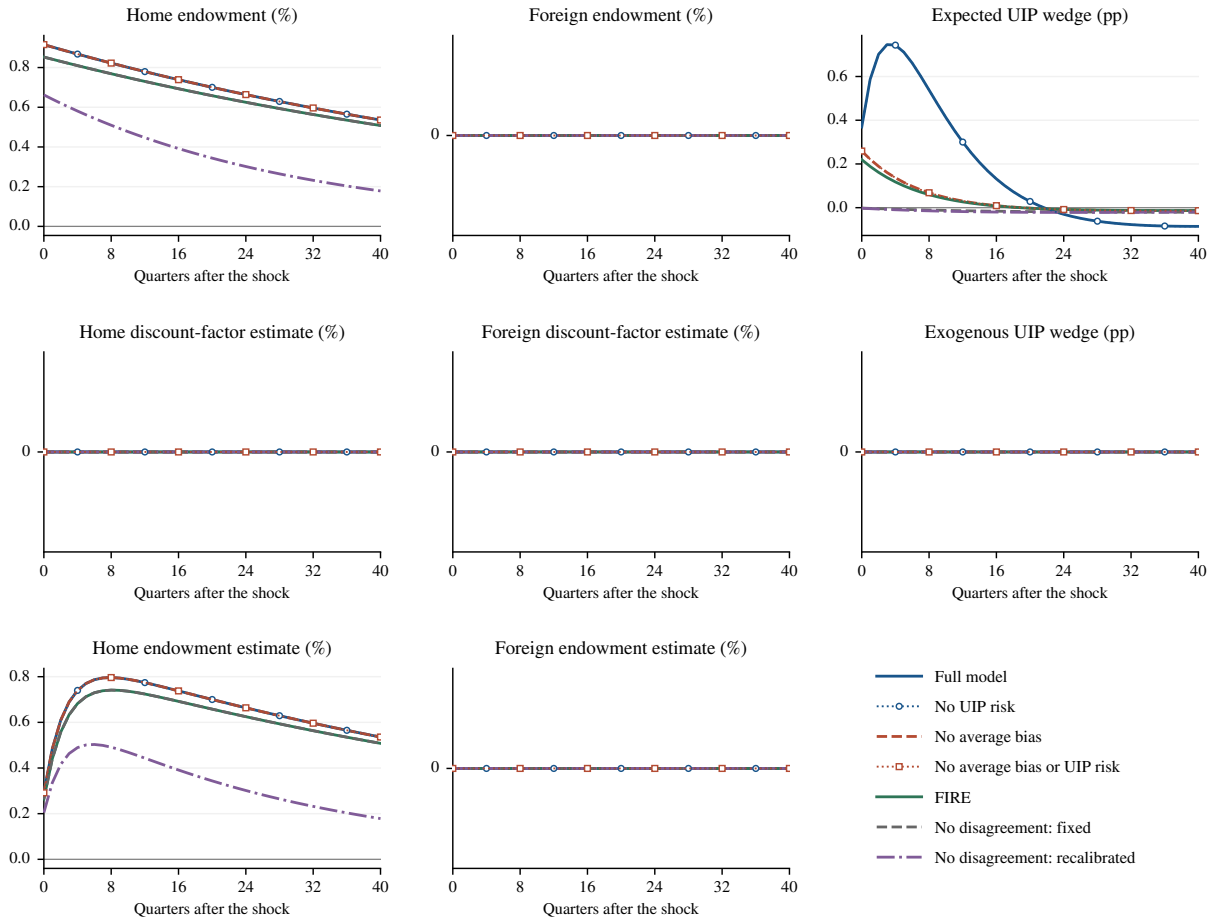


Figure F8: Home endowment increase: endowments, fundamental estimates, and UIP wedges
Notes: All seven model variants are shown over 40 quarters. The full model and No UIP risk coincide, as do No average bias and No average bias or UIP risk. Fundamental estimates are posterior estimates of persistent components. The expected UIP wedge is the Home-minus-Foreign short rate less consensus expected depreciation. No Foreign endowment or UIP-risk innovation is imposed.

F.3 Historical inversion and recursive accounting

F.3.1 Measurement and historical decomposition

The historical inversion uses the same U.S. and Fed-weighted advanced-foreign-economy data as the calibration. Both country output series are in logs per capita. I remove a common linear trend formed from the two fitted country trends, using the baseline economy's steady-state output shares, approximately 0.502 and 0.498. The trend is estimated over the original 1991Q2-2020Q1 calibration window and held fixed. Long yields are expressed in quarterly units. Macro observations and disagreement retain their original 1999Q1-2019Q2 centering.

The main inversion begins in 1999Q1 with inherited endogenous states at steady state. Initial shocks match the first observations, and the inferred states then evolve through

2019Q4. Blue Chip disagreement is the mean of the ten highest minus the mean of the ten lowest forecaster-quarter averages of the within-issue log ratio of next-quarter to current-quarter dollar forecasts. The two demand innovations, two output innovations, and one antisymmetric beta-belief innovation match the five observations in the disagreement model. The representative-agent economy uses only the four macro observations. Exchange-rate realizations and survey forecast errors are excluded from this step.

The figures plot 100 times log deviations and use a dollar-appreciation convention: the plotted e_t is the reciprocal of the nominal exchange rate expressed in Home currency per unit of Foreign currency. This sign convention applies to the historical figures. Let q_t^{app} denote the model's log real exchange rate in the same appreciation convention. I construct the nominal path as $\log e_t^{\text{model}} = q_t^{\text{app,model}} + (p_t^* - p_t)^{\text{data}}$, up to the fixed index normalization, where p_t and p_t^* are log CPIs. The observed relative CPI path is added identically to the real data and each model path. Consequently, nominal and real pointwise errors are identical. The nominal presentation is conditional on observed prices; the model does not separately predict the two nominal price levels. Every plotted path retains its 1999Q1-2019Q2 mean alignment, with no smoothing or volatility rescaling.

The historical shock decomposition replays the baseline's inferred innovations one block at a time, holding the remaining innovations at zero. For an additive real exchange-rate component q_t^j, I report its covariance allocation $\text{Cov}(q_t^j, q_t)/\text{Var}(q_t)$. These allocations sum to one and allow the inferred shock blocks to be correlated. Demand and output account for 56.5 and 43.5 percent of model real level variance, respectively, while the disagreement contribution is approximately zero. This is a decomposition of the model's fitted history, separate from the simulated forecast-error variance decomposition. In quarterly changes, the corresponding demand and output allocations are 92.2 and 7.6 percent, so the large historical output contribution concerns levels. The observed CPI component is kept separate from the structural shocks when decomposing the conditional nominal path.

The EBP comparison imposes the same externally measured UIP path at the calibrated persistence and innovation scale, then re-infers the other shocks. In the main sample, adding EBP worsens the disagreement model's level fit, although its quarterly-change correlation rises from approximately -0.009 to 0.099. The conclusion is specific to this proxy and the level comparison.

F.3.2 Extended sample

Figure F9 extends the baseline through 2024Q4, the last quarter with the joint macro and survey inputs. Data begin in 1991Q1, using an initial index anchor recovered from the saved 1991Q2 level and quarterly growth. Model inference begins in 1991Q2, the first quarter with

the calibrated yield source, and continues through 2024Q4 without a reset in 1999. The main-sample paths retain their original 1999 initialization and are not extracted from this longer history.

The shaded period before 1999 uses reported legacy trade-weighted dollar forecasts. I apply the same within-issue log forecast ratio and top-ten-minus-bottom-ten construction. The usable cross-sections contain 38-47 respondents per quarter. Early analyst codes identify respondents within each quarter; ambiguous duplicate identifiers and invalid index entries are excluded according to the archive rules. The legacy dollar index uses a different currency basket, so its disagreement series is a historical proxy for the calibrated measure. No loading is fitted to realized exchange rates, and this period is not separately recentered.

From 2022Q4 onward, also shaded, some quarters have fewer than 20 forecasters. I replace the unavailable top-minus-bottom measure with cross-sectional standard deviation mapped into its original units using a regression estimated over 1999Q1-2019Q2:

$$\widehat{D}_t = 0.0000249799 + 2.10477218 \ \mathrm{SD}_i[\log(F_{i,t,t+1}/F_{i,t,t})] \, .$$

Here $F_{i,t,t+1}$ and $F_{i,t,t}$ are respondent i's next- and current-quarter forecasts, with log ratios averaged within respondent and quarter before measuring dispersion. The frozen mapping applies throughout 2022Q4-2024Q4, including quarters in which the original statistic is available. This is a measurement extension of the survey input.

I retain calibration observations through 2020Q1 and append the same Fed-weighted AFE macro series thereafter. Index constants are aligned over the overlap. The foreign-yield source switches from the saved v3 to v4 spread vintage after 2020Q1; their endpoint gap is approximately 0.0048 basis points. The output trend is extrapolated without re-estimation. Parameters, EBP scaling, CPI conversion, and centering remain fixed. The large Covid-era level miss limits the baseline's performance over the extended sample, and the EBP comparison varies across windows and fit measures. These exercises are conditional historical accounts, not real-time forecasts.

F.3.3 Demand and beta-disagreement shocks

The heterogeneous-agent restriction retains three innovations: Home demand, Foreign demand, and beta-disagreement. All supply and signal-noise innovations are zero. I infer this history anew from the two long yields and survey disagreement; output is held out. The representative-agent benchmark retains only its two demand innovations and matches the same two yields. The right panel separately adds the EBP-implied UIP path.

Figure F10 reports the 1999Q1-2019Q4 comparison. Restricting the shock set lowers

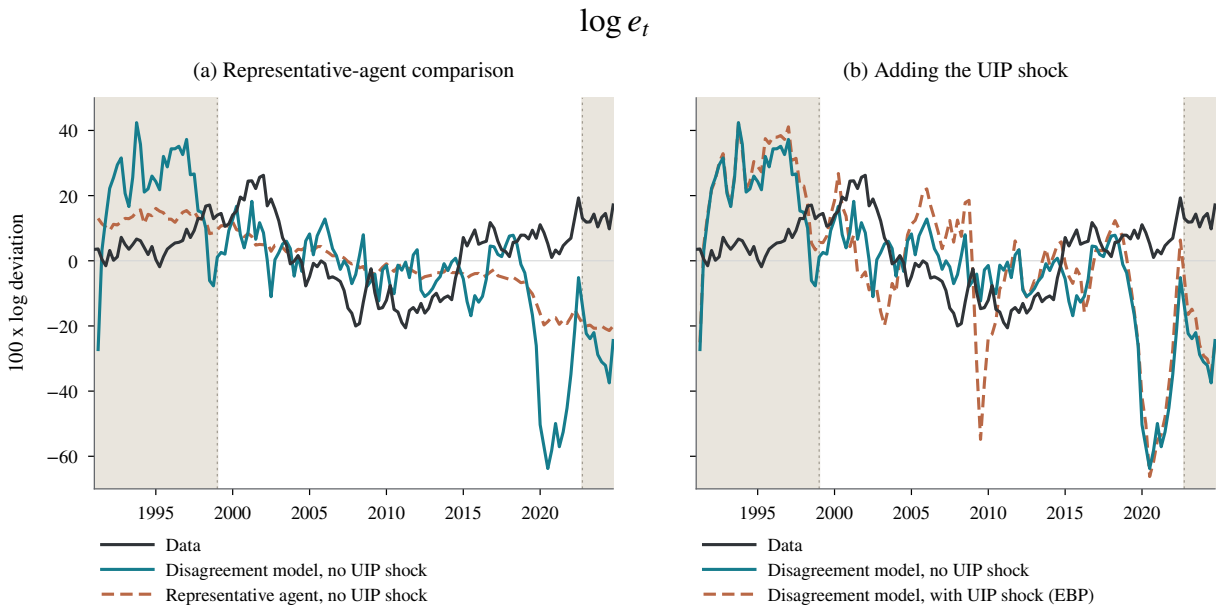


Figure F9: Historical inversion over the extended sample, 1991Q1-2024Q4

the disagreement model's level correlation from 0.497 to 0.298 and its model-to-data level standard-deviation ratio from 1.00 to 0.61. The representative-agent model has a higher correlation of 0.592 but a standard-deviation ratio of only 0.28. Adding EBP lowers the disagreement model's level correlation to 0.103. The restriction therefore preserves the representative-agent shortfall in amplitude, but it does not strengthen the disagreement model's level comovement. Figure F11 gives the corresponding extended history, subject to the same measurement qualifications as the baseline. Full and restricted figures share the same axes within each sample.

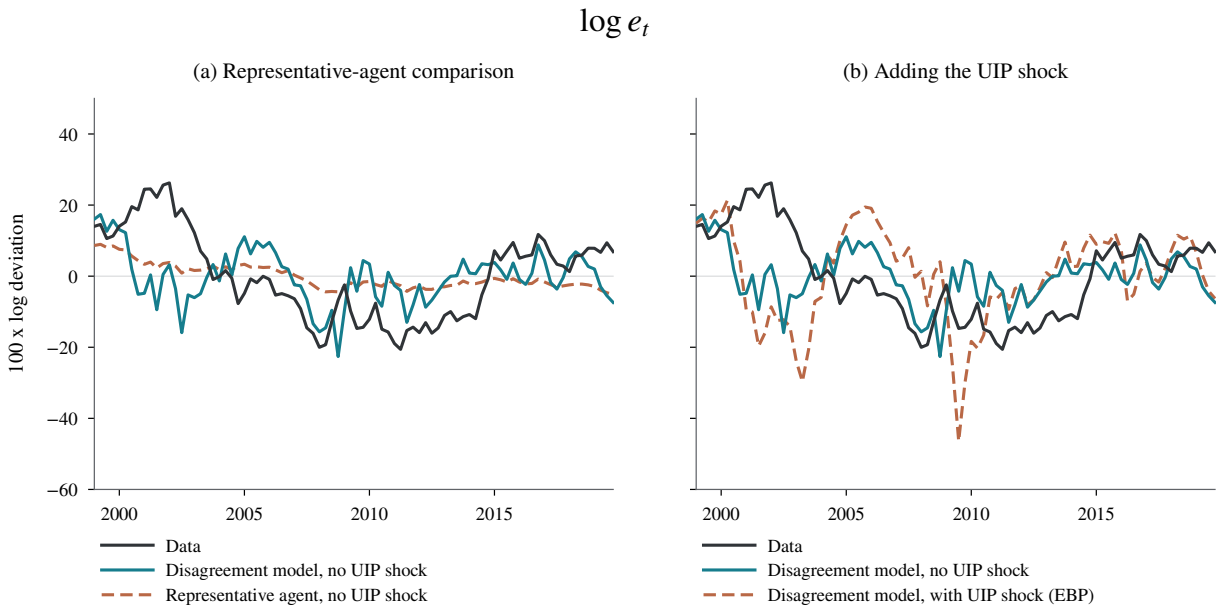


Figure F10: Demand and beta-disagreement inversion, 1999Q1-2019Q4

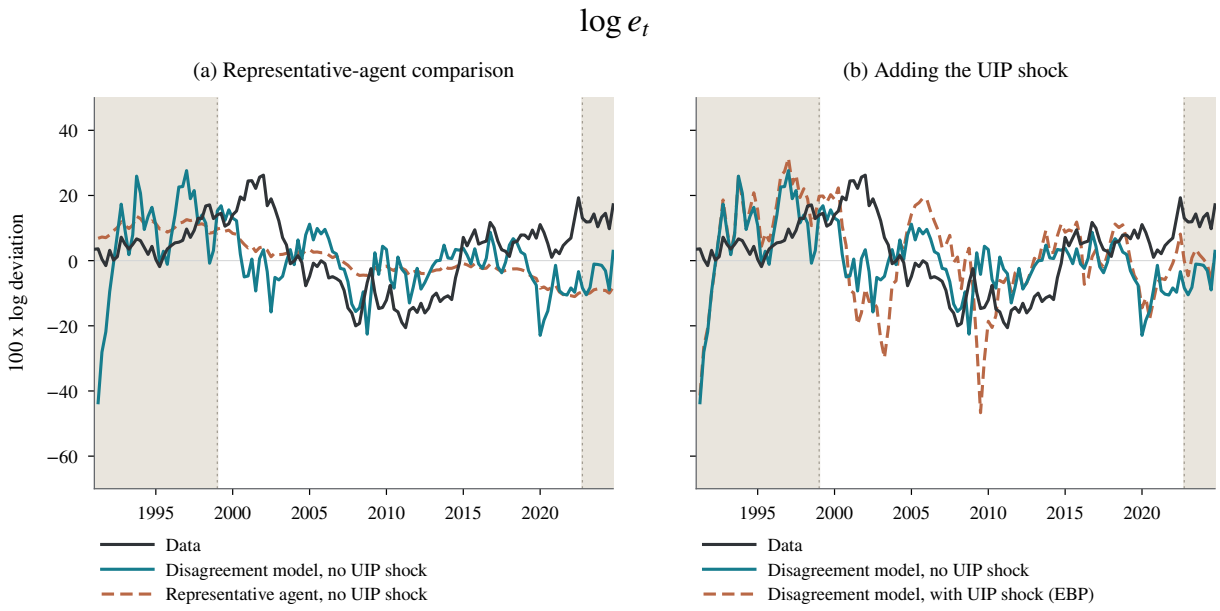


Figure F11: Demand and beta-disagreement inversion, 1991Q1-2024Q4

F.3.4 Recursive historical accounting

This complementary accounting exercise feeds observed output, interest rates, disagreement, and realized survey forecast errors into the model's core recursion. It examines how positions and wealth propagate observed news. The representative-agent/no-wealth recursion below is the earlier accounting benchmark, distinct from the solved representative-agent economy in Section 4.5. Because realized survey news contains exchange-rate realizations, the fit reported here is conditional and does not identify the origin of that news.

Specifically, it feeds observed Home and Foreign output, the rate differential, survey disagreement, and survey forecast errors into the model's core recursion[9]. In the full model, the modified UIP is written as,

$$\Delta\hat{e}_{t+1}^{\text{full}} = \hat{r}_t - \hat{r}_t^* + \varepsilon_{t+1}^e + \mathcal{N}_t - d_{ss}\widehat{\omega}_t^A - \delta_w \hat{d}_t, \tag{58}$$

and the wealth share $\hat{\omega}_t^A$ and the NFA terms $\mathcal{N}_t$ are implied from the model's equilibrium conditions, given the data.

The UIP risk version adds excess bond premium $\hat{\psi}_t$ additively in Equation 58, as in the full model and recent segmented asset market models with UIP shocks such as Itskhoki and Mukhin (2021); Kekre and Lenel (2024). Figure F12(a) first asks whether such a UIP-shock proxy improves this conditional historical fit. It does not. Without the proxy, the correlation between model and data in levels is 0.90; including it lowers the correlation to 0.63. The change correlation falls from 0.929 to 0.732 as well. The proxy helps selected unconditional and forecast-horizon moments, but it adds noise to this realized historical path.

The representative-agent/no-wealth comparison removes the wealth-share and disagreement:

$$\Delta\hat{e}_{t+1}^{\text{RA}} = \hat{r}_t - \hat{r}_t^* + \varepsilon_{t+1}^e + \mathcal{N}_t + \hat{\psi}_t. \tag{59}$$

Figure F12(b) then compares the preferred imputation, which omits the UIP-risk proxy, with the representative-agent/no-wealth recursion in equation (59). The full model raises the level correlation from 0.40 to 0.90 and lowers the level root-mean-squared error from 0.377 to 0.082, a 78 percent reduction. The corresponding correlations in quarterly changes are already 0.848 and 0.929 because both recursions share the realized survey-news term. The informative gain is instead in the low-frequency path: endogenous wealth and external positions prevent the cumulated exchange rate from drifting away from the data.

The historical accounting exercise gives the same message as the simulations. Realized

[9]Since I feed survey forecast errors into the model's recursion, this exercise must be seen as a validation of the model's core mechanism, rather than the model's ability to fit the exchange rate given non-exchange rate variables.

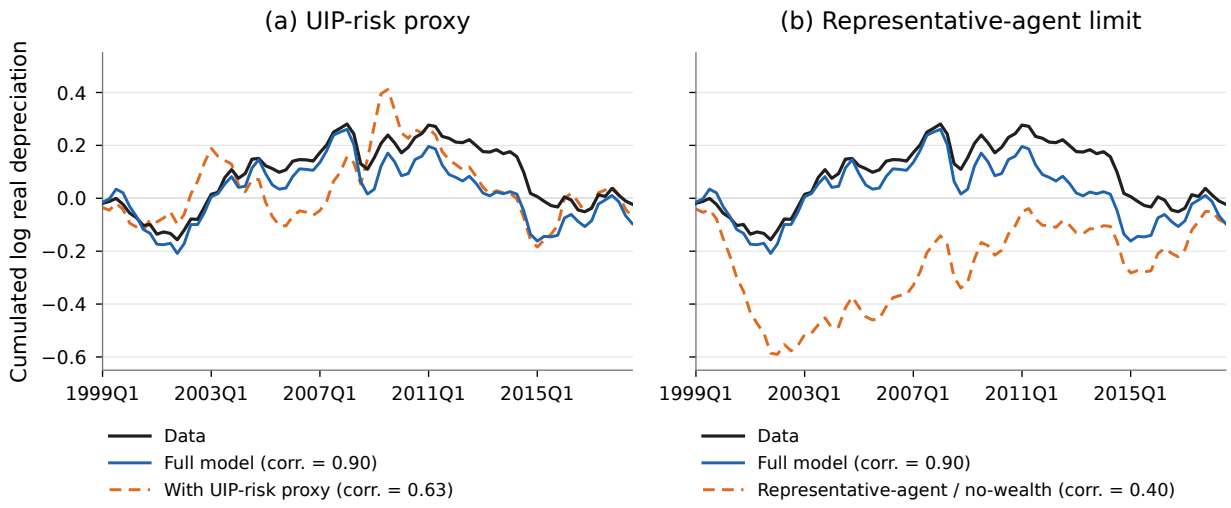


Figure F12: Historical exchange-rate imputation

survey news anchors much of the quarter-to-quarter movement in both economies. What disagreement adds is an endogenous stock-flow correction: positions redistribute wealth, wealth changes the marginal investor, and the resulting NFA and wealth-share terms keep the cumulated exchange rate from drifting away from the data. Disagreement does not matter primarily by adding another shock. It is the indispensable state through which ordinary news becomes a persistent stock–flow adjustment in the exchange rate.

The recursion figure uses the trade-weighted real dollar exchange rate. The blue line is the full recursion excluding the UIP-risk proxy; the dashed orange line adds the proxy in panel (a) and removes wealth and NFA accumulation in panel (b). All specifications share the observed rates and survey news, and paths are cumulated relative to 1998Q4. These conventions and the earlier accounting benchmark are retained for this complementary exercise.

G Additional Empirical Results

G.1 Disagreement and exchange-rate changes

The modified UIP condition, equation (5), gives disagreement a first-order role in expected currency returns. The belief component of the wedge is $-d_{ss}\hat{\omega}_t^A - \delta_w d_t$: prices depend both on the distribution of wealth across belief groups and, when their steady-state wealth shares differ, on time-varying disagreement itself. The direct disagreement term vanishes at equal wealth shares. This structure motivates testing whether survey disagreement adds information beyond observed fundamentals. It does not impose a common coefficient sign

across currencies, because survey dispersion does not identify which belief group holds more wealth.

Following the monthly explanatory-regression approach of Engel and Wu (2024), I estimate separately for each currency

$$\Delta e_{i,t} = a_i + B_i' X_{i,t} + \theta_{1i} \Delta PC_{1,i,t} + \theta_{2i} \Delta PC_{2,i,t} + u_{i,t}. \tag{60}$$

The controls $X_{i,t}$ include the change in the three-month real interest differential, foreign and U.S. year-on-year inflation, the lagged real exchange rate, changes in VIX and the three-month government-bond CIP deviation, and the U.S. goods-and-services trade balance scaled by industrial production. Real rates use nominal yields and inflation in common annual units; inflation is computed from exact twelve-month changes in the retained CPI series. This specification adapts the Engel and Wu approach to the available data. I compare the regression with and without the two disagreement terms on identical currency-month observations, using Bartlett HAC standard errors with six calendar-month lags.

Table G5 reports the additional explanatory power. Disagreement raises R^2 by a median of 0.91 percentage points across the nine currencies. The increase is 2.03 percentage points for CAD and 3.42 points for CHF, corresponding to 3.1 and 3.8 percent of the variation left unexplained by the controls. These are also the two currencies for which the disagreement terms are jointly significant at the unadjusted five-percent level. Their incremental fit remains visible when nominal short- and long-yield changes and more flexible inflation controls replace the real-rate specification, although inference is sensitive to the control set. Across the remaining currencies, the additional explanatory power is small and imprecisely estimated.

Table G5: Disagreement adds information to monthly exchange-rate regressions

Currency	N	Baseline R^2	With disag. R^2	Δ adj. R^2 (pp)	p-value
AUD	178	0.484	0.488	-0.18	0.540
CAD	178	0.351	0.371	1.33	0.025
CHF	178	0.096	0.130	2.49	0.025
EUR	178	0.340	0.348	0.04	0.458
GBP	178	0.270	0.284	0.53	0.317
JPY	175	0.141	0.150	-0.20	0.402
SEK	172	0.314	0.317	-0.51	0.726

Notes: Separate OLS regressions for nine currencies. Baseline and augmented regressions use the same months and controls listed in the text; the augmented regression adds changes in two currency-specific disagreement principal components. The sample ends in February 2021. The penultimate column reports the change in adjusted R^2 in percentage points. The final column tests both disagreement coefficients jointly using a HAC Wald F test with six calendar-month lags and a finite-sample covariance correction. The reported p-values are conditional on the estimated principal components and are unadjusted for multiple comparisons.

This evidence is consistent with disagreement carrying information about currency prices that is not fully captured by standard observables. It is a supporting diagnostic rather than a uniform rejection of models without disagreement: none of the nine currency tests rejects at five percent after a Holm adjustment for multiple comparisons. The regressions describe contemporaneous exchange-rate changes and do not identify a causal disagreement shock or establish out-of-sample predictability. The next exercise examines the link from measured beliefs to intermediaries' positions.